

SURFACE WATER AND DISTRIBUTION



Staff

Northeast Wisconsin Technical College

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This text was compiled on 04/23/2026

TABLE OF CONTENTS

Licensing

1: Source Water

- 1.1: Water Resources and the Origin of Contaminants
- 1.2: Source Water Quality

2: Surface Water Regulations

- 2.1: Microbiology Regulations
- 2.2: Water Chemistry and Standards
- 2.3: Microorganisms

3: Primary Standards

4: Secondary Standards

5: Coagulation and Flocculation

6: Sedimentation

7: Filtration

8: Disinfection

9: Chlorine

Index

Detailed Licensing

Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Source Water

[1.1: Water Resources and the Origin of Contaminants](#)

[1.2: Source Water Quality](#)

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1.1: Water Resources and the Origin of Contaminants

Learning Objectives

After reading this section, you should be able to:

- Understand and describe water sources
- Describe the water cycle
- Compare the term water contaminants with water pollutants

This chapter presents a brief introduction to the water sources that are available for potable supply; detailed information is presented in Water 132 (Water Supply). Understanding drinking water sources is important to assess the variety of contaminants that can be present in water.

Water Sources

According to the United States Geological Survey (USGS), 71% of the Earth's surface is covered in water, and of that 97% are the oceans. The Earth's water can be broken into two different categories: freshwater and salt water. Fresh water is simply water that is not salty and can be found in the planet's surface water such as streams, lakes, and frozen as ice, but also underground in aquifers. Fresh water is stored on the surface as **surface water** or stored underground in aquifers as **groundwater**. Salt water can be found in the world's oceans.

Figure 1.1.1 shows that our planet's water sources are limited. Freshwater is much easier to use for potable (drinkable) applications because it requires simpler treatment. However, it represents only a small fraction of the Earth's water portfolio. A large proportion of the freshwater is locked in glaciers and is not accessible. As a result, less than 1% of the water present on the planet is groundwater and surface water that can be more easily used as potable water source. Although desalination plants have become more prevalent, they are very costly, and their overall environmental impacts are unknown.

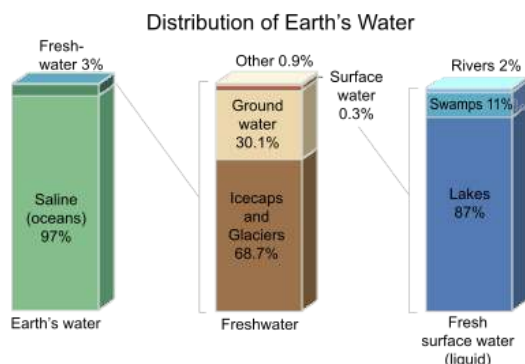


Figure 1.1.1: Distribution of Earth's Water.

The origin of a water source determines its characteristics and the presence of specific contaminants. The concentration and composition of contaminants in water is also influenced by the movements of water, which is illustrated by the water hydrologic cycle (Figure 1.1.2). The Water Cycle is both driven by gravity (i.e., water flows downhill) and sunlight (i.e., water evaporates and condenses).

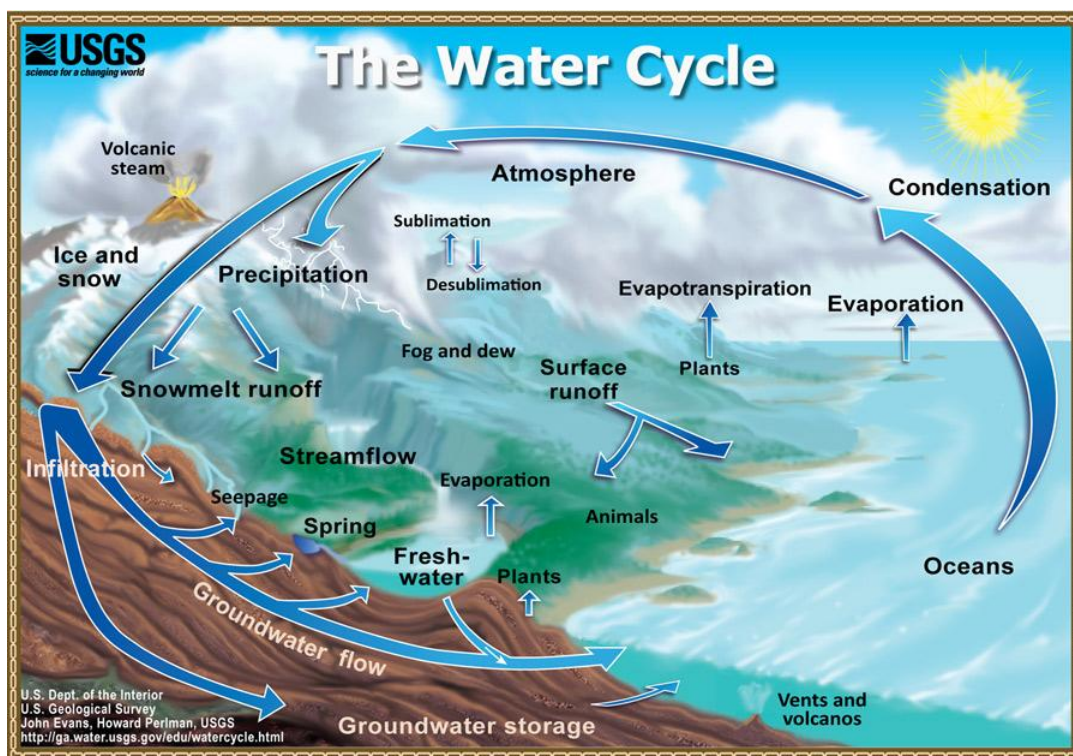


Figure 1.1.2: Water Hydrologic Cycle

Classifications of Contaminants

According to the Safe Drinking Water Act, a contaminant is any physical, chemical, biological, or radiological substance in water. While some contaminants are harmless, others are deadly. Contaminants can even mix with water molecules in the atmosphere. Technically, any compound that is not a water molecule is a contaminant when it comes to water.

Pin It! Misconception Alert

Many people think that all contaminants are manmade. They are not! Some have natural sources in the soil, such as arsenic and uranium. These can be expensive to remove because they occur naturally in the rocks around the water source.

Water can be contaminated by various human-caused (**anthropogenic**) activities or by existing natural features (**natural contaminants**) like mineral-rich geologic formations. It is not uncommon to find naturally occurring fluoride or arsenic in water.

Agricultural activities, industrial operations, landfills, animal operations, and small and large scale sewage treatment processes, among many other things, all can potentially contribute to contamination. As water runs over the land or infiltrates into the soil and **groundwater**, it dissolves material left behind by these potential contaminant sources.

There are three major groups of contamination: inorganic chemicals, organic chemicals, and biological agents. The risks and type of remediation for a contaminant depend on the type of chemicals present.

Pin It! Misconception Alert

People use the words contaminants and pollutants interchangeably. What is the difference between a contaminant and a pollutant?

- A contaminant is a substance that is not normally expected.
- A pollutant is a substance found at a concentration that has reached a level that adversely affects the suitability of the water for its intended purpose.

Key Terms

- **anthropogenic contaminants** – human-caused (manmade) contaminants
- **contaminant** – a substance that is not normally expected
- **groundwater** – water stored in the ground
- **natural contaminants** – contaminants in soil geology, erosion of soils, etc. Examples of contaminants include fluoride and arsenic
- **pollutant** – a substance found at a concentration that has reached a level that adversely affects the suitability of the water for its intended purpose
- **surface water** – water on the surface of the earth

Review Questions

1. Outline the water cycle highlighting water sources.
2. List the major contaminants of water in the environment.

Chapter Quiz

1. How much water on planet earth is easily treated for drinking water?
 1. 96%
 2. 20.9%
 3. 32%
 4. < 1.0%
2. Most of the water on earth is found in the _____.
 1. atmosphere
 2. ocean
 3. ground
 4. rivers
3. Which of the following is a natural contaminant?
 1. Pesticides
 2. Nitrates
 3. Phosphates
 4. Fluoride
4. Anthropogenic contaminants refers to _____.
 1. turbidity
 2. geological contaminants
 3. inorganic contaminants
 4. manmade contamination
5. The _____ is(are) essential for the maintenance of most life and ecosystems on the planet.
 1. climate
 2. water cycle
 3. aesthetic characteristic of water
 4. atmospheric circulation
6. The _____ drive(s) the water cycle.
 1. types of precipitation
 2. oxygen concentration
 3. nutrient levels
 4. sun

7. What portion of surface water is found in the permafrost and ice?

1. 69%
2. 68.7%
3. 96.5%
4. 30.1%

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1.2: Source Water Quality

Learning Objectives

After reading this chapter you should be able to identify and explain the following:

- The Earth's Hydrologic Cycle
- Sources of groundwater
- Sources of Surface Water
- Water math: Area and Unit Conversions

A very finite amount of water on our planet (0.34%) is available to treat for human consumption. Knowing where the water comes from assists certified operators in treating raw source water to make it potable. Newer technology has been developed to treat salinized or salty water found in the ocean. These treatment methods are still extremely expensive and not widely accessible. Supplying water to the public is an extremely important function in society as water is the basic building block of life. Because water quality is of the utmost importance new regulations and water quality standards are continually changing and evolving to make sure the public has safe sources of drinking water. The first drinking water standards were written in the Safe Drinking Water Act (SDWA), which was signed into law by President Ford in 1974. Throughout the text, new concepts will be introduced and an acronym will be given and used subsequently. You are going to learn to love acronyms if you become a certified operator. We use them quite frequently!

The Hydrologic Cycle

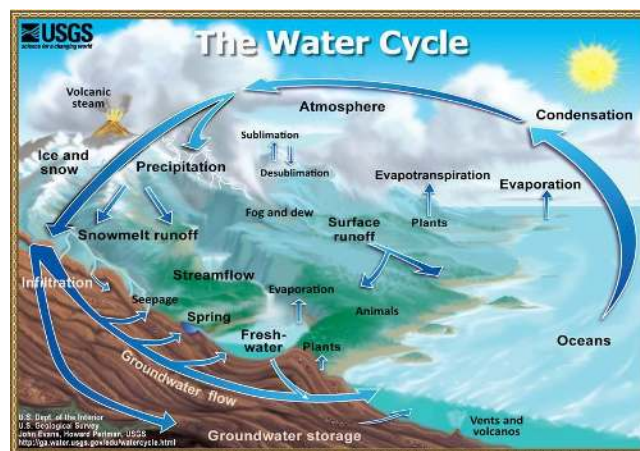


Figure 1.2.1: The Water Cycle by the USGS is in the public domain

The Hydrologic cycle is the continual movement of water on the surface of the planet. The water moves above, below, and across the Earth's surface as a liquid, gas, or solid. The 12 elements of the hydrologic cycle are defined as follows:

- Evaporation—Water moves in the gas state from the Earth's surface to the atmosphere
- Transpiration—Water moves in the gas state from plants to the atmosphere.
- Advection—Water in the gas state moves through air currents in the atmosphere
- Condensation—Water vapor converts from gas to liquid state in the form of water droplets.
- Precipitation—Water as a liquid and/or solid falls from the atmosphere. We commonly refer to this as rain, snow, sleet, and hail.
- Interception—Water in the liquid state that is captured by plants
- Infiltration—Water in the liquid state that soaks into the surface of the ground. This will be what is known as groundwater which will be detailed later in this chapter.
- Subsurface Flow—Water in the liquid state flows below the Earth's surface. The movement is generated by gravity and obstructions below the surface of the Earth.
- Runoff—Water in the liquid form travels to bodies of water. This would include the ocean, lakes, rivers, and streams. This is also known as surface water which will be detailed more later in this chapter.
- Channel flow—Water in the liquid form that flows from small channels into rivers and streams

- **Storage**—Water is naturally stored in liquid form in lakes, ponds, wetlands, and groundwater aquifers. In solid form, water is naturally stored in ice, snow, and glaciers. This natural storage provides much of the water treated for human consumption.
- **Snowmelt**—Water in the solid form converts to water in the liquid form and is returned to the hydrologic cycle. Note: In California, snowpack is a critical water supply indicator as it is the melting snowpack that recharges rivers and the general supply of water

There are other important terms to note in regard to the hydrologic cycle. The surface to atmosphere movement of water is known as evapotranspiration. This is the combination of evaporation and transpiration from plant life to the atmosphere. The most widely known movement of water is precipitation which is when water in various physical states falls from the atmosphere to the surface of the Earth. The movement of water from the surface to subsurface is known as percolation. Percolation might also be referred to as infiltration or recharge. These terms are commonly associated with underground aquifers. Finally, surface to surface flow is called runoff.

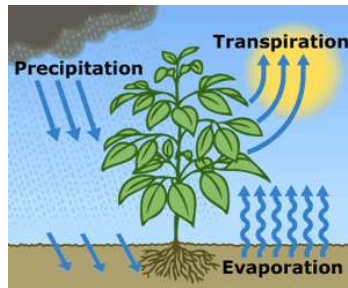


Figure 1.2.2: [Hydrologic Cycle Image](#) by the [USGS](#) is in the public domain - Evapotranspiration is the sum of evaporation from the land surface plus transpiration from plants. Precipitation is the source of all water

Groundwater

Groundwater is one of the two main sources of storage used by municipalities to produce potable water. It is formed by the **percolation** (infiltration and/or recharge) of water from the surface of the earth to the subsurface. Water moves through holes and cracks in the subsurface and collects in an underground aquifer. An aquifer is a geologic formation that accumulates water due to its porousness. Important characteristics of groundwater include consistent water quality and the ability to remain safe from surface contamination. With greater technology and testing methods chemicals and constituents known to be harmful to humans have been found in numerous well sites throughout the United States. Wells close to industrial areas have been contaminated with harmful chemicals and substances. New regulations are continually being updated to ensure source groundwater is safe to drink. In normal circumstances, very little treatment is required to yield groundwater.

The **Groundwater Treatment Rule (GWTR)** was enacted in 2006 to prevent microbial contamination from underground water supplies. The purpose of the new rule was to classify water systems that were at a greater risk for fecal contamination. These systems must employ a multiple-barrier protection similar to the surface water treatment rule which will be covered in more detail in the next section.

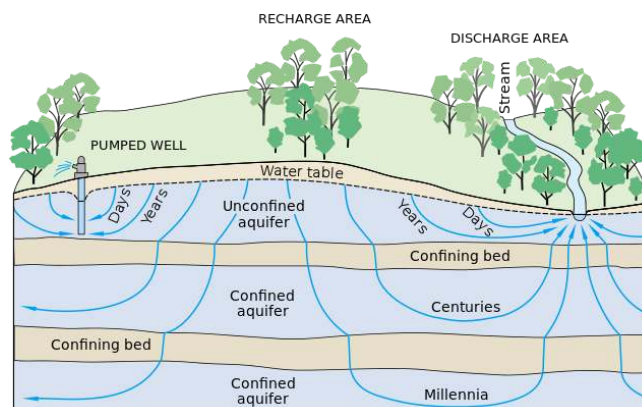


Figure 1.2.3: [Groundwater Image](#) by the [USGS](#) is in the public domain

There are two kinds of aquifers, confined and unconfined. In an unconfined or water table aquifer the water table is free to rise and fall. The water table in unconfined aquifers rise and fall depending on the amount of precipitation that recharges the aquifer. Confined aquifers, also known as artesian wells, contain water that is confined due to layers of low permeability. These layers, which restrict movement, are comprised of rock or hard clay and are referred to as confining beds, aquitards, or aquicludes. Artesian wells are generally under pressure when drilled. Once drilled the water level at which the column of water will rise is known as the piezometric surface. Sometimes, the water will rise to the surface or past the surface but in the instance, the water remains below it is known as a non-flowing artesian well.

Wells

The construction of wells is critical in the extraction of water from underground water supplies. The placement of wells is very important because proper location will produce the greatest yield. Important terms related to underground wells:

- Static Water Level—The level in the well when no water is being removed from the aquifer. This level can be measured in feet or elevation.
- Pumping water level—The level when water is being removed from the aquifer. This level can vary depending on the rate of flow from the well.
- Drawdown—The difference between the pumping water level and the static water level
- Cone of depression—The shape or “cone” created by the movement of water in all directions during pumping.
- Zone of influence—The area of water affected by the drawdown of water during pumping. It is important to note that wells cannot be placed too closely together because their zones of influence may affect each other.
- Well yield—The amount of water drawn from an aquifer over a specific period of time
- Specific capacity—The amount of water produced per drawdown expressed in gpm/ft
- Safe Yield/Perennial yield—The amount of water that can be pulled from an aquifer per year without a drop in the water table
- Overdraft—Too much water removed. Greater than the safe yield of a well
- Subsidence—The permanent drop in the water table due to overdraft

Specific Capacity Calculation

$$\text{Specific capacity} = \text{Flow (GPM)} \div \text{Drawdown (ft)} \quad (1.2.1)$$

Example 1.2.1

A well has a yield of 600gpm and the drawdown is 50 ft. What is the specific capacity of the well?

Solution

This is a direct application of Equation 1.2.1

$$\text{Specific Capacity} = 600 \text{ GPM} \div 50 \text{ ft} = 12$$

Surface Water

Throughout the United States, surface water is the most widely used source of water for large cities and other municipalities. Groundwater is not as widely available so is not a sufficient water supply for major cities. Surface water includes lakes, ponds, rivers, and streams. California is unique as the southern half of the state has 2/3 of the population but only 1/3 of the available water and the northern half of the state has 1/3 of the population but 2/3 of the water. Snowpack in northern California is critical to the state's water supply. Most of Southern California is very arid and densely populated so water travels from Northern California through the State Water Project to Southern California.

Surface runoff supplies water for all surface water sources. Influences of surface runoff include intensity of rainfall, duration of rainfall, composition of soil, amount of moisture in soil, slope of the ground, vegetation coverage, and human influences. One would think that a lot of rain would be ideal. However, if the rainfall density is too great, more water can be lost because the ground will no longer absorb the water. The same applies to the duration of rainfall. A prolonged rain even makes the soil too moist and unable to capture water. Vegetation coverage and the slope of the ground are very important to stopping runoff. If the slope of the ground is steep, the speed of runoff is increased. Vegetation slows the speed of runoff and allows more water to absorb or

infiltrate into the ground. Human influences have a great impact on water runoff. Impervious surfaces like concrete entirely prevent infiltration.

Natural Watercourses

Key Terms

Natural water courses include rivers, creeks, streams, washes, and arroyos. They flow in one of three ways:

- Perennial streams—Watercourses which flow continuously throughout the year
 - EX: Colorado River
- Ephemeral streams—Watercourses that flow sporadically, generally after rainfall
 - EX: Santa Clara River which runs through Santa Clarita and Ventura County
- Intermittent—Watercourses which flow somewhere between ephemeral and perennial streams. Rainfall and high groundwater levels will affect how often these streams flow.

Rivers and Streams are a good water supply source but are not necessarily the best source for public water supplies.



Figure 1.2.4: Image of the Colorado River by Paul Hermans is licensed under CC BY-SA 3.0



Figure 1.2.5: Santa Clara River Map by Shannon1 is in the public domain

Lakes

Lakes are the most widely used public water supply source. However, very few “natural” lakes exist. Most lakes used for public water supply are man-made and use a dam to create the lake and contain the water. This is known as an impoundment. Water from these lakes is piped to treatment facilities. Due to variances in temperature lakes develop “layers” also known as stratification. Denser, colder water will drop to the bottom (Benthic zone) of a lake during the summer. There are three layers:

- Epilimnion—The strata closest to the surface
- Hypolimnion—The strata near the bottom
- Thermocline—Middle strata with the greatest variance in temperature

Lake turnover will occur during seasonal temperature changes. When the temperature of a lake is uniform it is known as isothermal. Algae growth is a serious problem that causes taste and odor issues in treated water. Copper sulfate can be added to a lake to help remedy algae blooms. In severe cases, the water undergoes eutrophication which is the loss of oxygen. Complete or extreme oxygen depletion can kill all living creatures in the water including animals and fish.



Figure 1.2.6: Image of Castaic Lake by Rehman is in the public domain

Introduction to Water Math

Mathematics is a key component of water treatment. Operators use conversion tables and basic algebra to complete many daily tasks. Charts are available on the State Water Resources Control Board website that assist operators with most calculations you would find on the state exams or while on the job. Below is a list of basic units and their respective conversion factors:

Table 1.1: Units and Conversion Factors

Measurement	Equivalent
1 cubic foot of water	62.3832lbs
1 gallon of water	8.34lbs
1 liter of water	1,000 grams
1 mg/L	1 part per million (ppm)
1 ug/L	1 part per billion (ppb)
1 mile	5,280 feet
1 yard	3 feet
1 yard ³	27ft ³
1 acre	43,560 square feet or ft ²
1 cubic foot or ft ³	7.48 gallons
1 gallon	3.785 Liters or L
1 L	1,000 milliliters (ml)
1 pound	454 grams

Working with Fractions

NumeratorDenominator

$\frac{4}{4} = \frac{11}{11}$ or $\frac{1}{1}$ $\frac{42}{21} = \frac{2}{1}$ or $\frac{2}{1}$

Rounding

Rounding the number 324.179	
3 = Hundreds	The above number (324.179) rounded to the nearest tenth would be 324.2. Since the number in the hundredth place is at or above 5, the number in the tenth place is rounded up. If you were rounding this number to the nearest whole number it would be 323 since the number in the tenth place is below 5.
4 = Tens	
9 = Units	
1 = Tenth	
7 = Hundredth	
9 = Thousandth	

Unit Dimensional Analysis

This is the most important function in most water math problems. Make sure to always place your factors in the proper place or the equation will be impossible to solve correctly. The example below will use unit measurements. Water operators will commonly convert between different units of measurement.

Example 1

Convert 48 inches into feet. (There are 12 inches in a foot.)

$$48 \text{ inches} \times \frac{1 \text{ foot}}{12 \text{ inches}} = 4 \text{ ft}$$

Drop the one from the denominator and the answer is 4 ft.

The same process will be to convert between gallons and cubic feet.

Example 2

Convert 22.44 gallons into cubic ft. or ft³ (There are 7.48 gallons in one cubic foot.)

$$22.44 \text{ gallons} \times \frac{1 \text{ ft}^3}{7.48 \text{ gallons}} = 3 \text{ ft}^3$$

Drop the 7.48 from the denominator and your answer is 3 ft³

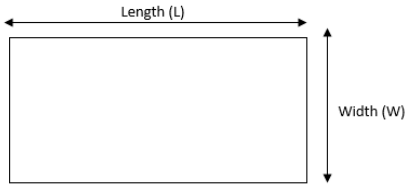
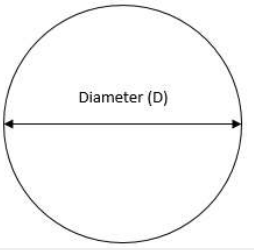
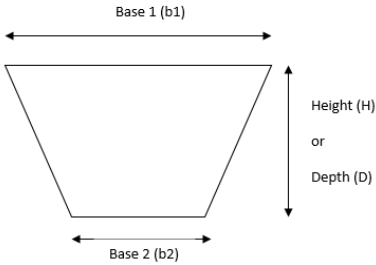
Example 3

How many gallons are there in a tank which holds 300ft³ of water?

$$300 \text{ ft}^3 \times \frac{7.48 \text{ gal}}{1 \text{ ft}^3} = 2,244 \text{ gallons}$$

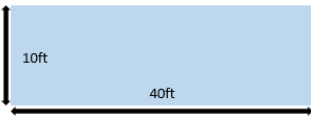
Area

Area will be important for many applications in water math. It may be necessary to calculate the area of a tank that requires painting or an area of ground cover near equipment.

Rectangles 	$\text{Area} = L \times W$
Circle 	$\text{Area} = 0.785 \times D^2$
Trapezoid 	$\text{Area} = \frac{b1 + b2}{2} \times H$

Example 1

A wall that is 10ft wide and 40ft in length needs to be painted. What is the total square feet of the wall?

	$\begin{aligned}\text{Area} &= \text{Length} \times \text{Width} \\ \text{Area} &= 10\text{ft} \times 40\text{ft} \\ \text{Area} &= 400\text{ft}^2\end{aligned}$
---	--

Example 2

What is the area of the top of a circular storage tank that is 100 feet in diameter? (Note: Use the formula $.785 \times d^2$. In most cases in water math, we will always be dealing with the diameter and not the radius. These short cuts will help as the problems become more difficult. In this problem I will demonstrate why we will use this formula and not the standard mathematical equation for solving area problems.)

Equation #1 $\text{Area} = \pi (3.14) \times r^2$ (100ft $\div$ 2 = 50 to find radius)

$\text{Area} = 3.14 \times 50 \times 50$

$\text{Area} = 7,850 \text{ ft}^2$

Equation #2 $\text{Area} = .785 \times d^2$

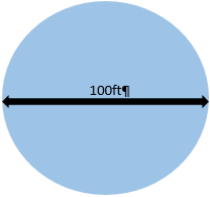
$\text{Area} = .785 \times 100 \times 100$

$$\text{Area} = 7,850 \text{ ft}^2$$

In this example, the first equation was easy to solve because we were working with a pretty friendly number. As the equations become more difficult we do not want to take the extra step to divide the diameter by two to solve for the radius. If the diameter of the tank was 357ft, this problem would have been slightly more difficult using the first equation.

Example 3

The top of a circular storage tank needs to be painted. It is 100 ft. in diameter. Each gallon of paint covers approximately 200 square feet. How many gallons of paint will you need to buy?

	$\text{Area} = .785 \times d^2$ $\text{Area} = .785 \times 100\text{ft} \times 100\text{ft}$ $\text{Area} = 7,850 \text{ ft}^2$ $7,850 \text{ ft}^2 \div 200 \text{ ft}^2/\text{gal} = 39.25 \text{ gal}$
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Chapter Review

- What is the middle layer of a stratified lake called?
 - Thermocline
 - Benthic Zone
 - Epilimnion
 - Hypolimnion
- What is the conversion of liquid water to gaseous water known as?
 - Advection
 - Condensation
 - Precipitation
 - Evaporation
- Water weighs _____.
 - 7.48 lbs/gal
 - 8.34 lbs/gal
 - 62.4 lbs/ft³
 - Both 2 and 3
- What is the static level of an unconfined aquifer also known as?
 - Drawdown
 - Water Table
 - Pumping Water Level
 - Aquitard
- What is the cause of taste and odor problems in raw surface water?
 - Copper sulfate
 - Blue-green algae
 - Oxygen
 - Lake turnover
- What chemical reduces blue-green algae growth?
 - Chlorine
 - Caustic Soda
 - Copper Sulfate
 - Alum

7. A water-bearing geologic formation that accumulates water due to its porousness.
 1. Aquifer
 2. Lake
 3. Aquiclude
 4. Well
8. What kind of stream flows continuously throughout the year?
 1. Ephemeral
 2. Perennial
 3. Intermittent
 4. Stratified
9. The surface to atmosphere movement of water is known as _____.
 1. Precipitation
 2. Percolation
 3. Stratification
 4. Evapotranspiration
10. An aquifer that is underneath a layer of low permeability is known as _____.
 1. Confined aquifer
 2. Water Table aquifer
 3. Unconfined aquifer
 4. Unreachable groundwater
11. What is the middle layer of a stratified lake known as?
 1. Hypolimnion
 2. Benthic Zone
 3. Thermocline
 4. Epilimnion
12. The amount of water that can be pulled from an aquifer without depleting.
 1. Drawdown
 2. Safe yield
 3. Overdraft
 4. Subsidence

Math Questions

Please show all work. On the State exams, you will not get credit if work is not shown.

1. What is the area of the top of a storage tank that is 75 feet in diameter?
 1. 4,000 ft²
 2. 4416 ft²
 3. 1104 ft²
 4. 17,663 ft²
2. What is the area of a wall 175 ft. in length and 20 ft. wide?
 1. 3,000 ft²
 2. 2,500 ft²
 3. 3,500 ft²
 4. 4,000 ft²
3. You are tasked with filling an area with rock near some of your equipment. One (1) bag of rock covers 250 square feet. The area that needs rock cover is 400 feet in length and 30 feet wide. How many bags do you need to purchase?
 1. 40 Bags
 2. 42 Bags

3. 45 Bags

4. 48 Bags

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CHAPTER OVERVIEW

2: Surface Water Regulations

[2.1: Microbiology Regulations](#)

[2.2: Water Chemistry and Standards](#)

[2.3: Microorganisms](#)

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2.1: Microbiology Regulations

Learning Objectives

After reading this section, you should be able to:

- Explain the Total Coliform Rule
- Compare the Groundwater Treatment Rule and Surface Water Treatment Rule
- Describe the concern with Cryptosporidium

Total Coliform Rule

The **Total Coliform Rule** (TCR), a National Primary Drinking Water Regulation (NPDWR), was published in 1989 and became effective in 1990. The rule sets a health goal (Maximum Contaminant Level Goal (MCLG)) and legal limits (Maximum Contaminant Levels, MCLs)) for the presence of total coliforms in drinking water.

The Environmental Protection Agency (EPA) published the Revised Total Coliform Rule (RTCR) in the Federal Register (FR) in February 2013 (78 FR 10269) and minor corrections in February 2014 (79 FR 10665). The RTCR is the revision to the 1989 Total Coliform Rule (TCR) and is intended to improve public health protection. All public water systems (PWSs), except aircraft PWSs subject to the Aircraft Drinking Water Rule (ADWR) (40 CFR 141 Subpart X), must comply with the RTCR starting April 2016, or an earlier state effective date.

Total coliforms are a group of related bacteria that are, with few exceptions, not harmful to humans. This is very important to note. The EPA considers total coliforms a useful indicator of other pathogens for drinking water. Total coliforms are used to determine the adequacy of water treatment and the integrity of the distribution system.

Key provisions of the RTCR include:

1. Setting a maximum contaminant level goal (MCLG) and maximum contaminant level (MCL) for E. coli for protection against potential fecal contamination.
2. Setting a total coliform treatment technique (TT) requirement.
3. Requirements for monitoring total coliforms and E. coli according to a sample siting plan and schedule specific to the PWS.
4. Provisions allowing PWSs to transition to the RTCR using their existing Total Coliform Rule(TCR) monitoring frequency, including PWSs on reduced monitoring under the existing TCR.
5. Requirements for seasonal systems (such as, Non-Community Water Systems not operated on a year-round basis) to monitor and certify the completion of a state-approved start-up procedure.
6. Requirements for assessments and corrective action when monitoring results show that PWSs may be vulnerable to contamination.
7. Public notification (PN) requirements for violations.
8. Specific language for CWSs to include in their Consumer Confidence Reports (CCRs) when they must conduct an assessment or if they incur an E. coli MCL violation.

Revised Total Coliform Rule

Overview of the Rule

Revised Total Coliform Rule (RTCR) 78 FR 10269, February 2013, Vol. 78, No. 30

Purpose: Increase public health protection through the reduction of potential pathways of entry for fecal contamination into distribution systems.

General Description

The rule establishes a maximum contaminant level (MCL) for E. coli and uses E. coli and total coliforms to initiate a find and fix approach to address fecal contamination that could enter into the distribution system. It requires public water systems (PWSs) to perform assessments to identify sanitary defects and subsequently take action to correct them.

The Total Coliform Rule applies to all Public Water Systems.

Implementation of the Total Coliform Rule will result in:

1. A decrease in the pathways by which fecal contamination can enter the drinking water distribution system.
2. Reduction in fecal contamination should reduce the potential risk from all waterborne pathogens including bacteria, viruses, parasitic protozoa, and their associated illnesses.

Requirements:

1. Public Water Systems must develop a written sample siting plan that identifies the system's sample collection schedule and all sample sites, including sites for routine and repeat monitoring.
2. PWSs monitoring quarterly or annually must also identify additional routine monitoring sites in their sample siting plans.
3. Sample siting plans are subject to state review and revision.

Routine Sampling Requirements

1. Total coliform samples must be collected by Public Water Systems at sites which are representative of water quality throughout the distribution system according to a written sample siting plan subject to state review and revision.
2. For Public Water Systems collecting more than one sample per month, collect total coliform samples at regular intervals throughout the month, except that ground water systems serving 4,900 or fewer people may collect all required samples on a single day if the samples are taken from different sites.

Repeat Sampling Requirements

Within 24 hours of learning of a Total Coliform positive routine sample result, at least 3 repeat samples must be collected and analyzed for total coliform:

1. One repeat sample must be collected from the same tap as the original sample.
2. One repeat sample must be collected from within five service connections upstream.
3. One repeat sample must be collected from within five service connections downstream.
4. The PWS may propose alternative repeat monitoring locations that are expected to better represent pathways of contamination into the distribution system.

If one or more repeat sample is Total Coliform Positive:

1. The TC+ sample must be analyzed for the presence of *E. coli*.
2. If any repeat TC+ sample is also EC+, then the EC+ sample result must be reported to the state by the end of the day that the PWS is notified.
3. The PWS must collect another set of repeat samples, unless an assessment has been triggered and the PWS has notified the state.

Routine Sampling Requirements

1. Each total coliform-positive (TC+) routine sample must be tested for the presence of *E. coli*.
2. If any coliform-positive (TC+) sample is also *E. coli*-positive (EC+), then the EC+ sample result must be reported to the state by the end of the day that the PWS is notified.
3. If any routine sample is TC+, repeat samples are required.
4. Public Water Systems on quarterly or annual monitoring must take a minimum of three additional routine samples (known as additional routine monitoring) the month following a TC+ routine or repeat sample.
5. Reduced monitoring may be available for Public Water Systems using only ground water and serving 1,000 or fewer persons that meet certain additional Public Water System criteria.

Assessments and Corrective Action

The RTCR requires PWSs that have an indication of coliform contamination (e.g., as a result of TC+ samples, *E. coli* MCL violations, performance failure) to assess the problem and take corrective action. Two levels of assessments exist (i.e., Level 1 and Level 2) based on the severity or frequency of the problem.

Purpose of Level 1 and Level 2 Assessments

To find sanitary defects at the Public Water System including:

1. Sanitary defects that could provide a pathway of entry for microbial contamination
2. Sanitary defects that indicate failure (existing or potential) of protective barriers against microbial contamination.

Deadline for Completing Corrective Actions

When sanitary defects are identified during a Level 1 or Level 2 Assessment, they should be corrected as soon as possible to protect public health. The Public Water System must complete corrective actions by one of the following timeframes:

1. No later than the time the assessment form is submitted to the state, which must be within 30 days of triggering the assessment,
or
2. Within state-approved timeframe which was proposed in the assessment form.

Level 1 Assessments

Conducting Level 1 Assessments

1. Performed by the PWS owner or operator each time a Level 1 Assessment is triggered.
2. Upon trigger of a Level 1 Assessment, the Level 1 Assessment form must be submitted within 30 days to the state.

Level 1 Assessment Triggers

Level 1 Assessment is triggered if any one of the following occurs:

1. A Public Water System collecting fewer than 40 samples per month has 2 or more TC+ routine/ repeat samples in the same month.
2. A PWS collecting at least 40 samples per month has greater than 5.0 percent of the routine/repeat samples in the same month that are TC+.
3. A PWS fails to take every required repeat sample after any single TC+ sample.

Level 2 Assessments

Conducting Level 2 Assessments

Performed by the state or state-approved entity each time a Level 2 Assessment is triggered.

1. The Public Water System is responsible for ensuring that the Level 2 Assessment is conducted regardless of the entity conducting the Level 2 Assessment.
2. (3) Upon trigger of a Level 2 Assessment, the Level 2 Assessment form must be submitted within 30 days to the state.

Level 2 Assessment Triggers

Level 2 Assessment is triggered if any one of the following occurs:

1. A Public Water System incurs an E. coli MCL violation.
2. A Public Water System has a second Level 1 Assessment within a rolling 12-month period.
3. A Public Water System on state-approved annual monitoring has a Level 1 Assessment trigger in 2 consecutive years.

Seasonal System Provisions

The Total Coliform Rule defines seasonal systems and specifies additional requirements for these types of PWSs:

1. A seasonal system is defined as a non-community water system that is not operated as a PWS on a year round basis and starts up and shuts down at the beginning and end of each operating season.

Start-up Procedures for Seasonal Systems

At the beginning of each operating period, before serving water to the public, seasonal water systems must:

1. Conduct state-approved start-up procedures.
2. Certify completion of state-approved start-up procedures.

3. An exemption from conducting state-approved start-up procedures may be available for seasonal systems that maintain pressure throughout the distribution system during non-operating periods.

Examples of state-approved start-up procedures, which need to be completed prior to serving water to the public, may include one or more of the following:

1. Disinfection.
2. Distribution system flushing.
3. Sampling for total coliform and E. coli.
4. Site visit by state.
5. Verification that any current or historical sanitary defects have been corrected.

Routine Monitoring for Seasonal Systems

1. The baseline monitoring frequency for seasonal systems is monthly.
2. A reduced monitoring frequency may be available for seasonal systems that use ground water only and serve fewer than 1,000 persons.

Other Provisions for the State Drinking Water Agency Special Monitoring Evaluation

The state must perform a special monitoring evaluation at all ground water systems serving 1,000 or fewer persons during each sanitary survey to review the status of the Public Water System and to determine whether the sample sites and monitoring schedule need to be modified.

Major Violations

E. coli MCL Violation

A Public Water System will receive an E. coli MCL violation when there is any combination of an E.coli positive sample result with a routine/repeat Total Coliform positive or E. coli positive sample result:

E. coli MCL Violation Occurs with the Following Sample Result Combination

Routine Repeat

EC+ TC+

EC+ Any missing sample

EC+ EC+

TC+ EC+

TC+ TC+ (but no E. coli analysis)

Treatment Technique Violation

A PWS will receive a Treatment Technique violation when any of the following occur:

1. Failure to conduct a Level 1 or Level 2 Assessment within 30 days of a trigger.
2. Failure to correct all sanitary defects from a Level 1 or Level 2 Assessment within 30 days of a trigger or in accordance with the state-approved timeframe.
3. Failure of a seasonal system to complete state-approved start-up procedures prior to serving water to the public.

Key Points for Public Water Systems to Remember

Find and correct sanitary defects as soon as you become aware of them. This can help reduce E. coli MCL violations, which trigger a Level 2 Assessment. This can help reduce TC+ sample results, which may trigger a Level 1 Assessment.

Make sure to collect all routine and repeat samples as required. Timely and correct monitoring can help reduce triggering a Level 1 or Level 2 Assessment because:

1. Failure to conduct repeat monitoring triggers a Level 1 Assessment.
2. A Level 1 Assessment triggered twice within a certain timeframe triggers a Level 2 Assessment.

Surface Water Treatment Rule

The purpose of the **Surface Water Treatment Rules (SWTRs)** is to reduce illnesses caused by pathogens in drinking water. The disease-causing pathogens include *Legionella*, *Giardia lamblia*, and *Cryptosporidium*.

The SWTRs requires water systems to filter and disinfect surface water sources. Some water systems are allowed to use disinfection only for surface water sources that meet criteria for water quality and watershed protection. This combination of rules is designed to provide protection from microbial pathogens.

Simultaneously, the rules minimize health risks to the population from disinfection byproducts.

For PWSs that use surface water sources and are not required to filter, existing regulations do not require any treatment for *Cryptosporidium*. New survey data suggest that typical *Cryptosporidium* levels in the treated water of unfiltered PWSs are higher than in the treated water of filtered PWSs. Thus, *Cryptosporidium* treatment by unfiltered PWSs is needed to achieve comparable public health protection.

Further, results from recent treatment studies has allowed EPA to develop standards for the inactivation of *Cryptosporidium* by ozone, ultraviolet (UV) light, and chlorine dioxide. Based on these developments, the EPA has established requirements for all unfiltered PWSs to treat for *Cryptosporidium*, with the required degree of treatment depending on the source water contamination level. Additionally, the long-term rule addresses risks in uncovered finished water storage facilities, in which treated water can be subject to significant contamination as a result of runoff, bird and animal wastes, human activity, algal growth, insects, fish, and airborne deposition. Existing regulations prohibit the building of new uncovered finished water storage facilities. Under the long-term rule, PWSs must limit potential risks by covering or treating the discharge of such storage facilities. The requirements in the Surface Water Treatment Rule reflect consensus recommendations from the Stage 2 Microbial and Disinfection Byproducts (M-DBP) Federal Advisory Committee. These recommendations are set forth in the Stage 2 M-DBP Agreement in Principle.

Long-Term Surface Water Treatment Rule

Source Water Monitoring. The rule requires PWSs using surface water or groundwater under the direct influence (GWUDI) of surface water to monitor their source water (the influent water entering the treatment plant) to determine an average *Cryptosporidium* level. Monitoring results determine the extent of *Cryptosporidium* treatment requirements.

Large PWSs (serving at least 10,000 people) must monitor for *Cryptosporidium* (plus *E. coli* and turbidity in filtered PWSs) for a period of two years. To reduce monitoring costs, small filtered PWSs (serving fewer than 10,000 people) initially monitor just for *E. coli* for one year as a screening analysis and are required to monitor for *Cryptosporidium* only if their *E. coli* levels exceed specified trigger values. Small filtered PWSs that exceed the *E. coli* trigger, as well as all small unfiltered PWSs, must monitor for *Cryptosporidium* for one or two years, depending on the sampling frequency.

Under the long-term rule, specific criteria are set for sampling frequency and schedule, sampling location, using previously collected data, providing treatment instead of monitoring, sampling by PWSs that use surface water for only part of the year, and monitoring of new plants and sources. The long-term rule also establishes requirements for reporting of monitoring results, using analytical methods, and using approved laboratories. The rule also requires a second round of monitoring to begin approximately 6.5 years after the first round concludes in order to determine if source water quality has changed to a degree that should affect treatment requirements.

The long-term rule establishes risk-targeted treatment technique requirements to control *Cryptosporidium* in PWSs using surface water or GWUDI. These treatment requirements supplement those established by existing regulations. Filtered PWSs are classified in one of four treatment categories (or bins) based on the results of the source water *Cryptosporidium* monitoring. This bin classification determines the degree of additional *Cryptosporidium* treatment, if any, that filtered PWS must provide. PWSs classified in Bins 2, 3, or 4 must achieve 1.0 to 2.5-log of treatment (90 to 99.7 percent reduction) for *Cryptosporidium* over and above that provided with conventional treatment. Different additional treatment requirements may apply to PWSs using other than conventional treatment, such as direct filtration, membranes, or cartridge filters.

Filtered PWSs must meet the additional *Cryptosporidium* treatment required in Bins 2, 3, or 4 by using one or more treatment or control processes from a microbial toolbox of options. The long-term rule requires all unfiltered PWSs to provide at least 2-log (99 percent) inactivation of *Cryptosporidium*. If the average source water *Cryptosporidium* level exceeds 0.01 oocysts/L, the unfiltered PWS must provide at least 3-log (99.9 percent) inactivation of *Cryptosporidium*. Further, under the long-term rule, unfiltered PWSs

must achieve their overall inactivation requirements (including *Giardia lamblia* and virus inactivation) using a minimum of two disinfectants.

PWSs with uncovered finished water storage facilities must take steps to address contamination risks. Existing regulations require PWSs to cover all new storage facilities for finished water but do not address existing uncovered finished water storage facilities. Under the long-term rule, PWSs using uncovered finished water storage facilities must cover the storage facility or treat the storage facility discharge to achieve inactivation and/or removal of 4-log virus, 3-log *Giardia lamblia*, and 2-log *Cryptosporidium* on a state-approved schedule.

The rule applies to all PWSs using surface water or GWUDI, including large and small PWSs, community and non-community PWSs, and non-transient and transient PWSs. Wholesale PWSs must comply with the requirements of the rule based on the population of the largest PWS in the combined distribution system.

Consecutive PWSs that purchase treated water from wholesale PWSs that fully comply with the monitoring and treatment requirements of the rule are not required to take additional steps for that water.

Existing Regulations for Microbial Pathogens in Drinking Water

1. The Surface Water Treatment Rule (SWTR) (54 FR 27486, June 1989) applies to all PWSs using surface water or ground water under the direct influence (GWUDI) of surface water as sources (Subpart H PWSs). It established MCLGs of zero for *Giardia lamblia*, viruses, and *Legionella*, and includes the following treatment technique requirements to reduce exposure to pathogenic microorganisms:
 1. Filtration, unless specific avoidance criteria are met
 2. Maintenance of a disinfectant residual in the distribution system
 3. Removal and/or inactivation of 3-log (99.9%) of *Giardia lamblia* and 4-log (99.99%) of viruses
 4. Maximum allowable turbidity in the combined filter effluent (CFE) of 5 nephelometric turbidity units
 5. (NTU) and 95th percentile CFE turbidity of 0.5 NTU or less for plants using conventional treatment or direct filtration
 6. Watershed protection and source water quality requirements for unfiltered PWSs
2. The Total Coliform Rule (TCR) (54 FR 27544, June 1989) applies to all PWSs. It established an MCLG of zero for total and fecal coliform bacteria and an MCL based on the percentage of positive samples collected. Coliforms are used as an indicator of fecal contamination and to determine the integrity of the water treatment process and distribution system. Under the Total Coliform Rule, no more than 5 percent of distribution system samples collected in any month may contain coliform bacteria (no more than 1 sample per month may be coliform positive in those PWSs that collect fewer than 40 samples per month). The number of samples to be collected in a month is based on the number of people served by the PWS.
3. Interim Enhanced Surface Water Treatment Rule (63 FR 69478, December 16, 1998) (USEPA 1998a) applies to PWSs serving at least 10,000 people and using surface water or GWUDI sources. Key provisions established by the interim rule include the following:
 1. An MCLG of zero for *Cryptosporidium*
 2. *Cryptosporidium* removal requirements of 2-log (99 percent) for PWSs that filter
 3. More stringent turbidity performance standards of 1.0 NTU as a maximum and 0.3 NTU or less at the 95th percentile monthly for treatment plants using conventional treatment or direct filtration
 4. Requirements for individual filter turbidity monitoring
 5. Disinfection benchmark provisions to assess the level of microbial protection that PWSs provide as they take steps to comply with new disinfection by-products standards
 6. Inclusion of *Cryptosporidium* in the definition of GWUDI and in the watershed control requirements for unfiltered PWSs
 7. Requirements for covers on new finished water storage facilities
 8. Sanitary surveys for all surface water systems regardless of size

The interim surface water treatment rule was developed in conjunction with the Stage 1 Disinfectants and Disinfection Byproducts Rule (Stage 1 DBPR) (63 FR 69389, December 1998), which reduced allowable levels of certain DBPs, including trihalomethanes, haloacetic acids, chlorite, and bromate.

4. Long Term Enhanced Surface Water Treatment Rule (67 FR 1812, January 2002) builds on the microbial control provisions established by the interim rule for large PWSs through extending similar requirements to small PWSs. The long-term interim rule applies to PWSs that use surface water or GWUDI as sources and that serve fewer than 10,000 people. Like the interim rule, the long-term rule established the following:
 1. 2- log (99 percent) *Cryptosporidium* removal requirements by PWSs that filter
 2. Individual filter turbidity monitoring and more stringent combined filter effluent turbidity standards for conventional and direct filtration plants
 3. Disinfection profiling and benchmarking; inclusion of *Cryptosporidium* in the definition of GWUDI and in the watershed control requirements for unfiltered PWSs
 4. The requirement that new finished water storage facilities be covered.
5. Filter Backwash Recycle Rule (66 FR 31085, June 2001) requires PWSs to consider the potential risks associated with recycling contaminants removed during the filtration process. The provisions of the filter backwash rule apply to all PWSs that recycle, regardless of population served. In general, the provisions include the following:
 1. PWSs must return certain recycle streams to a point in the treatment process that is prior to primary coagulant addition unless the State specifies an alternative location
 2. Direct filtration PWSs recycling to the treatment process must provide detailed recycle treatment information to the State
 3. Certain conventional PWSs that practice direct recycling must perform a one-month, one-time recycling self-assessment

Introduction to *Cryptosporidium*

Concern with *Cryptosporidium* in Drinking Water

EPA is promulgating the Long-Term Surface Water Treatment Rule to reduce the public health risk associated with *Cryptosporidium* in drinking water.

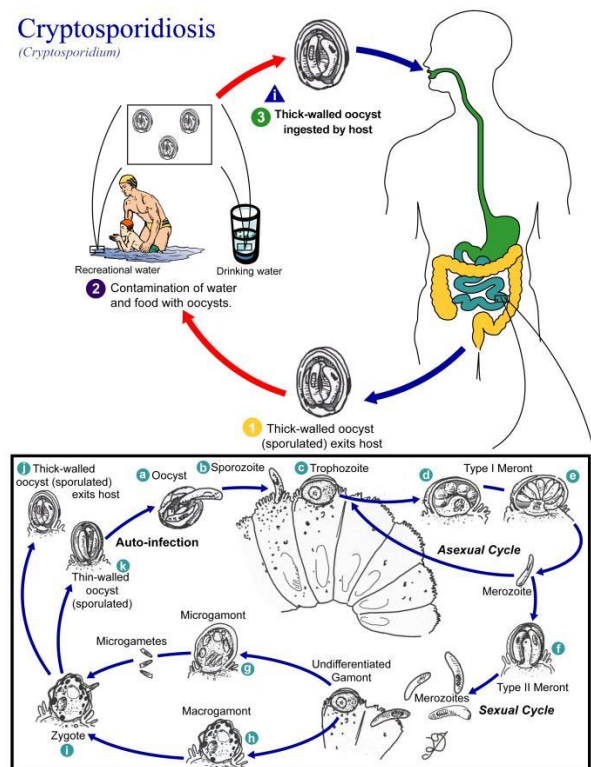


Figure 2.1.1: *Cryptosporidiosis*.

Cryptosporidium is a protozoan parasite that lives and reproduces entirely in one host. Ingestion of *Cryptosporidium* can cause cryptosporidiosis, a gastrointestinal (GI) illness. *Cryptosporidium* is excreted in feces. Transmission of cryptosporidiosis occurs

through consumption of water or food contaminated with feces or by direct or indirect contact with infected persons or animals. In the environment, *Cryptosporidium* is present as a thick-walled oocyst containing four organisms (sporozoites).

The oocyst wall insulates the sporozoites from harsh environmental conditions. Oocysts are 4–5 microns in length and width. Upon a host's ingestion of oocysts, enzymes and chemicals produced by the host's digestive system cause the oocyst to excyst, or break open. The excysted sporozoites embed themselves in the surfaces of the epithelial cells of the lower small intestine. The organisms then begin absorbing nutrients from their host cells. When these organisms sexually reproduce, they produce thick and thin-walled oocysts. The host excretes the thick-walled oocysts in its feces. The thin-walled oocysts excyst within the host and contribute to further host infection.

The exact mechanism by which *Cryptosporidium* causes GI illness is not known. Factors may include damage to intestinal structure and cells, changes in the absorption/secretion processes of the intestine, toxins produced by *Cryptosporidium* or the host, and proteins that allow *Cryptosporidium* to adhere to host cell surfaces. Upon excretion, *Cryptosporidium* oocysts may survive for months in various environmental media, including soil, river water, seawater, and human and cattle feces at ambient temperatures. *Cryptosporidium* can also withstand temperatures as low as 20 °C for periods of a few hours, but they are susceptible to desiccation.

Cryptosporidium is a widespread contaminant in surface water used as drinking water supplies. *Cryptosporidium* contamination can come from animal agriculture, wastewater treatment plant discharges, slaughterhouses, birds, wild animals, and other sources of fecal matter. *Cryptosporidium parvum* (*C. parvum*) has been the primary species of concern to humans.

Cryptosporidium infection is characterized by mild to severe diarrhea, dehydration, stomach cramps, and/or a slight fever. Incubation is thought to range from 2 to 10 days. Symptoms typically last from several days to 2 weeks. In a small percentage of cases, the symptoms may persist for months or longer in healthy individuals. Symptoms may be more severe in immunocompromised persons. Such persons include those with AIDS, cancer patients undergoing chemotherapy, organ transplant recipients treated with drugs that suppress the immune system, and patients with autoimmune disorders.

Cryptosporidium oocysts are highly resistant to standard disinfectants like chlorine and chloramines. Consequently, control of *Cryptosporidium* in most treatment plants is dependent on physical removal processes. However, due to their size (4–5 microns), oocysts can sometimes pass through filters. Monitoring data on finished water show that *Cryptosporidium* is sometimes present in filtered, treated drinking water.

Certain alternative disinfectants can be more effective in treating for *Cryptosporidium*. Ozone and chlorine dioxide have been shown to inactivate *Cryptosporidium* at doses higher than those required to inactivate *Giardia*, which has typically been used to set disinfectant doses. Studies have also demonstrated a synergistic effect of treatment using ozone followed by chlorine or monochloramine. UV light has recently been shown to achieve high levels of *Cryptosporidium* inactivation at feasible doses. Other processes that can help reduce *Cryptosporidium* levels in finished water include watershed management programs, pretreatment processes like bank filtration, and additional clarification and filtration processes during water treatment. Further, optimizing treatment performance and achieving very low levels of turbidity in the finished water has been shown to improve *Cryptosporidium* removal in treatment plants.

Cryptosporidium has caused a number of waterborne disease outbreaks since 1984 when the first was reported in the United States. Data from the Centers for Disease Control and Prevention include ten outbreaks caused by *Cryptosporidium* in drinking water between 1984 and 2000, with approximately 421,000 cases of illness. The most serious outbreak occurred in 1993 in Milwaukee; an estimated 403,000 people became sick, and at least 50 *Cryptosporidium*-associated deaths occurred among the severely immunocompromised.

A PWS that begins using a new surface water source after the date the PWS is required to conduct source water monitoring under the Long-Term 2 Surface Water Treatment Rule must monitor the new source on a schedule approved by the State. This applies to new plants that begin operation and previously operating plants that bring a new source on-line after the required monitoring date for the PWS. The State may determine that monitoring should be conducted before a new plant or source is brought on-line or initiated within some time period afterward.

The new source monitoring must meet all long-term rule requirements. The PWS must also determine its treatment bin classification and comply with any additional *Cryptosporidium* treatment requirements based on the monitoring results on a

schedule approved by the State.

Background and Analysis Monitoring requirements in the rule are designed to ascertain *Cryptosporidium* levels with suitable accuracy for making treatment bin classifications and in a time frame that does not delay the installation of *Cryptosporidium* treatment where needed. A mean *Cryptosporidium* concentration of 0.01 oocysts/L determines whether unfiltered PWSs are required to provide 2 or 3-log *Cryptosporidium* inactivation. No *E. coli* concentration was effective in determining whether PWSs were likely to fall above or below this level. Consequently, the rule requires all unfiltered PWSs to monitor for *Cryptosporidium*, unless they choose to provide 3-log *Cryptosporidium* inactivation.

Sampling location

The requirements in the rule for the source water sample collection location are designed to achieve two objectives:

1. Characterize the influent water to the treatment plant at the time each sample is collected.
2. Ensure that samples are not affected by treatment chemicals that could interfere with *Cryptosporidium* analysis.

The first objective is the basis for requiring PWSs that use multiple sources to analyze a blended source sample or calculate a weighted average of sources that reflects the influent at the time of sample collection. It is also the reason that PWSs are required to sample after certain pretreatment processes like bank filtration that do not involve chemical addition.

The second objective is why PWSs are generally required to sample upstream of chemical addition and prior to backwash addition (for PWSs that recycle filter backwash). However, EPA recognizes that in some situations, sampling prior to chemical addition will not be feasible and discontinuing chemical addition for a period of time prior to sampling will not be advisable. This situation could occur when a treatment chemical is added at an intake that is difficult to access. Further, some treatment chemicals may not interfere with *Cryptosporidium* analyses when present at very low levels. Consequently, the rule allows States to approve PWSs sampling after chemical addition when the State determines that collection prior to chemical treatment is not feasible, and the treatment chemical is not expected to interfere with the analysis of the sample. EPA believes that States should review source water monitoring locations for their PWSs.

Sampling Schedule

The requirement will help to ensure that monitoring determines the mean concentration of *Cryptosporidium* in the treatment plant influent. To achieve this objective, the timing of sample collection must not be adjusted in response to fluctuations in water quality, such as the avoidance of sampling when the influent water is expected to be of poor quality. EPA believes that the 5-day window for sample collection and associated allowances for sampling outside this window provide sufficient flexibility. If circumstances arise that prevent the PWS from sampling within the scheduled 5-day window, such as a weather event or plant emergency, the PWS must collect a sample as soon as feasible. In this case, feasibility includes the ability of the PWS to safely collect a sample and the availability of an approved laboratory to conduct the analysis within method specifications. In addition, the rule allows States to authorize a different date for collecting the delayed sample.

Failing to Monitor

For most monitoring and testing procedure violations under the Long-Term Rule 2, PWSs must provide a Tier 3 public notification, which is standard for this type of violation under an NPDWR. However, if a PWS fails to collect three or more *Cryptosporidium* samples, the violation is elevated to a Tier 2 special public notice. The reason for elevating the public notice at this point is the persistence of the violation and the difficulty the PWS will have in collecting the required number of samples for bin classification by the compliance date. The rule requires bin classification within six months following the end of the monitoring period specified for the PWS. This six-month period provides some opportunity for collecting and analyzing missed samples. The number of samples that can be made up in this period is limited, though, due to the need for samples to be evenly distributed throughout the year, as well as for PWSs and States to spend time during this period evaluating monitoring results to determine bin classification.

Failure by a PWS to collect the required number of *Cryptosporidium* samples for bin classification by the compliance date is a treatment technique violation with a required Tier 2 public notice. This violation reflects the inability of the PWS to determine and comply with its *Cryptosporidium* treatment requirements under the rule and provide the appropriate level of public health protection. The violation ends when the State determines that the PWS is carrying out a monitoring plan that will lead to bin classification.

A PWS that has already provided a Tier 2 public notice for missing three sampling dates and is successfully meeting a State-approved sampling schedule is not required to issue another public notice for missing the bin classification date. Alternatively, the PWS can choose to provide the highest level of *Cryptosporidium* treatment required under the rule, which is 5.5-log for filtered PWSs and 3-log for unfiltered PWSs.

Requirements for grandfathering previously collected monitoring data in the rule based on the principle that to be eligible for grandfathering, previously collected data must be equivalent in quality to data that will be collected under the rule.

The Stage 2 M-DBP Advisory Committee recommended that EPA accept previously collected *Cryptosporidium* data that are equivalent in sample number, frequency, and data quality (volume analyzed, percent recovery) to data that would be collected under the long-term rule to determine bin classification in lieu of further monitoring.

Ongoing watershed assessment treatment requirements under the rule are based on source water quality. Consequently, the rule requires watershed assessment and, a second round of monitoring following initial bin classification to determine if source water quality has changed to the degree that the treatment level should be modified. States are allowed to use programs other than the sanitary survey to assess changes in the watershed. The rule leverages the existing requirement for States to perform sanitary surveys on surface water PWSs. During the source water review in the sanitary survey, the rule requires States to determine if significant changes have occurred in the watershed that could lead to increased contamination by *Cryptosporidium*. The State can also choose to make this determination through an equivalent review of the source water under a program other than the sanitary survey, such as a Source Water Protection Assessment. If the State determines that significant changes have occurred, the State may specify that the PWS conduct additional source water monitoring or treat the potential contamination. This approach allows the PWS and State to respond to a significant change in source water quality prior to initiating a second round of monitoring or any time thereafter.

Second Round of Monitoring

A more rigorous reassessment of the source water occurs through a second round of monitoring that begins six years after initial bin classification. If EPA does not develop and finalize modifications to the rule prior to the date when PWSs must begin the second round of monitoring, then this second round must conform to the same requirements that applied to the initial round of monitoring. PWSs may be classified in a different treatment bin, depending on the results of the second round of monitoring.

Process Summary

Source water monitoring to determine additional treatment requirements for *Cryptosporidium*.

1. Sampling parameters and frequency, sampling location, sampling schedule, monitoring plants that operate only part-year, failing to monitor, providing treatment instead of monitoring, grandfathering previously collected data, ongoing source water assessment, second round of monitoring, and new source monitoring.

The Rule requires filtered PWSs using surface water or GWUDI sources to provide greater levels of treatment if their source waters have higher concentrations of *Cryptosporidium*. Filtered PWSs are classified in one of four treatment bins based on results from the source water monitoring. PWSs classified in the lowest concentration bin are subject to no additional treatment requirements, while PWSs assigned to higher concentration bins must reduce *Cryptosporidium* levels. All PWSs must continue to comply with the requirements of the SWTR, IESWTR, and LT1ESWTR, as applicable. This section addresses procedures for classifying filtered PWSs in *Cryptosporidium* treatment bins and the treatment requirements associated with each bin. Microbial toolbox that PWSs must use to meet additional *Cryptosporidium* treatment requirements.

- Bin classification-after completing initial source water monitoring, filtered PWSs must calculate a *Cryptosporidium* bin concentration for each treatment plant where *Cryptosporidium* monitoring is required. This *Cryptosporidium* bin concentration is used to classify filtration plants in one of the four treatment bins.

In general, the *Cryptosporidium* bin concentration is calculated by averaging individual sample results from one or more years of monitoring. Specific procedures vary, however, depending on the frequency and duration of monitoring. These procedures are as follows:

1. For PWSs that collect a total of at least 24 but not more than 47 *Cryptosporidium* samples over two or more years, the *Cryptosporidium* bin concentration is equal to the highest arithmetic mean of all sample concentrations in any 12

consecutive months of *Cryptosporidium* monitoring.

2. For PWSs that collect a total of at least 48 samples, the *Cryptosporidium* bin concentration is equal to the arithmetic mean of all sample concentrations.
3. For PWSs that serve fewer than 10,000 people and monitor for *Cryptosporidium* for only one year (collect 24 samples in 12 months), the *Cryptosporidium* bin concentration is equal to the arithmetic mean of all sample concentrations.
4. For PWSs with plants that operate only part-year that monitor for less than 12 months per year, the *Cryptosporidium* bin concentration is equal to the highest arithmetic mean of all sample concentrations during any year of *Cryptosporidium* monitoring.

In data sets with variable sampling frequency, PWSs must first calculate an arithmetic mean for each month of sampling and then apply one of these four procedures using the monthly mean concentrations. PWSs may grandfather previously collected *Cryptosporidium* data where the sampling frequency varies (one year of monthly sampling and one year of twice-per-month sampling). Filtered PWSs serving fewer than 10,000 people are not required to monitor for *Cryptosporidium* if they demonstrate a mean *E. coli* concentration less than or equal to 10/ 100 mL for lake/reservoir sources or 50/ 100 mL for flowing stream sources or do not exceed an alternative State-approved indicator trigger. PWSs that meet this criterion are classified in Bin 1.

When determining the *Cryptosporidium* bin concentration, PWSs must calculate individual sample concentrations as the total number of oocysts counted, divided by the volume as saved. In samples where no oocysts are detected, the result is assigned a value of zero for the purpose of calculating the bin concentration. Sample analysis results are not adjusted for analytical method recovery or the percent of *Cryptosporidium* oocysts that are infectious. PWSs must report their treatment bin classification to the State for approval following initial source water monitoring. The report must include a summary of the data and calculation procedure used to determine the bin concentration. PWSs must recalculate their bin classification after completing the second round of monitoring and report the results to the State for approval. If the State does not respond to a PWS regarding its bin classification after either report, the PWS must comply with the *Cryptosporidium* treatment requirements of the rule based on the reported bin classification.

- Bin treatment requirements. All filtered PWSs must comply with the treatment requirements based on their bin classification.

The total *Cryptosporidium* treatment required for plants in Bins 2, 3, and 4 is 4.0-log, 5.0-log, and 5.5-log, respectively. Conventional treatment (including softening), slow sand, and diatomaceous earth filtration plants in compliance with the IESWTR or LT1ESWTR, as applicable, receive a prescribed 3.0-log *Cryptosporidium* treatment credit toward these total bin treatment requirements. Accordingly, these plant types must provide 1.0 to 2.5-log of additional treatment when classified in Bins 2–4, respectively. Direct filtration plants in compliance with existing regulations receive a prescribed 2.5-log treatment credit and, consequently, must achieve 0.5-log greater treatment to comply with Bins 2–4.

For PWSs using alternative filtration technologies, such as membranes, bag filters, or cartridge filters, no prescribed treatment credit is available because the performance of these processes is specific to individual products. Consequently, when PWSs using these processes are classified in Bins 2–4, the State must determine additional treatment requirements based on the credit the State awards to a particular technology. The additional treatment requirements must ensure that plants classified in Bins 2–4 achieve total *Cryptosporidium* reductions of 4.0- to 5.5-log, respectively. PWSs can achieve additional *Cryptosporidium* treatment credit through implementing pretreatment processes like pre-sedimentation or bank filtration, by developing a watershed control program, and by applying additional treatment steps like ozone, chlorine dioxide, UV, and membranes. In addition, PWSs can receive a higher level of credit for existing treatment processes through achieving very low filter effluent turbidity or through a demonstration of performance.

A combination of options from the microbial toolbox can be used to gather additional credit. PWSs in Bin 2 can meet additional *Cryptosporidium* treatment requirements by using any option or combination of options from the microbial toolbox. For Bins 3 and 4, PWSs must achieve at least 1-log of the additional treatment requirement by using ozone, chlorine dioxide, UV, membranes, bag filtration, cartridge filtration, or bank filtration.

- The rule increases protection against *Cryptosporidium* and other pathogens in PWSs with the highest source water contamination levels. This targeted approach builds upon existing regulations under which all filtered PWSs must provide the same level of treatment regardless of source water quality. The rule is to ensure that PWSs with higher risk source waters achieve public health protection commensurate with PWSs with less contaminated sources.

The rule establishes risk-targeted *Cryptosporidium* treatment requirements and sets specific bin concentration ranges and treatment requirements that apply to filtered PWSs. The IESWTR established a *Cryptosporidium* MCLG of zero and requires large filtered PWSs to achieve 2-log *Cryptosporidium* removal. The LT1ESWTR extended this requirement to small PWSs. After these rules were promulgated, advances were made in analytical methods and treatment for *Cryptosporidium*, and EPA collected new information on *Cryptosporidium* occurrence and infectivity. Consequently, EPA assessed the implications of these developments for further controlling *Cryptosporidium* to approach the zero MCLG. The risk-targeted approach for filtered PWSs in the rule stems from four general findings based on new information on *Cryptosporidium*:

1. New data on *Cryptosporidium* infectivity suggest that the risk associated with a particular level of *Cryptosporidium* is most likely higher than EPA estimated at the time of earlier rules.
2. New data on *Cryptosporidium* occurrence indicate that levels are relatively low in most water sources, but a subset of sources has substantially higher concentrations.
3. The finding that UV light can readily inactivate *Cryptosporidium*, as well as other technology developments, makes achieving high levels of treatment for *Cryptosporidium* feasible for PWSs of all sizes.
4. EPA Methods 1622 and 1623 are capable of assessing annual mean levels of *Cryptosporidium* in drinking water sources.

These findings led EPA to conclude that most filtered PWSs currently provide sufficient treatment for *Cryptosporidium*, but additional treatment is needed in those PWSs with the highest source water *Cryptosporidium* levels to protect public health. Further, PWSs can characterize *Cryptosporidium* levels in their source waters with available analytical methods and can provide higher levels of treatment with available technologies. Consequently, risk-targeted treatment requirements for *Cryptosporidium* based on source water contamination levels are appropriate and feasible to implement.

Basis for bin concentration ranges and treatment requirements. To establish the risk-targeted treatment requirements in today's rule, EPA had to determine the degree of treatment that should be required for different source water *Cryptosporidium* levels to protect public health. This determination involved addressing several questions:

1. What is the risk associated with *Cryptosporidium* in a drinking water source?
2. How much *Cryptosporidium* removal do filtration plants achieve?
3. What is the appropriate statistical measure for classifying PWSs into treatment bins?
4. What degree of additional treatment is needed for higher source water *Cryptosporidium* levels?
5. How should PWSs calculate their treatment bin classification?

The risk of infection from *Cryptosporidium* in drinking water is a function of exposure (the dose of oocysts ingested) and infectivity (likelihood of infection as a function of ingested dose). Primary (direct) exposure to *Cryptosporidium* depends on the concentration of oocysts in the source water, the fraction removed by the treatment plant, and the volume of water consumed (secondary exposure occurs through interactions with infected individuals). Thus, the daily risk of infection (DR) is as follows:

$$DR = (\text{oocysts/L in source water}) \times (\text{fraction remaining after treatment}) \times (\text{liters consumed per day}) \times (\text{likelihood of infection per oocyst dose}).$$
 Assuming 350 days of consumption per year for people served by community water systems (CWSs), the annual risk (AR) of infection is as follows: $AR = 1 \text{ } \forall \text{ } (1 \text{ } \forall \text{ } DR)350.$

EPA has estimated the mean likelihood of infection from ingesting one *Cryptosporidium* oocyst to range from 4 to 16 percent. Median individual daily water consumption is estimated as 1.07 L/day.

Key Terms

- ***Cryptosporidium*** - a protozoan parasite that lives and reproduces entirely in one host; ingestion of *Cryptosporidium* can cause cryptosporidiosis, a gastrointestinal (GI) illness; cryptosporidium is excreted in feces.
- **Ground Water Rule (GWR)** – the purpose is to improve drinking water quality and provide protection from disease-causing microorganisms; water systems that have groundwater sources may be susceptible to fecal contamination; applies to public water systems that use ground water as a source of drinking water
- **Surface Water Treatment Rules (SWTRs)** – the purpose is to reduce illnesses caused by pathogens in drinking water, including *Legionella*, *Giardia lamblia*, and *Cryptosporidium*; the SWTRs requires water systems to filter and disinfect surface water sources.

- **Total Coliform Rule (TCR)** - a National Primary Drinking Water Regulation (NPDWR), published in 1989 and became effective in 1990; sets a health goal (Maximum Contaminant Level Goal (MCLG)) and legal limits (Maximum Contaminant Levels, MCLs)) for the presence of total coliforms in drinking water

Review Questions

1. **What are the primary components of the Total Coliform Rule?**
2. **What is the basis of the Total Coliform Rule?**
3. **What is the primary requirement in the Groundwater Rule?**
4. **List four EPA approved laboratory methods for testing for coliforms.**
5. **What are the major provisions of the Long-Term Surface Water Treatment Rule?**

Chapter Quiz

1. Total coliforms are used as a(n) _____ of the possible presence of pathogens in drinking water. Total coliforms are used to determine the adequacy of water treatment and the integrity of the distribution system.
 1. indicator
 2. model
 3. log removal
 4. alternative
2. The Revised Total Coliform Rule establishes _____ for E. coli and uses E. coli and total coliforms to initiate a find and fix approach to address fecal contamination that could enter into the distribution system.
 1. a log removal
 2. a maximum contaminant level (MCL)
 3. a secondary maximum contaminant level (SMCL)
 4. a model
3. Within 24 hours of learning of a Total Coliform positive routine sample result, at least ____ repeat samples must be collected and analyzed for total coliform.
 1. 1
 2. 2
 3. 3
 4. 4
4. States are required to conduct _____ of all groundwater systems in order to identify significant deficiencies, including deficiencies which may make a system susceptible to microbial contamination.
 1. hydrologic studies
 2. microbial examinations
 3. a sanitary survey
 4. laboratory studies
5. The Surface Water Treatment Rule requires water systems to _____ surface water sources.

1. filter and disinfect
 2. conserve
 3. aerate
 4. disinfect
6. In order not to be subject to triggered source water monitoring under the Groundwater Rule, a groundwater system can notify the state that it provides at least _____ treatment of viruses using virus inactivation, or removal, before or at the first customer.
1. 2-log
 2. 3-log
 3. 4-log
 4. 6-log
7. The Long-Term Surface Water Rule establishes risk-targeted treatment technique requirements to control _____ in PWSs using surface water or groundwater under the influence of surface water.
1. *E.coli*
 2. *Giardiasis lambdia*
 3. *Cryptosporidium*
 4. Coliforms
8. What is the maximum allowable turbidity in the combined filter effluent?
1. 0.5 NTU
 2. 1.0 NTU
 3. 3.0 NTU
 4. 5.0 NTU
9. EPA promulgated the Long-Term Surface Water Treatment Rule to reduce the public health risk associated with _____ in drinking water.
1. *Giardiasis lambdia*
 2. *Cryptosporidium*
 3. Coliforms
 4. *E. coli*
10. _____ is a protozoan parasite that lives and reproduces entirely in one host,
1. *E. coli*
 2. Coliforms
 3. *Cryptosporidium*
 4. Fecal streptococcus

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2.2: Water Chemistry and Standards

Learning Objectives

After reading the chapter the student should be able to identify:

- Matter, elements, and compounds
- Public water systems: community and non-community
- Primary and secondary standards
- Water treatment violations
- Volume math problems

Water treatment is a complex process that involves parts that the human eye cannot see. In this chapter, you will learn about the basic scientific principals related to the water treatment. We will also discuss drinking water standards in the United States and different community standards.

Matter, Elements, and Compounds

Matter

The smallest parts of an element are comprised of particles known as atoms. Even though they are so small, atoms still retain the characteristics of the element. Even with technological advancements, microscopes are still unable to capture atoms. The multiple arrangements of atoms make each element unique. The atom ever so small, is comprised of three particles known as the proton, neutron, and electron. Each particle is associated by different charges.

- Proton-positive
- Neutron-no charge
- Electron- Negative charge

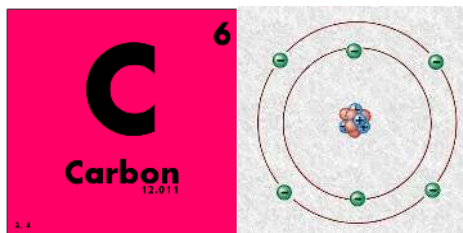


Figure 2.2.1: (left) Carbon by [Science Activism](#) is licensed under [CC BY 2.0](#). Figure 2.2.2: (right) Carbon atom by [Alejandro Porto](#) is licensed under [CC BY-SA 3.0](#)

The defining characteristic of an atom is identified by the proton. The proton, located in the nucleus, has a distinctive number. For example, carbon has six protons located in the nucleus. No other element has six protons in the nucleus. The number of protons is represented as the atomic number. The atomic weight is the number of protons and neutrons. The number of protons that exist for a given element is always the same, but the number of neutrons can vary. When there is a varying number of neutrons of a given element, it is known as an isotope. When an atom has a difference in electrons, it is called an ion. When the charges of the atom are not balanced, they become unstable. An atom that has more protons than neutrons is called a cation. An atom that has more electrons than protons is called an anion.

Most common elements in the water treatment profession:

- Aluminum (Al)
- Antimony (Sb)
- Arsenic (As)
- Barium (Ba)
- Beryllium (Be)
- Boron (B)

- Bromine (Br)
- Cadmium (Cd)
- Calcium (Ca)
- Carbon (C)
- Chlorine (Cl)
- Chromium (Cr)
- Copper (Cu)
- Fluorine (F)
- Hydrogen (H)
- Iodine (I)
- Iron (I)
- Lead (Pb)
- Magnesium (Mg)
- Manganese (Mn)
- Mercury (Mn)
- Nickel (Ni)
- Nitrogen (N)
- Oxygen (O)
- Phosphorus (P)
- Potassium (K)
- Radium (Ra)
- Selenium (Se)
- Silicon (Si)
- Silver (Ag)
- Sodium (Na)
- Strontium (Sr)
- Sulfur (S)
- Thallium (Tl)

There are elements that are pure in form, such as oxygen. Since elements are unstable, they often combine with other elements to form compounds. Water (H_2O), for example, is a compound. It is a combination of two hydrogen atoms and one oxygen atom. A compound is two or more elements that are bonded together due to their attraction by reverse charges. The combining elements form a molecule. When two chemicals are mixed together without a chemical reaction it is called a mixture. The difference between a mixture and a compound is bonded together by a chemical reaction. A good example of a mixture is saltwater. The salt can be removed from the water through distillation. Water chemistry will be covered in greater detail in the [Water Quality text](#) and course.

Drinking Water Standards

The first drinking water standards created in the United States occurred in 1974 and were called the Safe Drinking Water Act (SDWA). The US EPA sets drinking Water Standards that all Water Municipalities must adhere to. Although States are able to come up with their own Standards, they must meet the minimum requirements set forth by the federal EPA. In California, drinking water standards are much more stringent than the federal requirements and therefore have primacy. As of 2014, the State regulatory group responsible for drinking water is the State Water Resources Control Board. Revisions to the safe drinking water act include approval techniques for treatment plants, specifying criteria for filtration of public water supplies, distinguishing different treatment techniques for surface and groundwater, and prohibiting lead products in drinking water systems. There are two sets of drinking water Standards identified as primary standards and secondary standards. Primary Standards affect human health and are mandated with **Maximum Contaminant Levels (MCL)**. The MCL is the official safe level at which a human can consume the given contaminant without adverse health effects. Secondary Standards do not affect human health and are controlled with **Maximum Contaminant Level Goals (MCLG)**. When a contaminant is reported, there is no such thing as zero. Levels are set to protect human health but are also set based on the best available technology. As technology improves MCL's may be reexamined.

Instrumentation to measure contaminant levels are not always capable of reading to absolute zero. Because of this, a Detection level for reporting is required.

Public Water Systems

There are three different categories of public water systems:

- **Community Public Water System:** A community public water system has 15 or more service connections and serves at least 25 or more people year-round. These would include municipalities, mobile home parks, condos, and apartment buildings.
- **Nontransient, NonCommunity System:** A nontransient, NonCommunity public water system owns its own system and serves an average of 25 people for at least six months. Schools, hospitals, and office buildings are included in this category.
- **Transient, NonCommunity System:** A transient, NonCommunity public water system owns their own water system and serves an average of 25 people per day. In this category people consume the water for a short period of time. This category includes churches, parks, restaurants, and motels.

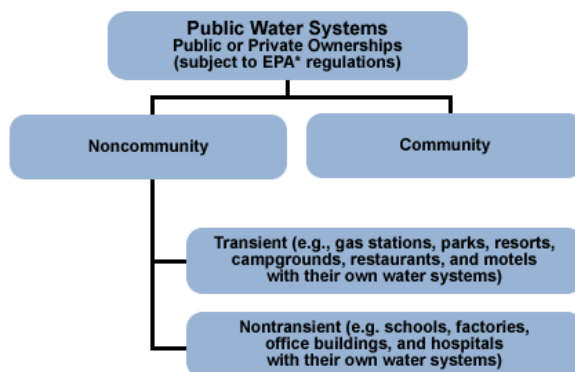


Figure 2.2.2: Image by the CDC is in the public domain

Primary Drinking Water Standards are split into five categories. The categories include Inorganics, Organics, Turbidity, Microbiological, and Radiological.

- **Inorganics:** Metals, Nitrate, and Fluoride
- **Organics:** Pesticides, solvents, and Disinfectant byproducts (DBP's)—The combination of Chlorine and natural organic material. This topic will be discussed in greater detail later in the text.
- **Turbidity:** The cloudiness of the water. Turbidity has the ability to shield microbiological material.
- **Microbiological:** Coliform testing (This will be covered in greater detail later in the text. Water operators do not test for specific microbiological agents. We test for the indicator organism coliform. They colonize in greater numbers so if a sample comes back positive there is a greater likelihood of fecal contamination.)
- **Radiological:** Gross alpha, beta, and radon

Secondary drinking water standards are solely based on the aesthetic quality of drinking water. The main focus of secondary standards is taste, odor, and color. A glass of water that smells like fish and is orange in color may be “safe” to drink but wouldn’t be well received or readily consumed. California, as well as some other states, has endorsable secondary standards.

Table 2.1: Recommended Levels for Contaminants and Characteristics

Contaminant/characteristic	Recommended level
Aluminum	0.05 to 0.2 mg/L
Chloride	250 mg/L
Color	15 color units
Copper	1 mg/L
Corrosivity	Non-corrosive
Fluoride	2 mg/L
Foaming agents	0.5 mg/L
Iron	0.3 mg/L
Manganese	0.05 mg/L
Odor	3 threshold odor number
pH	6.5 to 8.5
Silver	0.10 mg/L
Sulfate	250 mg/L
Total dissolved solids	500 mg/L
Zinc	5 mg/L

Source: Graph by the [EPA](#) is in the public domain

Public Notification

In the event that a treatment plant does not meet requirements of the SDWA, the public must be notified. There are three different tiers of notification with tier I being the worst of violations and tier III being the least. The EPA provides very specific language for public notifications in the event of a violation. Violating a SDWA compliance is bad enough, but failing to report violations brings even stiffer penalties and fines. In 2014 in Flint Michigan 15 people were criminally charged due to their negligence in the water treatment profession. It all started when the city of Flint changed their source water without properly testing. Officials knew the water was not safe to drink and numerous violations had been made. The public was not properly informed of these violations which resulted in 10 deaths and 77 others becoming severely ill. It is something that is rarely discussed because people in industrialized countries never really worry about the quality of their drinking water. The most important thing you can do as an operator is say something if there appears to be a problem with the quality of the drinking water. Otherwise, you may end up in jail. Notification options include radio or television announcements, newspapers, hand delivery, posting in public places, loudspeakers, texting, and reverse 911. The notification will vary based on the severity of the violation.

Violations

Tier I

- Any positive fecal coliform positive test and failure to sample after an initial positive test
- Nitrate or Nitrite violation
- Chlorine Dioxide Maximum residual disinfection limit
- Exceed treatment plants allowable turbidity level. Can be a tier II if the primacy agency does not elevate violation.
- Waterborne emergency or outbreak of waterborne illness

Tier II

- MCL, MRDL and, Treatment Technique (TT) violations if treatment plant does not perform corrective actions to fix issues in treatment plant or fails to inform public.
- Water quality monitoring violation—not taking required water quality samples. Can also be a tier III violation but can be elevated for gross negligence
- Noncompliance of a variance or an exemption

Tier III

- Water testing and monitoring violation
- Any time the treatment system is running under a variance or exception. Primacy agency may give a treatment plant a variance or exemption for a short period of time. The public notification is to inform the public that a water agency is not running in accordance with an approved treatment technique. It doesn't mean the water is unsafe to drink, but the operating manuals are very specific.

Water Quality monitoring

Continuous monitoring of drinking water ensures quality; reliable drinking water is being delivered to the public. The number of samples taken, frequency of sampling, sampling location, testing procedures, and requirements for record-keeping are all specified by state and federal requirements. If sampling requirements are not met, it can lead to public notification. The tier violation is based on the contaminant and whether or not the contaminant causes acute health effects.

The type of monitoring is based on the source of the water, the treatment technique, and the size of the system. Reporting and record keeping is based on the primacy agency's regulations. The state regulations must meet federal requirements at a minimum but may be more stringent as is the case in California.

Record Keeping

Below is a list of records that must be kept and the amount of time the records must be retained on file.

- Bacteriological and Turbidity—5 years
- Chemical analysis—10 years
- Corrective actions from violations—3 years
- Sanitary surveys—10 years
- Exemptions—5 years after expiration

Variances and Exemptions

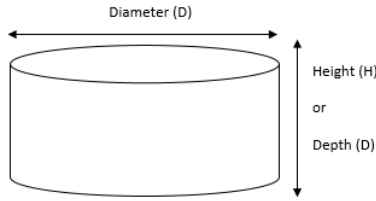
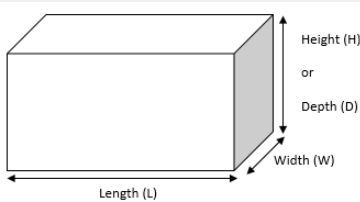
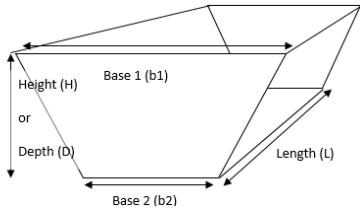
In the event a water system is unable to meet a MCL because of the source water, a primacy agency can grant a variance or exemption. The variance is only given when the agency has incorporated the best available technology and there is zero risk to public health. In the case of Flint Michigan, the new source water was not properly tested before it was used in the system. Flint Water plant was unable to properly treat the source water which had elevated levels of lead and the City was not using the best available technology. Even if the City of Flint would have applied for a variance or exemption, it would not have been granted because the lead levels in the water create a significant risk to public health.

In future chapters of the text, we will examine regulations added to the SDWA. Several changes and enhancements have been made since the exception of the original legislation. We will discuss in greater detail the Total Coliform Rule, Surface Treatment Water Rule, Long Term 2 Enhanced Surface Water Treatment Rule, Lead and Copper Rule, Ground Water Rule, and Stage 1 & 2 Disinfectant Bi-Product Rule.

Volume

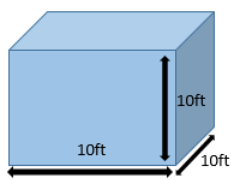
Volume calculations will become very common for water treatment operators. Operators use volume calculations to solve math questions with circles, triangles, and rectangles. If you look around a water treatment facility, it is full of geometric shapes. Tanks

can be cylindrical or rectangular in nature. Settling ponds may have a triangular nature to them. The basic volume questions will become more involved later on in the text as you may have to solve a volume question first, and then solve a flow-related question.

Cylinder 	Volume = $0.785 \times D^2 \times H$
Rectangle 	Volume = $L \times W \times H$
Trapezoid 	Volume = $\frac{b1 + b2}{2} \times H \times L$

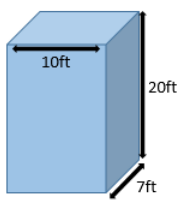
✓ Example 2.2.1

What is the volume of a cubed tank that is 10 feet high? (Remember cubes have the same unit of length on all three sides. This is the easiest problem to solve.)

	Volume = Length $\times$ Width $\times$ Height Volume (ft³) = 10ft $\times$ 10ft $\times$ 10ft Answer is 1,000 ft³
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✓ Example 2.2.1

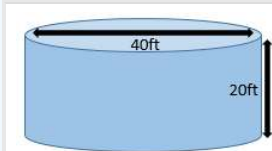
What is the volume of a rectangular tank that is 20 ft. high, 10 ft wide, and 7ft in length?

	Volume= Length $\times$ Width $\times$ Height Volume= 7ft $\times$ 10ft $\times$ 20ft Answer is 1,400ft³
---	--

There are two different ways to solve cylindrical math problems. The easiest formula to use is $\text{volume} = .785 d^2 H$. This formula is easiest because in water math we are generally looking at the diameter of a cylindrical shape and not the radius. The radius of a circle is half the diameter. It is a straight line measured from the center of the circle to the edge of the circle. The diameter is a straight line that passes through the center of the circle with endpoints on the circle.

✓ Example 2.2.2

What is the volume of a storage tank that measures 40 ft. in diameter and is 20 ft. deep?



$$\begin{aligned}\text{Volume} &= .785 \times d^2 \times H \\ \text{Volume} &= .785 \times 40\text{ft} \times 40\text{ft} \times 20\text{ft} \\ \text{Answer is } &25,120\text{ft}^3\end{aligned}$$

Often the water math word problems will require multiple steps to solve. Typically, we want to know how many gallons a given tank will hold. Once we find the volume of the tank in cubic feet, we will then convert the cubic feet into gallons.

✓ Example 2.2.3

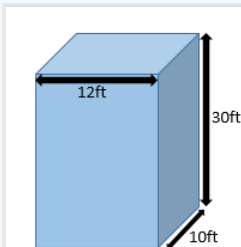
Convert the answer from Example 2.2.3 into gallons.

$$25,120 \text{ ft}^3 \times 17.48 \text{ gal} \times 1 \text{ ft}^3 = 187,898 \text{ gal} \times 1$$

Answer: 187,898 gallons

✓ Example 2.2.4

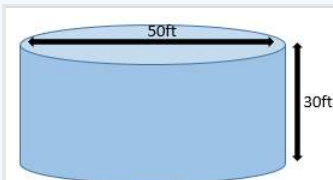
A rectangular water tank is full. The dimensions are 30ft high 12ft wide and a length of 10 ft. Convert to gallons.



$$\begin{aligned}\text{Volume} &= 10\text{ft} \times 12\text{ft} \times 30\text{ft} \\ \text{Volume} &= 3,600\text{ft}^3 \\ 3,600 \text{ ft}^3 \times 17.48 \text{ gal} \times 1 \text{ ft}^3 &= 26,928 \text{ gal} \times 1 \\ \text{Answer: } &26,928 \text{ gallons}\end{aligned}$$

✓ Example 2.2.5

A cylindrical storage tank has a diameter of 50ft and a height of 30ft. The tank is half full. How many gallons are in the tank?



$$\begin{aligned}\text{Volume} &= .785 \times 50\text{ft} \times 50\text{ft} \times 15\text{ft} \\ \text{Volume} &= 29,437.5 \text{ rounded to } 29,438\text{ft}^3 \\ 29,438 \text{ ft}^3 \times 17.48 \text{ gal} \times 1 \text{ ft}^3 &= 220,196 \text{ gal} \times 1 \\ \text{Answer: } &220,196 \text{ gallons}\end{aligned}$$

Chapter Review

1. The smallest part of an element is known as _____.

1. Proton

2. Neutron
 3. Atom
 4. Nucleus
2. The atomic weight is comprised of _____.
1. Neutrons and electrons
 2. Protons and neutrons
 3. Nucleus and neutrons
 4. Atom and nucleus
3. An atom with a negative charge is known as _____.
1. Proton
 2. Neutron
 3. Nucleus
 4. Electron
4. What does the symbol mg/L stand for?
1. Micrograms per liter
 2. Milligrams per/L
 3. parts per million
 4. Both 2 and 3
5. What does the acronym MCL stand for?
1. Minimum contaminant level
 2. Micron contaminant level
 3. Maximum contaminant Level
 4. Milligrams counted last
6. How long do sanitary surveys have to be retained for records?
1. 3 years
 2. 5 years
 3. 7 years
 4. 10 years
7. The most severe water system violation that requires the fastest public notification is _____.
1. Tier I
 2. Tier II
 3. Tier III
 4. Tier IV
8. The primacy agency may grant a variance or exemption as long as _____.
1. The agency is using the Best Available Technology
 2. There is no threat to public health
 3. There is never a scenario for a variance or exemption
 4. Both 1 and 2
9. A public water system that serves at least 25 people six months out of the year is known as _____.
1. Nontransient noncommunity
 2. Transient noncommunity
 3. Community public water system
 4. None of the above
10. Regulations based on the aesthetic quality of drinking water are known as _____.
1. Primary Standards
 2. Secondary Standards
 3. Microbiological Standards

4. Radiological Standards

11. The lowest reportable limit for a water sample is _____.
1. 0.5 mg/L
 2. Zero
 3. Public health goal
 4. Detection Level for reporting
12. Primary Standards are based on _____
1. Color and Taste
 2. Aesthetic quality
 3. Public Health
 4. Odor
13. A circular clearwell is 150 feet in diameter and 40 feet tall. The Clearwell has an overflow at 35 feet. What is the maximum amount of water the clearwell can hold in Million gallons rounded to the nearest hundredth?
1. MG
 2. 4.62 MG
 3. 18.50 MG
 4. 7.50 MG
14. A sedimentation basin is 400 feet length, 50 feet in width, and 15 feet deep. What is the volume expressed in cubic feet?
1. 100,000 ft³
 2. 200,000 ft³
 3. 300,000 ft³
 4. 400,000 ft³
15. A clearwell holds 314,000 ft³ of water. It is 100 ft in diameter. What is the height of the clearwell?
1. 25 ft
 2. 30 ft
 3. 35 ft
 4. 40 ft

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2.3: Microorganisms

The most important job of a Water Treatment Operator is providing reliable and quality water to the public. This is accomplished through chemical deactivation or physical removal of disease-causing microorganisms in the water. Microorganisms are deactivated by the addition of a chemical such as Chlorine or Ozone while physical removal is accomplished with the use of a filtration system.

Surface Water Treatment Rule

The **Surface Water Treatment Rule** (SWTR) was created in 1990 to further protect the public from waterborne illness. The specific diseases the SWTR seeks to prevent are those caused by viruses, legionella, and *Giardia lamblia*. Disease-causing microorganisms are also known as pathogens. The microbes that cause most waterborne illnesses are found in most surface water in the United States. Every public agency that uses surface water as source water must adhere to the SWTR. Surface Water is defined as any open body of water that is susceptible to surface runoff. Rainfall and snowmelt that enter open bodies of water such as lakes, rivers, and man-made reservoirs are susceptible to contamination from sewage treatment plants and animal feces.

At this time there are no simple inexpensive tests for viruses, legionella, and *Giardia lamblia*; therefore, other protective measures are used to test the source water and treated water. Water regulations provide detailed inactivation and removal regulations for Protozoa, also known as *Giardia*, and viruses. Bacteria fall somewhere in the middle of both microorganisms so there is no need to “regulate” them. Bacteria will be removed and/or deactivated within the parameters or regulations that are in place. *Giardia* and *Cryptosporidium* form hard cysts that are very difficult to deactivate chemically with Chlorine. They are larger than viruses so they or more easily removed through filtration. Newer technological advances such as ozone addition are excellent ways to deactivate *Giardia*.

Treatment plants must use a combination of disinfectant (chemical deactivation) and filtration and must achieve 99.9% removal or inactivation of *Giardia* cysts and 99.99% of viruses. You might also see the percentages expressed as 3 Log and 4 Log. 99.9% will be expressed as 3 Log and 99.99% will be expressed as 4 Log. Treatment plants, which use filtration, achieve their removal factor by monitoring the combined filter effluent turbidity. Turbidity is the measurement of the cloudiness in water. The reason it is important to measure turbidity is because microorganisms can hide behind the very small particles and render treatment techniques ineffective. Turbidity readings are expressed as nephelometric turbidity units (NTU). Most surface source water will not meet standards that would enable a treatment plant to refrain from using approved filtration techniques. The approved treatment techniques currently used by water operators are conventional treatment, direct filtration, slow sand, diatomaceous earth, reverse osmosis, and alternate treatment technologies approved on a case by case basis by the primacy regulating board. Newer technologies such as membrane filtration and UV treatment are being employed by some water agencies.

Table 2.3.1: Waterborne Pathogens

Bacteria	Viruses	Protozoa
• Campylobacter • Escherichia coli (E-coli) • Salmonella • Yersinia • Vibrio • Legionella • Aermmonas • Mycobacterium • Shigella • Pseudomonas	• Hepatitis A • Reovirus • Calicivirus • Enterovirus • Coxsackievirus • Adenovirus • Echovirus • Poliovirus	• <i>Giardia Lamblia</i> • <i>Cryptosporidium</i> • Entameoba • <i>Microsporidium</i>

A continuous disinfectant residual must always be in the distribution system to prevent waterborne illness. A disinfectant residual of .2 mg/L must be present at all times in the distribution system, however, most water operators will maintain somewhere between 2.5 mg/L-3.0 mg/L. The CT calculation is used to make sure proper disinfectant levels are being used during the treatment and distribution process. The effectiveness of the treatment process is calculated with the CT formula and uses data such as disinfectant

used, residual of disinfectant, length of time disinfectant is in contact with water, water temperature, and pH of the water. It is important to note that during the treatment process the goal is to disinfect the water (kill all pathogens) and not to sterilize (kill all organisms).

The “C” is the concentration of disinfectant while the “T” is the amount of time the disinfectant is in contact with water. On days with higher effluent plant rates, an operator may need to raise chlorine doses as the disinfectant will be in contact with the water for less time. Conversely, if plant flow rates are lower, a lower chlorine dose can be used because the disinfectant is in contact with the water for a greater time.

Case Study: Milwaukee April 1993

In April of 1993, the largest outbreak of *Cryptosporidium* on record occurred in Milwaukee, Wisconsin. Over 400,000 people reported illness and 100 people died. All of the people who died after drinking the tainted water had the AIDS virus. It can't be for certain that the *Cryptosporidium* was the cause of each of their deaths, but as will be discussed later in this chapter, people with weakened immune systems are much more susceptible to death if they drink tainted water.

The Milwaukee incident was the catalyst for enhancements to the SWTR. Since *Cryptosporidium* creates cysts and is not killed by chlorine, the “double barrier” treatment approach was created. Treatment plants were then required to monitor turbidity effluent levels and use alternative treatment techniques, such as ozone or ultraviolet light, that would deactivate the *Cryptosporidium*. An effective filtration plant using coagulation, flocculation, sedimentation, and filtration, should have been able to handle the outbreak, thus turbidity levels were not within standards.

The source of the *Cryptosporidium* was believed to be a sewage spill that was very close to the intake of the Southern Milwaukee treatment plant where the outbreak occurred. Others believe that the operators of the treatment plant were using higher quantities of waste stream water in the treatment process. Recycling washed filter water is a common practice, but after this event, it was established that only 10% of the sourced raw water can come from the wastewater.

The lesson everybody learned here is water quality is ever-changing and evolving. Mother Nature throws curve balls at us that change the quality of the source water. If something is wrong, it is your duty as an operator to say something. Regulations are all well and good but if they are not being followed, people can become very ill or possibly even die.

Total Coliform Rule

The **total coliform rule** was published in 1989 and was revised in 2014. It set up minimum requirements for the frequency and amount of coliform tests that would be taken by water agencies. As discussed earlier, the coliform bacteria exist in larger quantities than other bacteria. If there is a positive coliform test, there is a greater chance the treatment process is not functioning properly and there is an increased chance of pathogens in the water. Bacterial outbreaks in water cause gastroenteritis with the associated symptoms of nausea, diarrhea, vomiting, and cramps. Very young people, elderly, and people with weakened immune systems are at an even greater danger should there be waterborne illnesses in water. Each water agency is required to have an approved sampling site map approved by the primacy agency.

The Total Coliform Rule goal is zero positive coliform samples. Water systems that take less than 40 samples per month are only allowed one positive sample while systems that take greater than 40 samples must not find positive results in more than 5% of the coliform samples taken. If a positive coliform sample is found, it is not necessarily an indicator of a problem. Human sampling error could be the culprit. Once a positive is found the coliform sample is retaken within 24 hours at the positive sampling site and two additional samples are taken, one upstream and one downstream of the initial positive site. If there is a second positive coliform result, the water is then tested for E-Coli. If E-Coli is present in the sample, it is an immediate public health risk and is deemed an acute MCL violation. This would be considered a Tier I violation and public notification would have to occur within 24 hours.

Long-Term 1 Enhanced Surface Water Treatment Rule

The Long-Term 1 Enhanced Surface Water Treatment Rule (LT1ESWTR) applies to water systems that serve less than 10,000 people. Below is a list of requirements for water agencies that fall under this category:

- Must achieve 99% or 2 log removal of *Cryptosporidium*

- Most systems are required to meet turbidity standards of 0.5 NTU from 95% of combined filter effluent turbidity readings. Smaller systems will get 2 log removal credit of *Giardia* with a stricter .03 NTU combined filter effluent.
- Monitoring of individual filter is required. Higher turbidity reading from individual filters will require corrective action.
- Some states might require a disinfectant byproduct profile
- Any change in primary disinfectant must be profiled and approved
- All treated water clearwells must be covered and not open to atmosphere

Long-Term 2 Enhanced Surface Water Treatment Rule

The Long Term 2 Enhanced Surface Water Treatment Rule (LT2ESWTR) covers all water systems which serve more than 10,000 customers.

- This rule added enhancements to the SWTR. Water providers are required to test the source raw water for *Cryptosporidium* and E-coli for a 2 year period. This step was put in to ensure all treatment plant equipment was capable of properly deactivating and removing pathogens from water.
- After testing, systems were given a bin number which determined how susceptible the system was to contamination.
- If systems were susceptible, a time frame was given to correct deficiencies
- Continuous monitoring of the distribution system influent and maintaining at least .02 mg/L chlorine residual is required

Filter Back Wash Recycling Rule

As noted in the Milwaukee case study, water systems are able to use recycled wastewater as a raw water source. Many treatment plants utilize waste ponds, waste lagoons, and waste basins to capture recycled wash water and water from drains throughout treatment plant. In order for plants to use the recycled water, plants must put the wastewater through the same treatment process as raw untreated water. The amount of water that can be returned is based on the plants size and maximum plant flows.

Lead and Copper Rule

The lead and copper rule differs from most other water guidelines because these two constituents are usually found in water after it has gone through the treatment cycle. Most raw water has very low levels of lead and copper. Lead and copper are found in drinking water after a chemical reaction takes place in distribution pipes. Water that is more acidic can erode lead and copper pipes causing them to leach into water. The use of lead pipes in drinking water systems was banned in 1986 and they can no longer be used. Lead is known to cause several health problems in fetuses and young children. It can cause developmental problems. Lead can also have effects on the kidney, brain, red blood cells, and is known to cause anemia.

With optimal corrosion control, the ability of water to chemically react with lead and copper pipes is lowered. Many treatment plants add chemicals such as caustic soda to raise the pH of the water. Treated water with a pH around 8 will keep a thin layer of calcium on the inside of a distribution pipe which protects the pipe from corrosion. Corrosion control is tested by monitoring the conductivity, testing pH of water, water temperature, testing for calcium, testing for alkalinity, and testing for phosphate or silica corrosion inhibitor if used to control corrosion.

The Lead and Copper rule also differs from other rules because it bases its parameters on action levels. If 10 percent of the customers' water tests in the 90th percentile of the action level for lead and copper, then further preventative actions must be met. Most customers do not have to worry about this due to advances in corrosion control in the past few decades.



Figure 2.3.1: Image of lead pipes by Borsi112 is licensed under [CC BY-SA 3.0](#)

Stage 1 and 2 Disinfectant Bi-Product Rule

When natural organic matter mixes with chemical disinfectants, it is possible for disinfectant byproduct formation (DBP). Chemical disinfectants used in water treatment that form DBP's are chlorine, chlorine dioxide, chloramines, and ozone. The Stage 1 disinfectant byproduct rule established MCL's for trihalomethane (TTHM), haloacetic acid (HAA5), chlorite, and bromate. Compliance is set up on a running annual average for TTHM, HAA5, and Bromate and on a monthly average for chlorite. The rule also set up maximum disinfection levels for chlorine, chloramines, and chlorine dioxide.

The Stage 2 disinfectant byproduct rule applies to all community and nontransient noncommunity water systems that add a chemical disinfectant or purchase treated water that has a chemical disinfectant. The purpose of the Stage 2 rule is to monitor local sampling of individual connections. Some parts of the water system have less movement, therefore they are more susceptible to DBP formation. The Stage 2 rule covers TTHM and HAA5 and has the same MCL as the Stage 1 rule. See the chart below:

Disinfectant Residual	MRDL* as mg/L	Compliance Based On
Chlorine	4.0 mg/L	Running annual average RAA
Chloramines	4.0 mg/L	Running annual average RAA
Chlorine Dioxide	0.8 mg/L	Daily samples

*Maximum residual disinfectant level

Disinfection Byproduct	Maximum Contaminant Level	Compliance Based On
Trihalomethane	.080 mg/L (80 ppb)	Running annual average
Haloacetic acid	.060 mg/L (60 ppb)	Running annual average
bromate	.010 mg/L (10 ppb)	Running annual average
chlorite	1.0 mg/L (1 ppm)	Monthly average

*ppb- parts per billion ppm- parts per million

Consumer Confidence Reports

The public has a right to know what is in their drinking water. As you continue to read through the text you will notice that a lot of the terminology is quite a mouthful. In 1998 consumer confidence reports were made available to the public for transparency. Eight groups of information were added to the report:

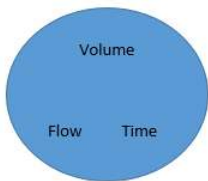
- System information including contact info
- Different sources of water (i.e. lakes, wells, rivers)
- Definitions non-water personnel can understand including MCL's, MCLG's and treatment techniques
- Any detected contaminants in the system along with a listing of the possible health effects if limits and goals are not met

- Non-regulated contaminants found
- List of violations if they occurred
- Variances and exemptions
- Educational tools for contaminants and affected populations

The internet has made it possible for water treatment facilities to post information monthly on company websites; however, a consumer confidence report will be mailed/electronically delivered to its customers on a yearly basis.

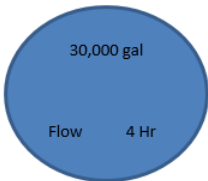
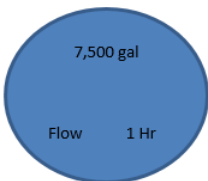
Flow Rate Calculation

The flow rate calculation will be one of the most frequently used in water math. This calculation tells us a lot about what is going on in our water system. Common flow rates you will see are Gallons per Minute (GPM), Cubic Feet Per Second (CFS), Million Gallons a Day (MGD), and Acre Feet per Year (AFY). Water operators use flow rates for different purposes. If you were running a small treatment plant, you might use GPM or CFS in your day to day operations but when looking at water produced throughout the entire month or year, calculating in Acre Feet may be more appropriate. Like any algebraic formula, there will be an unknown factor you are solving for. In the case of a flow question, there will be 3 values, two known and one unknown.

	Flow Rate Equation $\text{Flow Rate} = \text{Volume} \div \text{Time}$ The algebraic “Wheel” is a very effective way to solve many water math problems. Those students with more “mechanical” minds may find this way of solving math problems easier. Below is an example of how to solve a flow rate equation with both methods.
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Example 1

In 4 hours, a water tank's volume increases by 30,000 gallons. What was the flow rate of the water entering the tank? Express in gallons per minute.

Flow Rate = Volume/Time $\text{Flow Rate} = 30,000 \text{ gallons} / 4 \text{ hours}$ $7,500 \text{ gallons} / 1 \text{ hour}$ $7,500 \text{ gallons} / 1 \text{ hour} = 1 \text{ hour} / 60 \text{ min}$ $= 125 \text{ gallons/minute}$		
---	---	---

We can also use the Flow Rate calculation to solve for volume or time.

Example 2 - Solve for Time

How long will it take to drain a 100,000-gallon storage tank where the water is exiting at 2,500 Gallons per minute?

$$\text{Time} = 100,000 \text{ gal} / 2,500 \text{ GPM} = 40 \text{ minutes}$$

Always put the unknown in the left part of the formula so your factors cancel out properly.

Example 3 - Solve for Volume

Your water tank pump is set to run for 90 minutes. Your pump output is 3,000 GPM. How many gallons of water will enter the tank?

$$\text{Volume} = 3,000 \text{ GPM} \times 90 \text{ min}$$

$$\text{Volume} = 270,000 \text{ gallons}$$

Example 4

A water tank's full capacity is 500,000 gallons. The operator will bring the tank down to half full in an 8-hour period. What is the flow rate of the water exiting the tank?

(Note: The state gets very creative with their test questions. Remember to read the question multiple times before solving it. Don't be the test taker that uses 500,000 gallons in the equation instead of 250,000 gallons!)

Flow Rate = 250,000 gallons/8 hours

Flow Rate = 31,250 gallons/1 hour

31,250 gal/1 hour = 1 hour/60 min = 520.83 GPM

Chapter Review

1. A disease-causing microorganism:
 1. Pathogen
 2. Colilert
 3. Pathological
 4. Turbidity
2. According to the Surface Water Treatment Rule, what is the combined inactivation and removal for *Giardia*?
 1. 1.0 Logs
 2. 2.0 Logs
 3. 3.0 Logs
 4. 4.0 Logs
3. What is the equivalency expressed as a percentage for the SWTR inactivation and removal of viruses?
 1. 99.9%
 2. 99.99%
 3. 99.0%
 4. 99.999%
4. A water agency that takes more than 40 coliform samples must fall under what percentile?
 1. 10%
 2. 7%
 3. 5%
 4. No positive samples allowable
5. The multiple barrier treatment approach includes _____.
 1. Sterilization and filtration
 2. Disinfection and filtration
 3. Disinfection and sterilization
 4. Infection and filtration
6. The maximum disinfectant residual allowed for chlorine in a water system is _____.
 1. .02 mg/L
 2. 2.0 mg/L
 3. 3.0 mg/L
 4. 4.0 mg/L
7. How do water agencies monitor the effectiveness of their filtration process?
 1. Alkalinity
 2. Conductivity
 3. Turbidity
 4. pH
8. What is the disinfectant byproduct caused by ozonation?
 1. Trihalomethanes
 2. Bromate
 3. Chlorite

4. No DBP formation
9. Haloacetic Acids are also known as _____.
 1. TTHM
 2. HOCL
 3. Chlorite
 4. HAA5
10. What is the MCL for trihalomethanes?
 1. .10 mg/L
 2. .06 mg/L
 3. .08 mg/L
 4. .12 mg/L
11. What is the MCL for Haloacetic Acids?
 1. 100 ppb
 2. 60 ppb
 3. 80 ppb
 4. 120 ppb
12. What is the MCL for bromate?
 1. .010 mg/L
 2. .020 mg/L
 3. .030 mg/L
 4. .040 mg/L
13. A treatment plant operator must fill a clearwell with 10,000 ft³ of water in 90 minutes. What is the rate of flow expressed in GPM?
 1. 111 GPM
 2. 831 GPM
 3. 181 GPM
 4. 900 GPM
14. A water tank has a capacity of 6MG. It is currently half full. It will take 6 hours to fill. What is the flow rate of the pump?
 1. 3,333 GPM
 2. 6,333 GPM
 3. 8,333 GPM
 4. 16,666 GPM
15. A clearwell with a capacity of 2.5 MG is being filled after a maintenance period. The flow rate is 2,500 GPM. The operator begins filling at 7 AM. At what time will the clearwell be full?
 1. 10:00 PM
 2. 10:40 PM
 3. 11:00 PM
 4. 11:40 PM

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3: Primary Standards

Learning Objectives

After reading this section, you should be able to:

- Define primary standards
- Describe the primary metal contaminants
- Describe the primary organic contaminants
- Describe the primary synthetic organic contaminants
- Describe sampling guidelines
- Describe the Radiologic Rule
- Describe the Total Coliform Rule

Federal (Environmental Protection Agency, EPA)

The process by which EPA sets drinking water standards is long and complicated. It involves deciding which contaminants may endanger public health, conducting studies of the effects of these contaminants, defining the maximum safe level of each contaminant, estimating the costs and benefits of regulation, proposing a standard or limit, listening to and evaluating public reactions to the proposed standard, revising the standard (if needed), proposing the standard in final form, seeking and evaluating additional public input, and finally, publishing the standard. The **primary standards** are referred to as the Maximum Contaminant Level (MCL). When the level of a contaminant is above the MCL, it is considered to be a health hazard. MCLs are enforceable by law.

The entire process by which a standard is proposed and promulgated is governed by strict procedural guidelines. It often takes three or more years to produce a standard.

Types of Contaminants

Five types of primary contaminants are considered to be of public health importance:

1. **Inorganic contaminants**, such as lead and mercury;
2. **Organic contaminants**, which include pesticides, herbicides, THMs, solvents, and other synthetic organic compounds;
3. **Turbidity**, such as small particles suspended in water that interfere with light penetration and disinfection;
4. **Microbial contaminants**, such as bacteria, viruses, and protozoa; and
5. **Radiologic contaminants**, which include natural and manmade sources of radiation.

Identifying Contaminants to Be Regulated

The 1986 Safe Drinking Water Act (SDWA) required EPA to establish a priority list of contaminants that may have adverse health effects and may require regulation. The 1996 SDWA amendments retained the list of 83 contaminants developed earlier but revised the process by which contaminants are selected for regulation. Before selecting a contaminant for regulation, the EPA must now consult with the scientific community, solicit public comments, and demonstrate that the contaminant actually occurs in public water systems. For this latter purpose, the 1996 amendments required EPA to establish an occurrence database by August 1999. The database contains information concerning regulated and unregulated contaminants and the information is available to the public.

To regulate a substance, EPA must demonstrate that the contaminant meets three criteria:

1. The contaminant has an adverse effect on human health;
2. It occurs, or is likely to occur, in public water systems at a frequency and concentration of significance to public health; and
3. Regulation of the contaminant offers a meaningful opportunity to reduce health risks for people served by public water systems.

In addition to meeting these three criteria, EPA must also weigh the relative health risks of various contaminants being considered for regulation, the risk reduction that regulation would accomplish, and the costs of implementing the regulations.

Working on a 5-year cycle, the EPA is required to select at least five contaminants from its Drinking Water Contaminant Candidate List (DWCCCL) and decide whether there sufficient reason exists to regulate them. The first DWCCCL included 50 chemicals and 10

microbial contaminants. In June 2002, the EPA announced a preliminary finding that their research revealed no need to regulate ten of the listed contaminants. Based on public comments and continued research and evaluation, EPA will make a final determination about regulating any of these contaminants and will begin to develop proposed regulations as necessary.

Unregulated Contaminants

EPA uses the Unregulated Contaminant Monitoring (UCM) program to collect data for contaminants suspected to be present in drinking water, although they do not have health-based standards set under the SDWA. Every 5 years, EPA reviews the list of contaminants, largely based on the Contaminant Candidate List.

The SDWA Amendments of 1996 provide for the following:

- Monitoring no more than 30 contaminants per 5-year cycle
- Monitoring only a representative sample of public water systems
- serving fewer than 10,000 people
- Storing analytical results in a National Contaminant Occurrence Database (NCOD)

The UCM program has progressed in several stages. Currently, EPA manages the program directly as specified in the Unregulated Contaminant Monitoring Regulation (UCMR). EPA has established a timeline and plans for monitoring unregulated contaminants through 2015.

Under the UCMR, community water systems and non-transient non-community water systems serving more than 10,000 people are required to monitor for unregulated contaminants; smaller systems may also have to conduct monitoring if required by the state. Transient water systems are not affected by the UCMR.

Always keep in mind that EPA's regulatory process is basically just a process. No list is permanent; regulations change frequently as new information becomes available. Water treatment plant operators must continually seek opportunities to learn about the regulations that affect the water industry. Also, operators should feel free to express their concerns regarding the regulations and the regulatory process to EPA.

Primary Standards

Inorganic chemicals are metals, salts, and other chemical compounds that do not contain carbon. The health concerns about inorganic chemicals are not centered on cancer, but rather on their suspected links to several different human disorders. For example, lead is suspected of contributing to mental retardation in children. Waters exceeding the MCL for these elements for short periods of time will pose no immediate threat to health. However, studies show that these substances must be controlled because consumption of drinking water that exceeds these standards over long periods of time may prove harmful.

Antimony

In the Code of Federal Regulations Reference: 40 CFR 721.1930, antimony is used in the production of ceramics, glassware, and pigments. Most exposures to antimony occur in industrial settings where workers may inhale dust containing particles of the metal.

Purpose of Regulating Antimony: To improve public health protection by reducing exposure to the amount of antimony in drinking water.

Public Health Benefit: Ingestion of antimony can alter cholesterol and glucose levels and may cause chromosome damage.

MCL/MCLG: 0.006 mg/L/0.006 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Arsenic

In the Code of Federal Regulations Reference: 40 CFR 141 Subpart B, arsenic occurs naturally in the environment, especially in the western United States, and it is used in insecticides. Arsenic is found in foods, tobacco, shellfish, drinking water, and in the air in some locations. Anyone who drinks water that continuously exceeds the national standard by a substantial amount over a lifetime may experience fatigue and loss of energy. Extremely high levels of arsenic can cause poisoning.

The regulation of arsenic has been vigorously debated. In 1997, EPA released a health effects study plan. Under this plan, EPA worked with the Water Research Foundation (WRF) and the Association of California Water Agencies to assess the health risks from exposure to low levels of arsenic. Based on the results of these studies, EPA proposed an MCL for arsenic in June 2000, and a final rule was issued January 22, 2001. Technologies available for removing arsenic include ion exchange, iron hydroxide coagulation followed by microfiltration, and activated alumina.

Purpose of Regulating Arsenic: To improve public health protection by reducing exposure to arsenic in drinking water.

Public Health Benefit: The avoidance of bladder and lung cancer, dental problems, and a reduction in the frequency of non-carcinogenic diseases.

MCL/MCLG: 0.010 mg/L/zero

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Asbestos

In the Code of Federal Regulations Reference: 40 CFR 141.62, the general term asbestos refers to a family of fibrous silicate minerals that have been widely used in the manufacture of commercial products. Some examples of products containing asbestos include floor and ceiling tiles, paper products, paint and caulking, plastics, brake linings, insulation, cement, and filters.

Asbestos in cement pipes is a potential source of contamination of drinking water. Deterioration of this material with age and exposure to corrosive water is thought to release asbestos particles. Mining for asbestos minerals has also contributed to the contamination of some drinking water sources throughout the United States. Scientific evidence on the harmful effects of asbestos consumed in drinking water is less conclusive than the evidence relating to inhaled asbestos.

Purpose of Regulating Asbestos: To improve public health protection by reducing exposure to asbestos in drinking water.

Public Health Benefit: To reduce the risk of developing benign intestinal polyps.

MCL/MCLG: 7 million fibers per liter (MFL)/7 MFL

Monitoring Requirements: Once every 9 years.

Barium

Code of Federal Regulations Reference: 40 CFR 141.62, although not as widespread as arsenic, this element also occurs naturally in the environment in some areas of the US. Barium can also enter water supplies through industrial waste discharges. Small doses of barium are not harmful. However, it is quite dangerous when consumed in large quantities and will bring on increased blood pressure, nerve damage, and death.

Purpose of Regulating Barium: To improve public health protection by reducing exposure to large quantities of barium in drinking water.

Public Health Benefit: To reduce the risk of high blood pressure.

MCL/MCLG: 2 mg/L/2 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Beryllium

In the Code of Federal Regulations Reference: 40 CFR 141.62, the major source of beryllium in the environment is the combustion of coal and oil. Airborne particulates containing beryllium can be inhaled or washed by precipitation into drinking water supplies. Animal studies show that when large doses of beryllium are ingested, it is carried by the bloodstream from the stomach to the liver; however, it is gradually transferred to the animal's bones.

Purpose of Regulating Beryllium: To improve public health protection by reducing exposure to large quantities of beryllium in drinking water.

Public Health Benefit: To reduce the risk of intestinal lesions.

MCL/MCLG: 0.004 mg/L/0.004 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Bromate

In the Code of Federal Regulations Reference: 40 CFR Parts 9, 141, and 142, bromate is formed as a byproduct of ozone disinfection of drinking water. Ozone reacts with the naturally occurring bromide ion in the water to form bromate. Bromate is an inorganic ion that is tasteless, colorless, and has a low volatility. Bromate dissolves easily in water and is fairly stable. Bromate is not typically derived from natural sources. It is commonly found in raw water supplies such as streams, lakes, and reservoirs.

Purpose of Regulating Bromate: To improve public health protection by reducing exposure to bromate in drinking water.

Public Health Benefit: To reduce the risk of cancer.

MCL/MCLG: 0.010 mg/L/zero

Monitoring Requirements: Monthly (ozone systems only).

Cadmium

In the Code of Federal Regulations Reference: 40 CFR 141.62, only extremely small amounts of cadmium are found in natural waters in the United States. Waste discharges from the electroplating, photography, insecticide, and metallurgy industries can increase cadmium levels. The most common source of cadmium in drinking water is from galvanized pipes and fixtures.

Purpose of Regulating Cadmium: To improve public health protection by reducing exposure to cadmium in drinking water.

Public Health Benefit: To prevent kidney damage.

MCL/MCLG: 0.005 mg/L/0.005 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Chlorite

In the Code of Federal Regulations Reference: 40 CFR Parts 9, 141, and 142, the main source of chlorite in drinking water is chlorine dioxide used for disinfection. Chlorite forms as the chlorine dioxide breaks down during the disinfection process. Chlorite is an inorganic ion that is colorless, odorless, tasteless, and dissolves easily in water. Chlorite is one of the two or three chemicals involved in the process of generating chlorine dioxide for water treatment. Chlorite is not typically derived from natural sources.

Purpose of Regulating Chlorite: To improve public health protection by reducing exposure to chlorite in drinking water, especially for infants and young children.

Public Health Benefit: To prevent adverse effects of the nervous system and anemia.

MCL/MCLG: 1.0 mg/L/0.8 mg/L

Monitoring Requirements: For systems that add chlorine dioxide, daily samples are required at distribution system entry points.

Chromium

In the Code of Federal Regulations Reference: 40 CFR 141.62, chromium is found in cigarettes, some foods, and the air. It is also discharged from steel and pulp mills and the erosion of natural deposits. Some studies suggest that in very small amounts, chromium may be essential to human beings, but it has not been proven.

Purpose of Regulating Chromium: To improve public health protection by reducing exposure to chromium in drinking water.

Public Health Benefit: To prevent allergic dermatitis.

MCL/MCLG: 0.1 mg/L/0.1 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Copper

In the Code of Federal Regulations Reference: 40 CFR 141.86, copper in drinking water usually results from the reaction of water on copper plumbing. Treatment of surface water in storage reservoirs to control algae may also cause high levels of copper. Copper is an essential nutrient; adults require 2 mg daily and children of preschool age require about 0.1 mg daily for normal growth.

The presence of copper in drinking water is also undesirable because of its aesthetic effects. At low levels (0.5 mg/L in soft waters) blue or blue-green staining of porcelain occurs. At higher levels (4 mg/L), it will stain clothing and blond hair. Concentrations greater than 1 mg/L can produce insoluble green curds when reacting with soap.

The primary standard for copper includes a treatment technique of the taps sampled exceed the action level (AL) of 1.3 mg/L.

Action levels communicate health or physical hazards by categorizing these hazards into levels. Action levels are used by OSHA and NIOSH to determine how harmful a substance or activity is.

Purpose of Regulating Copper: To improve public health protection by reducing water corrosivity.

Public Health Benefit: To prevent gastrointestinal, kidney, and liver problems.

MCL/MCLG: A treatment technique (TT) is triggered at AL = 1.3 mg/L/1.3 mg/L

Monitoring Requirements: Residential sample taken at the kitchen or bathroom sink tap. ALs must be met in 90 percent of the samples. Follow-up monitoring every 6 months after corrosion controls initiated or optimized. Reduced monitoring for systems consistently meeting AL.

Cyanide

In the Code of Federal Regulations Reference: 40 CFR 141.62, cyanide is a common ingredient of rat poisons, silver and metal polishes, and photo processing chemicals. The major source of cyanide in drinking water is discharge from industrial chemical factories.

Ingestion of even very small amounts of sodium cyanide or potassium cyanide can cause death within a few minutes to a few hours. Typical symptoms of cyanide poisoning include nausea without vomiting, anxiety, vertigo, lower jaw stiffness, convulsions, paralysis, irregular heartbeat, and coma.

Purpose of Regulating Cyanide: To improve public health protection by reducing long-term exposure to cyanide.

Public Health Benefit: To prevent thyroid/neurological damage.

MCL/MCLG: 0.2 mg/L/0.2 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Fluoride

In the Code of Federal Regulations Reference: 40 CFR 141.62, fluoride is a natural mineral and many sources of drinking water contain some fluoride. Fluoride produces two effects, depending on its concentration. EPA has set primary and secondary limits to regulate it. Many communities add fluoride to their drinking water to promote dental health. The decision to fluoridate a water supply is made by the state or local municipality, and is not mandated by EPA or any other federal entity.

However, EPA has completed and peer-reviewed a quantitative dose-response assessment based on the available data for severe dental fluorosis as recommended by the National Research Council (NRC). At levels of 6 to 8 mg/L, fluoride may cause brittle bones and stiffening of the joints. On the basis of this health hazard, fluoride has been added to the list of primary standards.

At levels of 2 mg/L and greater, fluoride may cause dental fluorosis, which is discoloration and mottling of the teeth, especially in children. EPA has reclassified dental fluorosis as a cosmetic effect, raised the primary drinking water standard from 1.4–2 mg/L to 4 mg/L, and established a secondary standard of 2 mg/L for fluoride.

Purpose of Regulating Fluoride: To improve public health protection by preventing long-term exposure to fluoride.

Public Health Benefit: To prevent bone disease.

MCL/MCLG: 4.0 mg/L/4.0 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Lead and Copper

In the Code of Federal Regulations Reference: 40 CFR 141 Subpart I, is concerned with the control of lead and copper. Basic Requirements of the Lead and Copper Rule (LCR) are:

1. Require water suppliers to optimize their treatment system to control corrosion in customers' plumbing.
2. Determine tap-water levels of lead and copper for customers who have lead service lines or lead-based solder in their plumbing system.
3. Rule out the source water as a source of significant lead levels.
4. If lead ALs are exceeded, require the suppliers to educate their customers about lead and suggest actions they can take to reduce their exposure to lead through public notices and public education programs. If a water system, after installing and optimizing corrosion control treatment, continues to fail to meet the lead AL, it must begin replacing the lead service lines under its ownership.

Revisions to the Lead and Copper Rule (LCR)—EPA is promulgating a rule that makes several targeted regulatory revisions to the existing national primary drinking water regulations

(NPDWRs) for lead and copper. The revisions to the LCR will do the following:

1. Enhance the implementation of the LCR in the areas of monitoring, treatment, customer awareness, and lead service line replacement
2. Improve compliance with the public education requirements of the LCR to ensure drinking water consumers receive meaningful, timely, and useful information needed to help them limit their exposure to lead in drinking water

Lead is a naturally occurring metal that was used regularly in a number of industrial capacities and commonly used in household plumbing materials and water service lines. Lead was used as a component of paint, piping, solder, brass, and as a gasoline additive until the 1980s. The greatest exposure to lead is swallowing or breathing it in lead paint chips and dust. Research has confirmed that lead is highly toxic.

Lead is rarely found in source water. Mostly, lead contamination occurs after water has left the treatment plant due to corrosion of plumbing materials. Since water is naturally corrosive, it corrodes the pipes and plumbing through which it passes. Homes built before 1986 are more likely to have lead pipes, fixtures, and solder. However, new homes are also at risk because even legally lead-free plumbing may contain up to 8-percent lead. Grounding of electrical circuits in homes to water pipes and galvanic action between two dissimilar metals may increase corrosion that could cause lead to leach into the water.

Purpose of Regulating Lead: To protect public health by minimizing lead levels in drinking water, primarily by reducing water corrosivity.

Public Health Benefit: To prevent kidney problems, high blood pressure, and delays in physical and mental development in infants and children.

MCL/MCLG: A TT is triggered at AL = 0.015 mg/L/zero

Monitoring Requirements/Comments: Residential samples taken at the kitchen or bathroom sink tap. ALs must be met in 90 percent of the samples. Follow-up monitoring every 6 months after corrosion controls initiated or optimized. Reduced monitoring requirements for systems consistently meeting the AL. An AL exceedance is not a violation but can trigger other requirements that include monitoring for the following water quality indicators: pH, corrosion, control treatment, source water monitoring/treatment, public education, and lead service line replacement.

Copper is a metal found in natural deposits such as ores containing other elements. Copper is widely used in household plumbing materials.

Purpose of Regulating Copper: To protect public health by minimizing the risk of ingesting copper in drinking water, primarily from the corrosion of pipes.

Public Health Benefit: To prevent gastrointestinal distress due to short-term exposure and to prevent liver or kidney damage from long-term exposure.

MCL/MCLG: A TT is triggered at AL = 1.3 mg/L/1.3 mg/L

Monitoring Requirements/Comments: Residential samples taken at the kitchen or bathroom sink tap. ALs must be met in 90 percent of the samples. Follow-up monitoring every 6 months after corrosion controls initiated or optimized. There are reduced monitoring requirements for systems consistently meeting the AL. An AL exceedance is not a violation but can trigger other requirements that include monitoring for the following water quality indicators: pH, corrosion control treatment, source water monitoring/treatment, public education, and lead service line replacement.

All water systems are required to optimize corrosion control; that is to say, they must do the best they can to control corrosion. Small- and medium-sized systems are considered to have optimized corrosion control if they meet the lead and copper action levels during each of two consecutive six-month sampling periods. Large systems have optimized corrosion control if the difference between the source water lead level and the 90th percentile tap-water lead level is less than 0.005 mg/L, and the 90th percentile copper concentration is less than or equal to half the copper action level (0.65 mg/L) for two consecutive six-month periods.

Any system that is achieving optimal corrosion control and meeting the action limits is permitted to reduce monitoring and reporting frequencies. Large water systems that do not meet the guidelines for optimal corrosion control are required to conduct studies of various types of corrosion control treatment and then submit to the state a plan that would adequately control corrosion in their distribution system.

Small- and medium-sized systems exceeding the action levels must make a written recommendation to the state for a proposed treatment method, but may also be required to conduct similar studies before the state makes a final determination.

Once the state directs a water supplier to install treatment, the agency has 24 months to install the specified treatment and 12 months to collect follow-up samples to determine if the system is working. The state will also determine maximum acceptable levels for pH, alkalinity, calcium (if carbonate stabilization is used), orthophosphate (if an inhibitor with a phosphate compound is used), and silica (if an inhibitor with a silicate compound is used). The system must then continue to operate within these water quality guidelines.

If corrosion control methods fail to reduce lead and copper concentrations below the action levels, the water supplier must collect and analyze source water samples. Data from this sampling is sent to the state along with the water agency's recommended treatment plan for reducing contaminant levels. The state may accept the agency's recommended plan or may require some alternative treatment such as ion exchange, reverse osmosis, lime softening, or coagulation-filtration.

When required to install source water treatment, the agency is allowed 24 months for installation and 12 months to collect follow-up samples of specific water quality indicators. Based on the results of the follow-up sampling, the state determines the maximum allowable source water lead and copper concentrations and the water system is required to meet these state limits on an ongoing basis.

Water systems that continue to exceed the lead action level after installing optimal corrosion control treatment and source water treatment will have to begin a program to replace lead service lines within a maximum of 15 years.

Sampling Guidelines

The sampling required by the Lead and Copper Rule includes tap-water sampling at the consumer's faucet and distribution system sampling. Tap-water samples for initial and routine lead and copper monitoring must be collected at high-risk locations, which are defined as homes with lead solder installed after 1982, homes with lead pipes, and homes with lead service lines.

Tap-water samples for water quality indicators should be taken from representative taps throughout the distribution system. The representative taps can be the same as total coliform sampling sites. These samples will be analyzed for water quality indicator levels to identify optimal treatment and monitor compliance with the rule.

The rule calls for first-draw or first-flush samples, which are water samples taken after the water stands motionless in the plumbing pipes for at least six hours. This rule usually means taking a sample early in the day before water is used in the kitchen or bathroom. The rule permits water utilities to instruct consumers in sampling techniques so that they may collect the samples.

One-liter samples are needed.

Distribution system sampling conducted in connection with corrosion control studies requires samples from representative taps throughout the system as well as samples from entry points to the distribution system.

Initial and Base Monitoring Sample Sites

Compliance with the Lead and Copper Rule depends on meeting the action levels during two consecutive six-month monitoring periods. All public water systems are required to collect one sample for lead and copper analysis from a number of sites depending on population during each six-month monitoring period.

If a system meets the lead and copper action levels or maintains optimal corrosion control treatment for two consecutive six-month periods, tap-water sampling for lead and copper can be reduced to once per year and half the number of sites. After three consecutive years of acceptable performance, tap-water sampling can be reduced to once every three years at the reduced number of sites listed above.

Water Quality Indicator Frequency

All large systems (more than 50,000 persons) and any small- or medium-sized system that exceeds the lead or copper action level will be required to monitor for the following water quality indicators: pH, alkalinity, calcium, conductivity, orthophosphate, silica, and water temperature. Samples must be taken from representative taps throughout the system and at entry points to the distribution system. Two tap-water samples must be collected for each applicable water quality indicator.

In addition to the sampling from representative taps, one sample must be collected every two weeks for each applicable water quality indicator at each entry point to the distribution system.

All large water systems and any small- and medium-sized system that exceed the lead or copper action levels after installing optimal corrosion control treatment must continue to collect samples as follows:

1. 2 samples for each applicable water quality indicator at each of the sampling sites (taps) every 6 months.
2. 1 sample for each applicable water quality indicator at each entry point to the distribution system every 2 weeks.
3. After meeting water quality guidelines for 2 consecutive 6-month monitoring periods, systems may reduce the number of tap samples collected, and after 3 consecutive years of acceptable performance they will be permitted to collect annual samples at the reduced number of sampling points.

Calculation of the 90th Percentile

The action levels that trigger implementation of treatment techniques to reduce lead and copper concentrations are based on the principle that no more than 10 percent of the samples should exceed the limit. To determine whether a system meets this test, it is necessary to figure out the lead or copper concentration on the sample that falls at the 90th percentile.

To calculate the 90th percentile value, write down the results from a set of samples arranging them in order of increasing or decreasing concentration. Number each sample beginning with the number 1 for the smallest concentration. Multiply the total number of samples (which will be the same as the number assigned to the highest concentration) by 0.9. Then, use the value from this calculation to identify the sample with the same number.

The concentration of the sample whose number matches the calculated value is the 90th percentile and is used in determining compliance with the action level. For systems serving fewer than 100 people, the 90th percentile is the average of the highest concentration and the second highest concentration.

Public Education

A public education program developed by the EPA must be delivered to water system customers within 60 days whenever the system exceeds the lead or copper action levels. The information describes the harmful effects of lead in drinking water and tells customers what they can do to reduce their exposure. The education program should continue as long as the system fails to meet the lead or copper action levels. In conjunction with the public education program, the water system must offer to sample a customer's water if asked to do so. However, the water system is not required to pay for the sampling and analysis and is not required to collect and analyze the sample.

Lead Service Line Replacement

Any water system that continues to exceed the lead and copper action levels even after optimizing corrosion control and installing source water treatment will be required to replace its lead distribution pipes within a maximum of 15 years. A lead pipeline qualifies for replacement if it contributes more than 15 parts per billion to the total tap-water lead levels. Seven percent of the lines meeting this standard must be replaced annually. Customers must be notified 45 days before replacing the lines.

Mercury

In the Code of Federal Regulations Reference: 40 CFR 141.62, mercury is found naturally throughout the environment. Large increases in mercury levels in water can be caused by industrial and agricultural use. The health risk from mercury is greater from mercury in fish than simply from waterborne mercury.

Mercury poisoning may have an acute health effect in large doses, or a chronic health effect in lower doses taken over an extended time period.

Purpose of Regulating Mercury: To improve public health protection and reduce ingestion of mercury.

Public Health Benefit: To prevent kidney damage.

MCL/MCLG: 0.002 mg/L/0.002 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Nitrate

In the Code of Federal Regulations Reference: 40 CFR 141.62, nitrate (as N) in drinking water above the national standard of 10 mg/L poses an immediate threat to children under three months of age. In some infants, excessive levels of nitrate have been known to react with intestinal bacteria that change nitrate to nitrite. It is often difficult to pinpoint sources of nitrate because so many possibilities exist.

Sources of nitrogen may include runoff or seepage from fertilized agricultural lands, municipal and industrial wastewater, refuse dumps, animal feedlots, septic tanks and private sewage disposal systems, urban drainage, and decaying plant debris.

Purpose of Regulating Nitrate: To improve public health protection.

Public Health Benefit: To prevent methemoglobinemia (blue baby syndrome)/diuresis.

MCL/MCLG: 10 mg/L/10 mg/L

Monitoring Requirements: Groundwater annually; surface water quarterly initially, then annually.

Nitrite

Code of Federal Regulations Reference: 40 CFR 141.62, nitrite reacts with hemoglobin in the blood. This reaction will reduce the oxygen-carrying ability of the blood and produce an anemic condition commonly known as blue baby syndrome. Sources of nitrite may include runoff or seepage from fertilized agricultural lands, municipal and industrial wastewater, refuse dumps, animal feedlots, septic tanks and private sewage disposal systems, urban drainage, and decaying plant debris.

Purpose of Regulating Nitrite: To improve public health protection.

Public Health Benefit: To prevent methemoglobinemia (blue baby syndrome)/diuresis.

MCL/MCLG: 1 mg/L/1 mg/L

Monitoring Requirements: One sample during first 3-year compliance period. Repeat frequency determined by state.

Selenium

In the Code of Federal Regulations Reference: 40 CFR 141.62, this mineral occurs naturally in soil and plants, especially in western states. Selenium is found in meat and other foods. Although it is believed to be essential in the diet, indications exist that excessive amounts of selenium may be toxic.

Studies are underway to determine the amount required for good nutrition and the amount that may be harmful. If a person's intake of selenium came only from drinking water, it would take an amount many times greater than the standard to produce any ill effects.

Purpose of Regulating Selenium: To improve public health protection.

Public Health Benefit: To prevent hair or fingernail loss, numbness of fingers or toes, and circulatory problems.

MCL/MCLG: 0.05 mg/L/0.05 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Thallium

In the Code of Federal Regulations Reference: 40 CFR 141.62, thallium is a metal found in natural deposits such as ores containing other elements. An average of 23 ppb of thallium in surface water and 11 ppb in groundwater have been found at hazardous waste sites. Since thallium compounds mix easily in water, you can be exposed if you live near a chemical waste site where thallium emissions have contaminated the water.

The greatest use of thallium is in specialized electronic research equipment. It is also used for production of optical lenses, jewelry, semiconductors, dyes, and pigments. The most common source of thallium poisoning is ingestion of rat poisons and insecticides.

Purpose of Regulating Thallium: To improve public health protection.

Public Health Benefit: To protect nervous system, lung/heart/kidney/liver/intestinal problems.

MCL/MCLG: 0.002 mg/L/0.0005 mg/L

Monitoring Requirements: Once a year for surface water; once every 3 years for groundwater. To minimize monitoring costs, use historical data, waivers, susceptibility waivers, or make composite samples.

Organic Standards

Organic chemicals are either natural or synthetic chemical compounds that contain carbon. Synthetic organic chemicals (SOCs) are manmade compounds and are used throughout the world as pesticides, paints, dyes, solvents, plastics, and food additives. **Volatile organic chemicals (VOCs)** are a subcategory of organic chemicals. These chemicals are termed volatile because they evaporate easily.

The most commonly found VOCs are THMs, trichloroethylene (TCE), tetrachloroethylene, and 1,1-dichloroethylene. THMs were the first regulated VOCs when EPA finalized regulations in 1979. The most common sources of organic contamination of drinking water are pesticides and herbicides, industrial solvents, and DBPs (THMs and haloacetic acids).

Millions of pounds of pesticides are used on croplands, forests, lawns, and gardens in the United States each year. They drain off into surface waters or seep into underground water supplies. Spills, poor storage, improper application, and haphazard disposal of organic chemicals have resulted in widespread groundwater contamination. A critical problem occurs once groundwater is contaminated, it may remain that way for a long time. Many organic chemicals pose health problems if they get into drinking water and the water is not properly treated.

In 2011, EPA decided to address up to 16 volatile organic compounds (VOCs) as a group that may cause cancer. The agency determined that they represent a near-term opportunity and it is likely that the public health goal for all VOCs would likely be set at zero because they may cause cancer. A preliminary evaluation of occurrence indicates that some of these VOCs may co-occur. This group will include trichloroethylene (TCE) and tetrachloroethylene (PCE).

EPA determined in March 2010 that the drinking water standards for TCE and PCE need to be revised. Addressing these VOCs as a group will help reduce exposure to these contaminants.

Trichloroethylene (TCE)

In the Code of Federal Regulations Reference: 40 CFR 141.50, although the use of TCE is declining because of stringent regulations, it was, for many years, a common ingredient in household products (spot removers, rug cleaners, and air fresheners),

dry cleaning agents, industrial metal cleaners and polishes, refrigerants, and even anesthetics. Its wide range of use is perhaps why TCE is the organic contaminant most frequently encountered in groundwater.

Purpose of Regulating Trichloroethylene: To improve public health protection.

Public Health Benefit: To prevent cancer risks; liver problems.

MCL/MCLG: 0.005 mg/L/zero

Monitoring Requirements: Four consecutive quarterly samples during first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems. These requirements apply to community and non-community water systems.

1,1-Dichloroethylene

In the Code of Federal Regulations Reference: 40 CFR 141.50, this solvent is used in manufacturing plastics and, more recently, in the production of 1,1,1-trichloroethane.

Purpose of Regulating 1,1-Dichloroethylene: To improve public health protection.

Public Health Benefit: To prevent impacts to the liver, kidneys, heart, and central nervous system

MCL/MCLG: 0.007 mg/L/0.007 mg/L

Monitoring Requirements/Comments: For VOCs: Four consecutive quarterly samples during first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems.

With the completion of source water assessments, primacy agencies are allowed to develop alternative monitoring requirements. Applies to community water systems and non-transient

Non-community water systems.

Vinyl Chloride

In the Code of Federal Regulations Reference: 40 CFR 141.50, billions of pounds of this solvent are used annually in the United States to produce polyvinyl chloride (PVC), the most widely used ingredient for manufacturing plastics throughout the world. Also, evidence exists that vinyl chloride may be a biodegradation end-product of TCE and PCE under certain environmental conditions.

Purpose of Regulating Vinyl Chloride: To improve public health protection.

Public Health Benefit: To prevent cancer in humans.

MCL/MCLG: 0.002 mg/L/zero

Monitoring Requirements/Comments: For VOCs: Four consecutive quarterly samples during first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems.

With the completion of source water assessments, primacy agencies are allowed to develop alternative monitoring requirements. Applies to community water systems and non-transient non-community water systems.

Carbon Tetrachloride

In the Code of Federal Regulations Reference: 40 CFR 141.50, carbon tetrachloride was once a popular household solvent, a frequently used dry cleaning agent, and a charging agent for fire extinguishers. Since 1970, however, carbon tetrachloride has been banned from all use in consumer goods in the United States.

In 1978, carbon tetrachloride was banned as an aerosol propellant. Currently, its principal use is in the manufacture of fluorocarbons, which are used as refrigerants.

Purpose of Regulating Carbon Tetrachloride: To improve public health protection.

Public Health Benefit: To prevent liver problems and cancer.

MCL/MCLG: 0.005 mg/L/zero

Monitoring Requirements: Four consecutive quarterly samples during first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems. These requirements apply to community and non-community water systems.

Benzene

In the Code of Federal Regulations Reference: 40 CFR 141.50, benzene is used primarily in the synthesis of styrene (for plastics), phenol (for resins), and cyclohexane (for nylon). Other uses include the production of detergents, drugs, dyes, and insecticides. Benzene is still being used as a solvent and as a component of gasoline. It is also discharged from factories and leaches from gas storage tanks and landfills.

Purpose of Regulating Benzene: To improve public health protection.

Public Health Benefit: To prevent anemia, a decrease in blood platelets, and cancer.

MCL/MCLG: 0.005 mg/L/zero

Monitoring Requirements: Four consecutive quarterly samples during first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems. These requirements apply to community and non-community water systems.

Para-Dichlorobenzene (p-Dichlorobenzene)

In the Code of Federal Regulations Reference: 40 CFR 141.50, the principal uses of p-dichlorobenzene are in moth control (moth balls and powders) and as lavatory deodorants. It is also used as an insecticide and fungicide on crops; in the manufacture of other organic chemicals; and in plastics, dyes, and pharmaceuticals.

The major source of p-dichlorobenzene in drinking water is from surface water sources. Low levels at best since it does not dissolve easily in water and quickly evaporate into the air.

Purpose of Regulating Para-Dichlorobenzene (p-Dichlorobenzene): To improve public health protection.

Public Health Benefit: To prevent kidney/liver/spleen/circulatory system problems.

MCL/MCLG: 0.075 mg/L/0.075 mg/L

Monitoring Requirements: Four consecutive quarterly samples during the first compliance period. Compliance is based on annual average of quarterly samples. If no detections are found during the initial round, two quarterly samples are required each year for systems serving greater than 3,300; one sample is required every 3 years for smaller systems. These requirements apply to community and non-community water systems.

Disinfection By-Products

Disinfection of drinking water by the addition of chlorine has long been considered a highly effective yet relatively low-cost method of preventing widespread outbreaks of waterborne diseases. In addition to reacting with disease-causing organisms in water, however, chlorine also reacts with many other types of organic materials.

The EPA has published information on scientific research which indicates that certain byproducts of water disinfection are linked to increases cancer incidence. In some cases, the research results are contradictory, showing no adverse health effects. EPA regulators weigh the public health benefits of disinfection against the risks of potentially harmful disinfection byproducts and continue to conduct research. They have adopted enforceable regulations to limit occurrence of disinfection byproducts in drinking water for a group of four total trihalomethanes (TTHMs) and the individual byproducts chlorite and bromate.

THMs are an example of a compound formed by the reaction of chlorine with organic matter in water. THMs are suspected of being carcinogenic and have been regulated by EPA in the 1996 SDWA amendments. The MCL for TTHMs is 0.080 milligram per liter or 80 micrograms per liter.

In May 1996, EPA published the Information Collection Rule (ICR). This rule required large public water systems to undertake extensive monitoring of microbial contaminants and DBPs in their water systems. Also, some water systems conducted studies on the use of granular activated carbon and membrane processes. The data reported under the ICR were used by EPA to study the risks and tradeoffs involved in disinfecting drinking water and adopt additional regulations if necessary. The ICR data form the scientific basis for EPA's development of the Enhanced Surface Water Treatment Rule and the Disinfectants and Disinfection Byproducts Rule.

EPA issued the Stage 1 DBPR on December 16, 1998 (Federal Register 63, No. 241). This rule set new MCLGs and MCLs for TTHMs, HAA5, bromate, and chlorite. Maximum residual disinfectant level goals (MRDLGs) and maximum residual disinfectant levels (MRDLs) have also been set for chlorine, chloramine, and chlorine dioxide.

The Stage 1 DBPR attempts to further reduce potential formation of harmful DBPs by requiring the removal of THM precursors. A treatment technique of enhanced coagulation, enhanced softening, or use of granular activated carbon (GAC) applies to conventional filtration systems. In most cases, systems must reduce total organic carbon (TOC) levels based on specific source water quality factors.

For large systems (serving more than 10,000 persons) that use surface water or groundwater under the direct influence of surface water, the compliance date for the Stage 1 DBPR was January 1, 2002. Small systems (serving fewer than 10,000) that use surface water or groundwater under the direct influence of surface water and all groundwater systems must have complied by January 1, 2004.

The Stage 1 DBPR has very specific laboratory and monitoring requirements. The routine monitoring requirements include the following regulated contaminants/disinfectants:

1. TTHM/HAA5
2. Bromate
3. Chlorite
4. Chlorine/chloramines
5. Chlorine dioxide
6. DBP precursors (TOC/alkalinity/specific UV absorbance)

Also, the Stage 1 DBPR specifies the monitoring coverage in terms of surface water, groundwater, and groundwater under direct influence (GWUDI), population served, and the type of filtration system and disinfection system. Monitoring frequency depends on the type of source water, population served, and type of treatment and disinfection system. The routine monitoring requirements are based on the regulated contaminants/disinfectants and include the MCL, MRDL, analytical method, preservation/quenching agent, holding time for sample/extract, and sample container size and type.

On December 15, 2005, EPA promulgated the Stage 2 DBPR. This rule reduces potential cancer and reproductive and developmental health risks from DBPs in drinking water, which form when disinfectants are used to control microbial pathogens. This final rule strengthens public health protection for consumers by tightening compliance monitoring requirements for two groups of DBPs: TTHMs and HAA5.

The rule targets systems with the greatest risk and builds incrementally on existing rules. This regulation reduces DBP exposure and related potential health risks and provides more equitable public health protection. The Stage 2 DBPR was promulgated simultaneously with the Long Term 2 Enhanced Surface Water Treatment Rule to address concerns about risk tradeoffs between pathogens and DBPs.

Under the Stage 2 DBPR, systems will conduct an evaluation of their distribution systems, known as an Initial Distribution System Evaluation (IDSE), to identify the locations with high DBP concentrations. These locations are used by the systems as the sampling sites for Stage 2 DBPR compliance monitoring.

Compliance with the maximum contaminant levels for two groups of DBPs (TTHMs and HAA5) is calculated for each monitoring location in the distribution system. This approach, referred to as the locational running annual average (LRAA), differs from previous requirements, which determine compliance by calculating the running annual average of samples from all monitoring locations across the system.

The Stage 2 DBPR also requires each system to determine if they have exceeded an operational evaluation level, which is identified using their compliance monitoring results. The operational evaluation level provides an early warning of possible future MCL violations, which allows the system to take proactive steps to remain in compliance. A system that exceeds an operational evaluation level is required to review their operational practices and submit a report to their state that identifies actions that may be taken to mitigate future high DBP levels; particularly, those levels that may jeopardize their compliance with the DBP MCLs.

Radionuclide Standards

EPA promulgated the Radionuclides Rule on December 7, 2000. The purpose of the Radionuclides Rule is to reduce the exposure to radionuclides in drinking water, which reduces the risk of cancer. This rule also improves public health protection by reducing exposure to all radionuclides. The rule applies to all community water systems.

The public health benefits from the rule include reduced exposure to uranium, which can cause toxic kidney effects and cancer.

In the Code of Federal Regulations Reference: 40 CFR 141.24, radioactivity is the only contaminant that has been shown to cause cancer for which standards have been set. Three radioactive elements, radon, radium, and uranium, occur naturally in the ground and dissolve into groundwater supplies. However, the possible exposure to radiation in drinking water is only a fraction of the exposure from all natural sources. The main source of radioactive material in surface water is fallout from nuclear testing. Other sources could be nuclear power plants, nuclear fuel processing plants, and uranium mines. Those sources are monitored constantly and no great risk of contamination exists, barring accidents.

Alpha and radium radioactivity occur naturally in groundwater in parts of the West, Midwest, and Northeast. Standards for those types of radioactivity and for manmade, or beta radiation, have been set at levels of safety comparable to other contaminants. The MCLs for radiological contaminants are divided into two categories: natural radioactivity that results from well water passing through deposits of naturally occurring radioactive materials and manmade radioactivity such as that which might result from industrial wastes, hospitals, or research laboratories.

A pCi/L, picoCurie per Liter, is a measure of radioactivity. One picoCurie of radioactivity is equivalent to 0.037 nuclear disintegrations per second. For surface water systems serving more than 100,000 persons and systems determined by the state to be vulnerable. Monitoring for natural radioactivity contamination is required every four years for surface water and groundwater community systems. Routine monitoring procedures to follow are:

1. Test for gross alpha activity; if gross alpha exceeds 5 pCi/L, then
2. Test for radium 226; if radium 226 exceeds 3 pCi/L, then
3. Test for radium 228.

The MCL for gross beta particle activity is 4 mrem/yr for water systems using surface water and serving more than 100,000 people, and for any other community system determined by the state to be vulnerable to this type of contamination. After the initial monitoring period, beta radiation must be monitored every 4 years unless the MCL is exceeded or the state determines that more frequent monitoring is appropriate.

Regulation of radon continues to be a source of considerable debate. EPA proposed a maximum contaminant level in 1991 and then withdrew the proposed MCL in 1997 in response to the 1996 SDWA amendments. EPA published a proposed regulation for radon in 1999, but the final MCL has not been published.

The final Radionuclides Rule was published in 2000. The rule retains the current standards for radium 226 and 228, gross alpha activity, and gross beta particle activity, but establishes a new MCL 30 µg/L for uranium. The EPA has indicated that it will continue to review the standard for beta particle activity and may revise the MCL in the future. EPA must base its new rules on a scientific risk assessment and an analysis of the amount of health risk reduction that will be achieved and the cost of implementing the new rules.

Microbial Standards

Microbial contamination of drinking water can pose a potential public health risk in terms of acute outbreaks of disease. Bacteria, viruses, and other organisms have long been recognized as serious contaminants of drinking water. Organisms such as *Giardia*, *Cryptosporidium*, and *E. coli* (a type of coliform bacteria) cause almost immediate gastrointestinal illness when people consume

them in water. Waterborne diseases such as typhoid, cholera, infectious hepatitis, and dysentery have been traced to improperly disinfected drinking water.

In most communities, drinking water is treated to remove contaminants before being piped to consumers, and bacterial contamination of municipal water supplies has been largely eliminated by adding chlorine or applying other forms of disinfection to drinking water to prevent waterborne diseases. By treating drinking water and wastewater, diseases such as typhoid and cholera have been virtually eliminated.

Many types of coliform bacteria from human and animal wastes may be found in drinking water if the water is not properly treated. Often, the bacteria do not cause diseases transmitted by water, although certain coliforms have been identified as the cause of traveler's diarrhea. In general, however, the presence of coliform bacteria indicates that other harmful organisms may be present in the water.

Throughout the early 1990s, chemical contaminants in drinking water were the focus of EPA's regulatory efforts. With passage of the 1996 SDWA amendments, Congress directed EPA to focus on microbial contaminants and to promulgate a series of new regulations to improve treatment effectiveness. Significant new regulations of the microbial contaminant *Cryptosporidium* were promulgated in December 1998. The new regulations consist of mandatory treatment techniques similar to the current regulations for *Giardia* and coliform bacteria. Filtration and disinfection requirements for surface water systems are currently in place and the final Ground Water Rule was signed in 2006. These regulations significantly affect the water treatment processes of most water suppliers.

Total Coliform Rule

In the Code of Federal Regulations Reference: 40 CFR 141 Subpart C, coliforms are bacteria that are always present in the digestive tracts of animals, including humans, and are found in their wastes. They are also found in plant and soil material. The most basic test for bacterial contamination of a water supply is the test for total coliform bacteria. Total coliform counts give a general indication of the sanitary condition of a water supply.

Coliforms include the following:

1. Total coliforms—include bacteria that are found in the soil, in water that has been influenced by surface water, and in human or animal waste.
2. Fecal coliforms—are the group of the total coliforms that are considered to be present specifically in the gut and feces of warm-blooded animals. Because the origins of fecal coliforms are more specific than the origins of the more general total coliform group of bacteria, fecal coliforms are considered a more accurate indication of animal or human waste than the total coliforms.

Escherichia coli (*E. coli*) is the major species in the fecal coliform group. Of the five general groups of bacteria that comprise the total coliforms, only *E. coli* is generally not found growing and reproducing in the environment. Consequently, *E. coli* is considered to be the species of coliform bacteria that is the best indicator of fecal pollution and the possible presence of pathogens.

2012 Revised Total Coliform Rule (RTCR)

EPA has revised the Total Coliform Rule so that public water systems in the US take steps to ensure the integrity of the drinking water distribution system and monitoring for the presence of microbial contamination. Public water systems and the state and local agencies that oversee them must comply with the RTCR beginning April 1, 2016. The final RTCR requires the following:

1. Public water systems are to notify the public if a test exceeds the MCL for *E. coli* in drinking water
2. Public water systems that are at risk and therefore vulnerable to microbial contamination are required to identify and fix the problems, including the potential sources and the pathways of contamination
3. Criteria will be used for public water systems that are well operated to qualify for and stay on reduced monitoring, which could reduce water system burden
4. Provide incentives for better system operation, in particular small systems
5. Require additional monitoring requirements for seasonal systems such as campgrounds and state and national parks

The RTCR establishes an MCLG and an MCL for *E. coli* and eliminates the MCLG and MCL for total coliforms, replacing them with treatment techniques for coliforms that require assessment and corrective action. EPA has established an MCLG of zero for *E. coli*, which is a more specific indicator of fecal contamination and those pathogens that are potentially more harmful than total

coliform. EPA has removed the 1989 MCLG and MCL for total coliforms, although the acute total coliform MCL violation under the 1989 TCR has been maintained as the MCL for *E. coli* under the RTCR.

Under the new treatment technique, total coliforms serve as the indicator of a potential pathway of contamination into the system's distribution system. Public water supplies that exceed a specified frequency of total coliform occurrence must conduct an assessment to determine if any sanitary deficiencies exist; if found, they must be corrected. A public water system that experiences an *E. coli* MCL violation must conduct an assessment and correct any sanitary defects that are identified.

Monthly notification requirements based only on the presence of total coliforms will no longer be required. Instead, the RTCR will require public notification when an *E. coli* MCL violation occurs, indicating a potential health threat or when a water system fails to assess and take corrective action.

Sanitary Survey

In the Code of Federal Regulations Reference: 40 CFR 141.2, sanitary surveys are conducted to identify possible health risks that may not be discovered by routine coliform sampling. The

Total Coliform Rule requires community water systems serving fewer than 4,100 persons to complete an initial sanitary survey by June 1994 and to conduct subsequent surveys every 5 years.

Non-community water systems are required to complete an initial sanitary survey by June 1999 and subsequent surveys every 5 years with the exception of those systems with protected and disinfected groundwater sources, which are allowed 10 years for subsequent surveys.

A sanitary survey requires detailed planning, a thorough system survey, and reporting of the results. The planning portion involves a review of water quality records for compliance with applicable microbial, inorganic chemical, organic chemical, and radiological contaminant MCLs as well as the records of compliance with the monitoring requirements for those contaminants.

The actual field survey is a detailed evaluation and inspection of the source of the water supply and all conveyances, storage, treatment, and distribution facilities to ensure protection from all pollution sources.

The final report includes the date(s) of the survey, who was present during the survey, the survey findings, the recommended improvements to correct identified problems, and the target dates of completion for any improvements.

Sampling Plan

The Total Coliform Rule requires each water supply system to develop and follow a written sampling plan. Each plan must specifically identify sampling points throughout the distribution system. Sampling plans must be approved by the state regulatory agency, and it is necessary to check with the state agency to determine the details of their review process and what documents need to be submitted.

Laboratory Procedures

In the Code of Federal Regulations Reference: 40 CFR 141.21 and 40 CFR 141 Subpart C, with regard to coliform testing methodology, EPA maintains a current list of analytical methods that are approved for monitoring drinking water supplies that is published and updated in 40 CFR 141. Examples of the acceptable analytical methods for determining total coliforms include the following:

1. Multiple-tube fermentation technique (MTF)
2. Membrane filter technique (MF)
3. Presence-absence coliform test (P-A)
4. Colilert™ system (ONPG-MUG test)
5. Colisure test

Regardless of the method used, the standard sample volume for total coliform testing is now 100 ml. This volume is an increase over the past testing method using 50 mL for the MTF technique.

The multiple-tube fermentation method of testing for coliforms determines the presence or absence of coliforms by the multiple-tube dilution method. This test is a process whereby equal portions of a sample (100 mL) are added to 10 tubes containing a culture

medium and an inverted vial. If gas accumulates in the inverted vial, it indicates presumptive evidence of coliform organisms and the sample is considered coliform-positive. If no gas forms in any of the vials, the sample is coliform-negative.

The membrane filter method provides for filtering a 100 mL water sample through a thin, porous, cellulose membrane filter under a partial vacuum. The filter is placed in a sterile container and incubated in contact with a special liquid called a culture medium, which the bacteria use as a food source. Colonies of bacteria then grow on the media. The coliform colonies are visually identified, counted, and recorded as the number of coliform colonies per 100 mL of sample.

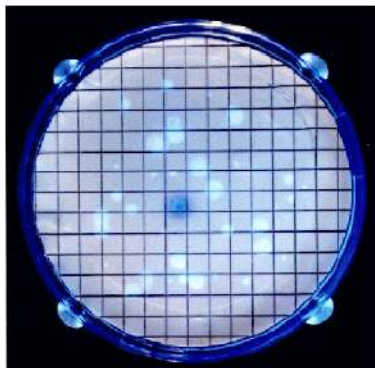


Figure 3.1: Membrane filter test for coliforms.

The presence-absence (P-A) test for the coliform group is a simple modification of the multiple-tube procedure. Even though this simplified test uses only one large test portion (100 mL) in a single culture bottle, the results still indicate the presence or absence of coliform bacteria and thus meet the testing requirements.

The Colilert™ system is a privately developed version of the ONPG-MUG test procedure that meets the EPA's laboratory methods requirements. The ONPG-MUG test indicates the presence or absence of coliform bacteria within 24 hours. Although the Colilert™ method can yield presence-absence results within 24 hours and is easier to perform than the Membrane Filtration (MF) method, operators should be aware of the limitations of these tests in evaluating samples for regulatory purposes. The results for the Colilert™ and MF tests are not always comparable, which may be due to the following:

1. Interferences in the sample that may suppress or mask bacterial growth
2. Greater sensitivity of the Colilert™ media
3. Added stress to organisms related to filtering
4. The fact that different media may obtain better growth for some bacteria

These test methods are approved by the EPA for reporting under the Safe Drinking Water Program. When these tests produce conflicting results, however, the safest course of action is to increase monitoring and treatment efforts until the results for each test is negative.

The Colisure test is a presence-absence test for coliform bacteria. A sample of water is added to a dehydrated medium and, after 24 to 48 hours, the medium is examined for the presence of coliforms.

Table 3.1: Total Coliform Indicator Color Chart

Positive Test	Negative Test
- Many gas bubbles present - Gel rises to surface - Liquid below gel is cloudy - Indicator turns yellow	- Liquid above gel is clear - Indicator remains red or turns yellow with no gas bubbles - Gel remains at bottom of tube

Water supply systems may contain various other types of bacteria besides coliforms. These organisms are nonpathogenic bacteria and are sometimes referred to as heterotrophic bacteria. During the analyses of water samples for coliforms, heterotrophic bacteria sometimes interfere with the test causing one of the following reactions:

1. MTF technique: a turbid culture with no gas production
2. P-A test: a turbid culture in the absence of an acid reaction
3. MF technique: confluent growth or a coliform colony number too numerous to count

The sample being tested can be considered invalid if any of these situations occurs and total coliforms are not detected. A replacement sample must be taken from the same location within 24 hours of receiving the laboratory results. If the replacement sample also shows heterotrophic interference but tests positive for total coliforms, the sample is considered valid.

Monitoring Frequency

The routine monitoring frequency for community water systems is based on the population served. Routine monitoring frequencies for non-community systems are based on the source of supply and, in some cases, the population served. Whenever a routine sample tests positive for total coliform, two provisions of the Total Coliform Rule take effect: repeat samples (formerly referred to as check samples) must be taken and the original coliform-positive sample must be tested for the presence of fecal coliforms or *E. coli* to determine whether an actual or potential violation of the coliform MCL exists.

The reason repeat samples are required is to investigate whether the original coliform-positive sample was caused by a contamination problem that exists throughout the distribution system or if it is a localized problem that exists only at that one sampling point. With this information, appropriate corrective action can be taken to eliminate the problem as quickly as possible.

The Total Coliform Rule specifies how many repeat samples must be taken, when, and from what locations. It also directs the water utility to collect a specific number of samples the following month, based on the number of routine samples ordinarily collected.

Non-community systems serving more than 1,000 people have the same requirements as community water systems. Non-community systems serving fewer than 1,000 people are required to collect one routine sample per quarter. When the routine sample tests positive, four repeat samples are required in the same quarter and five samples are required the following quarter.

Whenever a routine or repeat sample tests positive for total coliform, the water agency must collect a set of three or four repeat samples within 24 hours of receiving the laboratory results. At least one of the repeat samples must be taken from the same tap as the original coliform-positive sample; the remaining repeat samples in the set must be collected at nearby taps (within five service connections of the original sampling point), upstream and downstream of the original.

Repeat samples must be taken until no coliforms are detected or until the MCL is exceeded and the state is notified.

Determining Compliance

The MCLG for total coliforms (including fecal coliforms and *E. coli*) is zero. The MCL, based on the presence-absence concept, is as follows:

1. For water systems analyzing at least 40 samples per month, no more than 5.0 percent of the samples (including routine and repeat samples) may be positive for total coliforms.
2. For water systems analyzing fewer than 40 samples per month, no more than one sample per month may be positive for total coliforms.

The Revised Total Coliform Rule makes another significant change in the way compliance is calculated. All valid coliform-positive samples, routine and repeat samples, must be counted when calculating compliance with the monthly MCL. Under previous regulations, check or repeat samples were not included in the monthly MCL calculation. On a case-by-case basis, the state may declare a sample invalid for one of several reasons, including interference by heterotrophic bacteria during laboratory analysis, as previously discussed.

Total-coliform-positive samples may be invalidated by the state under any of the following conditions:

1. The analytical laboratory acknowledges that improper sample analysis caused the positive result
2. The state determines that the contamination is a local plumbing problem
3. The state has substantial grounds to believe that the positive result was unrelated to the quality of drinking water in the distribution system

Reporting and Notification Requirements

Reporting frequencies for coliform test results increase in step with the urgency of the problem. A water agency must report the results of monthly coliform testing to the state regulatory agency within the first 10 days of the following month.

Any time a water agency fails to collect a sample as required, the state must be notified within 10 days after the system learns of the violation. An invalid sample result is considered a failure to monitor and must be reported.

If the MCL is exceeded, the state must be notified no later than the end of the next business day and the public within 14 days. This requirement could occur when the water agency exceeds its monthly coliform-positive limit or when test results show the presence of fecal coliforms or *E. coli* in any sample.

The most critical situation exists when either of two situations occurs:

1. A routine sample tests positive for total coliforms and for fecal coliforms or *E. coli*, and any repeat sample tests positive for total coliforms.
2. A routine sample tests positive for total coliforms and negative for fecal coliforms or *E. coli*, and any repeat sample tests positive for fecal coliforms or *E. coli*.

These situations are considered an acute risk to health. This occurrence is a Tier 1 violation, which requires that the state and the public be notified within 24 hours.

Key Terms

- **Inorganic contaminants** – non-Carbon-based contaminants, such as lead and mercury;
- **Microbial contaminants** - bacteria, viruses, and protozoa
- **Organic contaminants** – Carbon-based contaminants, which include pesticides, herbicides, THMs, solvents, and other synthetic organic compounds
- **Primary standards – federal** and state standards of contaminants with health effects; the Maximum Contaminant Level (MCL) for a contaminant
- **Secondary standards** – federal and state standards for contaminants that affect the aesthetic qualities relating to the public acceptance of drinking water
- **Radiologic contaminants** - natural and manmade sources of radiation, including uranium
- **Volatile organic compounds** – a subcategory of organic chemicals. These chemicals are termed volatile because they evaporate easily.
- **Turbidity**, such as small particles suspended in water that interfere with light penetration and disinfection

Review Questions

1. List the metals in the Safe Drinking Water Act that are considered to be primary contaminants.

Contaminant	MCLG (mg/L)	MCL or TT (mg/L)	Potential Health Effects from Ingestion of Water	Sources of Contaminant in Drinking Water

2. List the radiologic contaminants that drinking water plants are required to be tested for in the Safe Drinking Water Act.
3. What is the action level for lead and copper in the Safe Drinking Water Act?

4. **What is the MCL for nitrate in drinking water?**
5. **Which contaminant has a primary and secondary contaminant limit?**
6. **List the disinfection by-products that are monitored and controlled in the Safe Drinking Water Act?**
7. **Describe a Sanitary Survey.**

Chapter Quiz

1. In the Code of Federal Regulations, 40CFR 141.2, a sanitary survey should be conducted every _____.
 1. one per year
 2. every two years
 3. every five years
 4. every six years
2. EPA uses the _____ to collect data for contaminants suspected to be present in drinking water, although they do not have health-based standards set under the Safe Drinking Water Act.
 1. Maximum Contaminant Level
 2. Maximum Contaminant Level Goal
 3. Unregulated Contaminant Monitoring
 4. Secondary Maximum Contaminant Level
3. Which of the primary contaminants has an action level rather than an MCL?
 1. THMs and HAA5s
 2. Arsenic
 3. Mercury
 4. Lead and Copper
4. In the Code of Federal Regulations Reference: 40 CFR 141.62, the amounts of cadmium that are found in natural waters in the United States is small. Waste discharges from the electroplating, photography, insecticide, and metallurgy industries can increase cadmium levels. The most common source of cadmium in drinking water is from galvanized pipes and fixtures. What is the MCL and MCLG for Cadmium?
 1. 5 ppb
 2. 50 ppb
 3. 5 ppm
 4. 50 ppm
5. In the Code of Federal Regulations Reference: 40 CFR 141.62, chromium is found in cigarettes, some foods, and the air. It is also discharged from steel and pulp mills and the erosion of natural deposits. Some studies suggest that in very small amounts, chromium may be essential to human beings, but it has not been proven. Chromium VI or Hexavalent Chromium was the subject of the Erin Brockovich case and movie. What is the MCL and MCLG for Chromium?
 1. 1.0 ppb
 2. 10 ppb
 3. 1.0 ppm

4. 0.1 ppm
6. In order to prevent methemoglobinemia (blue baby syndrome)/diuresis, _____ is monitored annually in groundwater and quarterly in surface water initially and then annually. The MCL is 10 mg/L.
1. nitrite
 2. nitrate
 3. sodium
 4. chloride
7. Microbiological contamination often causes flu-like symptoms; however, it can also cause serious diseases. Which of the following is not a water borne disease?
1. Hepatitis
 2. AIDS
 3. Giardiasis
 4. Legionnaire's Disease
8. The Revised Total Coliform Rule establishes an MCLG and an MCL for *E. coli* and eliminates the MCLG and MCL for total coliforms, replacing them with treatment techniques for coliforms that require assessment and corrective action. EPA has established an MCLG of _____ for *E. coli*, which is a more specific indicator of fecal contamination than those pathogens that are potentially more harmful than total coliform.
1. zero
 2. one
 3. three
 4. depends on the number of samples that are taken each month by the water purveyor
9. In the Code of Federal Regulations Reference: 40 CFR 141.24, _____ is the only contaminant that has been shown to cause cancer for which standards have been set.
1. Arsenic
 2. Chromium
 3. Radioactivity
 4. THMs
10. Which of the following contaminants can potentially cause mental deficiencies primarily in children, if found in drinking water above the maximum level allowed by Safe Drinking Water Act?
1. Copper
 2. Arsenic
 3. Mercury
 4. Lead

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4: Secondary Standards

Learning Objectives

After reading this section, you should be able to:

- Describe the secondary standards
- Outline the secondary contaminants
- Explain physical standards
- Outline levels of violations

Secondary Standards

Enforcement of Regulations

The National Secondary Drinking Water Regulations control contaminants in drinking water that primarily affect the aesthetic qualities relating to the public acceptance of drinking water. These regulations are called **secondary standards**. At considerably higher concentrations of these contaminants, health implications may also exist as well as aesthetic (or appearance) degradation. These regulations are not federally enforceable; however, some states have passed laws requiring the state health agency to enforce the regulations.

Secondary Maximum Contaminant Levels

Secondary Maximum Contaminant Levels (SMCLs) apply to public water systems and, in the judgment of the EPA Administrator, are necessary to protect the public welfare or for public acceptance of the drinking water. The SMCL means the maximum permissible level of a contaminant that is delivered to the free-flowing outlet of the ultimate user of a public water system.

Contaminants added to the water under circumstances controlled by the user, except those contaminants resulting from corrosion of piping and plumbing caused by water quality, are excluded by definition. The only exceptions are those resulting from corrosion of piping and plumbing caused by water quality. Currently, 15 secondary standards exist. States may establish higher or lower levels depending on local conditions, providing that public health and welfare are adequately protected.

Aesthetic qualities are important factors in public acceptance and confidence in a public water system. States are encouraged to implement SMCLs so that the public will not be driven to obtain drinking water from potentially lower-quality, higher risk sources. Many states have chosen to enforce Primary and SMCLs to ensure that the consumer is provided with the best quality water available.

Monitoring

In order to monitor, you must collect samples for secondary contaminants at a free-flowing outlet of water being delivered to the consumer. Monitor contaminants in these regulations at least as often as the monitoring performed for inorganic chemical contaminants listed in the Primary Drinking Water Regulations as applicable to community water systems. For surface water systems, the rule means yearly monitoring is required; for groundwater systems, monitor at least once every three years. Collect monthly distribution system physical water quality monitoring samples for color and odors. More frequent monitoring would be appropriate for specific contaminants such as pH, color, odor, or others, under certain circumstances as directed by the state.

Secondary Contaminants

Aluminum

Until relatively recently, most environmental aluminum was found in forms that did not greatly affect humans and other animals. Acid rain, however, has caused a significant increase in aluminum exposure. The average daily intake of aluminum in the general population is about 20 milligrams per day; the SMCL is 0.05 to 0.2 mg/L. Although aluminum is regulated as a secondary contaminant because precipitates of the metal can cause cloudy-looking water, aluminum compounds may also interfere with absorption of fluoride in the gastrointestinal tract and may decrease the absorption of calcium, iron, and cholesterol. Many over-

the-counter antacids contain aluminum and use of these products frequently causes constipation, which is thought to be the result of the gastrointestinal effects.

Chloride

The SMCL for chloride is 250 mg/L.

Undesirable Effects

- Objectionable salty taste in water
- Corrosion of the pipes in hot water systems and other pipelines

Studies Concerning the Mineralization Effects of Water Indicate

- Major taste effects are produced by anions (where TDS was studied).
- Chloride produces a taste effect somewhere between the milder sulfate and the stronger carbonate.
- Laxative effects are caused by highly mineralized waters.

Mineralized waters often contain chloride as well as high levels of sodium and magnesium sulfate.

Corrosion Effects

- Studies indicate that corrosion depends on concentration of TDS (TDS may contain 50 percent chloride ions).
- Domestic plumbing, water heaters, and municipal waterworks equipment will deteriorate when high concentrations of chloride ions are present.

Example

Where the TDS = 200 mg/L (Chloride = 100 mg/L), water heater life will range from 10 to 13 years. Water heater life declines uniformly as a function of TDS-1 year shortened life per 200 mg/L additional TDS.

Color

The SMCL for color is 15 color units. The level of this water quality indicator is not known to be a measure of the safety of water. However, high color content may indicate the following:

- High organic chemical contamination
- Inadequate treatment
- High disinfectant demand and the potential for production of excess amounts of DBPs

Color may be caused by:

- Natural color-causing solids such as aromatic, polyhydroxy, methoxy, and carboxylic acids
- Fulvic and humic acid fractions
- Presence of metals such as copper, iron, and manganese

Rapid changes in color levels may provoke more citizen complaints than relatively high, constant color levels.

Copper

The SMCL for copper is 1.0 mg/L. Soft water containing a low level (0.5 mg/L) of copper may cause blue or blue-green staining of porcelain. Higher levels (4 mg/L) of copper will stain clothing and blond hair. When soap is used with water having a copper concentration greater than 1 mg/L, insoluble green curds will form.

Corrosivity

The corrosivity of water indicates the rate at which water causes the gradual decomposition or destruction of a material (such as a metal or cement lining). The severity and type of corrosivity are dependent on the chemical and physical characteristics of the water and the material.

Corrosivity causes materials to deteriorate and go into solution (be carried by the water). Corrosion of toxic metal pipe materials such as lead can create a serious health hazard. Corrosion of iron may produce a flood of unpleasant telephone calls from consumers complaining about rusty water, stained laundry, and bad tastes.

Corrosivity can cause the reduction of the carrying capacity of a water main. This reduced carrying capacity can cause an increase in pump energy costs and may reduce distribution system pressures. Leaks in water mains due to corrosivity may eventually require replacement of a water main.

Fluoride

Fluoride produces two effects, depending on its concentration. At levels of 6 to 8 mg/L, fluoride may cause skeletal fluorosis (the bones become brittle) and stiffening of the joints. For this reason, fluoride has been added to the list of primary standards (having health effects).

At levels of 2 mg/L and greater, fluoride may cause dental fluorosis, which is discoloration and mottling of the teeth, especially in children. EPA has recently reclassified dental fluorosis as a cosmetic effect, raised the primary drinking water standard from 1.4 to 2 mg/L to 4 mg/L, and established a secondary standard of 2 mg/L for fluoride.

Foaming Agents

The SMCL for foaming agents is 0.5 mg/L.

Undesirable Effects

1. Causes frothing and foaming, which are associated with contamination (greater than mg/L).
2. Imparts an unpleasant taste (oily, fishy, perfume-like) (less than 1.0 mg/L).

Information

1. Because no convenient foamability test exists and because surfactants are one major class of substances that cause foaming, this property is determined indirectly by measuring the anionic surfactant (MBAS) concentration in the water.
2. Surfactants are synthetic organic chemicals and are the principal ingredient of household detergents.
3. The requirement for biodegradability led to the widespread use of linear alkylbenzene sulfonate (LAS), an anionic surfactant.
4. Concentrations of anionic surfactants found in drinking waters range from 0 to 2.6 mg/L in well supplies and 0 to 5 mg/L in surface water supplies.
5. LAS is essentially odorless. The odor and taste characteristics are likely to arise from the degradation of waste products rather than the detergents.
6. If water contains an average concentration of 10 mg/L surfactants, the water is likely to be entirely of wastewater origin.
7. From a toxicological standpoint, an MCL of 0.5 mg/L, assuming a daily adult human water intake of 2 liters, would give a safety factor of 15,000.

Iron and Manganese

1. Iron and manganese are frequently found together in natural waters and produce similar adverse environmental effects and color problems. Excessive amounts of iron and manganese are usually found in groundwater and in surface water contaminated by industrial waste discharges.
2. Before 1962, these elements were covered by a single recommended limit.
3. In 1962, the US Public Health Service recommended separate limits for iron and manganese to reflect more accurately the levels at which adverse effects occur for each.
4. Each are highly objectionable in large amounts in water supplies for domestic and industrial use.
5. Each element imparts color to laundered goods and plumbing fixtures.
6. Taste thresholds in drinking water are considerably higher than the levels that produce staining effects.
7. Each element is part of the daily nutritional requirements; however, these requirements are not met by the consumption of drinking water.

Iron

The SMCL for iron is 0.3 mg/L.

Undesirable Effects

1. At levels greater than 0.05 mg/L some color may develop, staining of fixtures may occur, and precipitates may form.
2. The magnitude of the staining effect is directly proportional to the concentration.

3. Depending on the sensitivity of taste perception, a bitter, astringent taste can be detected from 0.1 mg/L to 1.0 mg/L.
4. Precipitates that are formed create not only color problems but also lead to bacterial growth of slimes and of the iron loving bacteria, *Crenothrix*, in wells and distribution piping.

Nutritional Requirements

1. Daily requirement is 1 to 2 mg; however, intake of larger quantities is required as a result of poor absorption.
2. The limited amount of iron permitted in water (because of objectionable taste or staining effects) constitutes only a small fraction of the amount normally consumed and does not have toxicologic (poisonous) significance.

Manganese

The SMCL for manganese is 0.05 mg/L.

Undesirable Effects

1. A concentration of more than 0.02 mg/L may cause buildup of coatings in distribution piping.
2. If these coatings slough off, they can cause brown blotches in laundry items and black precipitates.
3. Manganese imparts a taste to water above 0.15 mg/L.
3. The application of chlorine, even at low levels, increases the likelihood of precipitation of manganese at low levels.
4. Unless the precipitate is removed, precipitates reaching pipelines will promote bacterial growth.

Toxic Effects

1. Toxic effects are reported as a result of inhalation of manganese dust or fumes. Liver cirrhosis has arisen in controlled feeding of rats.
2. Neurological effects have been suggested; however, these effects have not been scientifically confirmed.

Nutritional Requirements

1. Daily intake of manganese from a normal diet is about 10 mg.
2. Manganese is essential for proper nutrition.
3. Diets deficient in manganese will interfere with growth, blood and bone formation, and reproduction.

Odor

The SMCL for odor is a Threshold Odor Number (TON) of 3.

Important facts to remember when dealing with odors include:

1. Taste and odor go hand-in-hand.
2. Absence of taste and odor helps to maintain the consumers' confidence in the quality of their water, even though it does not guarantee that the water is safe.
3. Research indicates that there are only four true taste sensations: sour, sweet, salty, and bitter.
4. All other sensations ascribed to the sense of taste are actually odors, even though the sensation is not noticed until the material is taken into the mouth.
5. Odor tests are less fatiguing to people than taste tests when testing for tastes and odors.
6. Taste and odor tests are useful:
 1. As a check on the quality of raw¹⁰ and treated water
 2. To help control odor throughout the plant
7. Odor is a useful test:
 1. For determining the effectiveness of different kinds of treatment
 2. As a means for tracing the source of contaminants
8. Hydrogen sulfide is included under the odor SMCL.

pH

The SMCL for pH is defined as pH values beyond the acceptable range from 6.5 to 8.5. A wide range of pH values in drinking water can be tolerated by consumers.

Undesirable Effects

1. When the pH increases, the disinfection activity of chlorine falls significantly.
2. High pH may cause increased halogen reactions, which produce chloroform and other THMs during chlorination.
3. Both excessively high and low pHs may cause increased corrosivity, which can, in turn, create taste problems, staining problems, and significant health hazards.
4. Metallic piping in contact with low-pH water will impart a metallic taste.
5. If the piping is iron or copper, high pH will cause oxide and carbonate compounds to be deposited, leaving red or green stains on fixtures and laundry.
6. At a high pH, drinking water acquires a bitter taste.
7. The high degree of mineralization often associated with basic waters results in encrustation of water pipes and water-using appliances.

Silver

Silver is a nonessential element, providing no beneficial effects from its ingestion in trace amounts. Chronic toxicity causes an unsightly blue-gray discoloration of the skin, mucous membranes, and eyes. Apparently, beside cosmetic changes, no physiologic effects are known. Ingestion of trace amounts of silver or silver salts results in its accumulation in the body, particularly the skin and eyes. Some evidence exists that changes to the kidneys, liver, and spleen can occur.

The SMCL for silver is 0.10 mg/L.

Sulfate

The SMCL for sulfate is 250 mg/L.

Undesirable Effects

1. Tends to form hard scales in boilers and heat exchangers.
2. Causes taste effects.
3. Causes a laxative effect. This effect is commonly noted by newcomers or casual or intermittent users of water high in sulfate. Water containing more than 750 mg/L of sulfate usually produces the laxative effect while water with less than 600 mg/L sulfate usually does not. An individual can adjust to sulfate in drinking water.
4. Sodium sulfate and magnesium sulfate are more active as laxatives, whereas calcium sulfate is less active.
5. When the magnesium sulfate content is 200 mg/L, the most sensitive person will feel the laxative effect; however, magnesium sulfate levels between 500 mg/L and 1,000 mg/L will induce diarrhea in most individuals.
6. Tastes may sometimes be detected at 200 mg/L of sulfate, but generally are detected in the range of 300 to 400 mg/L.

Total Dissolved Solids (TDS)

The SMCL for total dissolved solids is 500 mg/L.

Undesirable Effects

1. TDS imparts adverse taste effects at greater than 500 mg/L.
2. Highly mineralized water influences the deterioration of distribution systems as well as domestic plumbing and appliances (the life of a water heater will decrease one year with each additional 200 mg/L of TDS above a typical 200 mg/L value).
3. Mineralization can also cause precipitates to form in boilers and other heating units, sludge in freezing processes, rings on utensils, and precipitates in food being cooked.
4. There may be a great difference between a detectable concentration and an objectionable concentration of the neutral salts. Many people can become accustomed to high levels.
5. Studies show that the temperature of mineralized waters influences their acceptability to the public.

Zinc

The SMCL for zinc is 5 mg/L.

Undesirable Effects

1. High concentrations of zinc produce adverse physiological effects.
2. Zinc imparts a bitter, astringent taste that is distinguishable at 4 mg/L. Also, at 4 mg/L a metallic taste will exist.
3. Zinc will cause a milky appearance in water at 30 mg/L.
4. Zinc may increase lead and cadmium concentrations.
5. The activity of several enzymes is dependent on zinc. Enzymes are proteins that catalyze chemical reactions. Zinc is an essential enzyme growth element in terms of nutrition at low doses (zinc must be present) and is toxic to enzymes at high doses.
6. Cadmium and lead are common contaminants of zinc used in galvanizing steel pipe.

Physiological Effects

1. A concentration of 30 mg/L can cause nausea and fainting.
2. Zinc salts act as gastrointestinal irritants. This symptom of illness is acute and transitory.
3. The vomiting concentration range is 675 to 2,280 mg/L.
4. A wide margin of safety exists between normal food intake and concentrations in water high enough to cause oral toxicity.

Dietary Requirements

1. The daily requirement for preschool children is 0.3 mg Zn/ kg of weight.
2. Total zinc in an adult human body averages two grams.
3. Zinc most likely concentrates in the retina of the eye and in the prostate.
4. Zinc deficiency in animals leads to growth retardation.

Key Terms

- **Secondary standards** – federal and state standards for contaminants that affect the aesthetic qualities relating to the public acceptance of drinking water
- **Secondary Maximum Contaminant Level** – levels that apply to public water systems and are necessary to protect public welfare or for public acceptance of the drinking water

Review Questions

1. Explain the Secondary Maximum Contaminant Levels found in the Safe Drinking Water Act.
2. What is the SMCL for chloride?
3. What is the SMCL for color?
4. What is the MCL and the SMCL for fluoride?
5. What is the SMCL for TDS (total dissolved solids)?

Chapter Quiz

1. _____ apply to public water systems and, in the judgment of the EPA, are necessary to protect the public welfare or for public acceptance of the drinking water. The rule means the maximum permissible level of a contaminant that is delivered to the free-flowing outlet of the ultimate user of a public water system. This maximum is not enforceable under EPA, but recommended: however, in some states, these levels are enforceable by law.
 1. Maximum Contaminant Levels
 2. Maximum Contaminant Level Goals
 3. Secondary Maximum Contaminant Levels
 4. Unregulated Contaminant Levels

2. _____ are important factors in public acceptance and confidence in a public water system. States are encouraged to implement SMCLs so that the public will not be driven to obtain drinking water from potentially lower-quality, higher risk sources. Many states have chosen to enforce Primary and Secondary Contaminant Levels to ensure that the consumer is provided with the best quality water available.
 1. Biological contaminants
 2. Chemical contaminants
 3. Aesthetic qualities
 4. Radiologic contaminants
3. Although _____ is regulated as a secondary contaminant because precipitates of the metal can cause cloudy-looking water, this contaminant may interfere with absorption of fluoride in the gastrointestinal tract and may decrease the absorption of calcium, iron, and cholesterol.
 1. aluminum
 2. zinc
 3. silver
 4. manganese
4. _____ may be caused by aromatic, polyhydroxy, methoxy, and carboxylic acids, as well as fulvic and humic acid fractions, and the presence of metals such as copper, iron, and manganese. Changes to these levels may provoke citizen complaints.
 1. Taste
 2. Odor
 3. Color
 4. Laxative effects
5. The SMCL for _____ are a TON of 3. This contaminant is a useful test for determining the effectiveness of different kinds of treatments and as means for tracing the source of contaminants.
 1. tastes
 2. odors
 3. colors
 4. Laxative effects
6. Sulfates tend to form hard scales in boilers and heat exchangers, cause taste effects, and _____ are commonly noted by newcomers or casual or intermittent users of water high in sulfate.
 1. tastes
 2. odors
 3. colors
 4. laxative effects
7. The SMCL for _____ is 5 mg/L. High concentrations produce adverse physiological effects. It imparts a bitter, astringent taste that is distinguishable at 4 mg/L and will cause a milky appearance in water at 30 mg/L. The daily requirement for preschool children is 0.3 mg / kg of weight. A deficiency in animals leads to growth retardation.
 1. silver
 2. TDS
 3. zinc
 4. manganese
8. The SMCL for _____ is 250 mg/L. The undesirable effect of this contaminant is an objectionable salty taste in water and corrosion of the pipes in hot water systems and other pipelines. Domestic plumbing, water heaters, and municipal waterworks equipment will deteriorate when high concentrations are present.
 1. chloride
 2. TDS
 3. zinc
 4. manganese

9. The SMCL for _____ is 0.05 mg/L. Undesirable effects include coatings in distribution piping, brown blotches in laundry items and black precipitates, and a taste to water above 0.15 mg/L. Unless the precipitate is removed, precipitates reaching pipelines will promote bacterial growth. Liver cirrhosis has arisen in controlled feeding of rats. It is essential for proper nutrition, and diets deficient in manganese will interfere with growth, blood and bone formation, and reproduction.
1. chloride
 2. TDS
 3. zinc
 4. manganese
10. The SMCL for _____ is 500 mg/L. The undesirable effects include adverse taste effects at greater than 500 mg/L, the deterioration of distribution systems as well as domestic plumbing and appliances, and it causes precipitates to form in boilers and other heating units.
1. chloride
 2. TDS
 3. zinc
 4. manganese

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5: Coagulation and Flocculation

Learning Objectives

After reading this chapter you should be able to identify and explain the following:

- Coagulant types
- Coagulation chemistry
- Mixing systems for both coagulation and flocculation
- Flocculation process theory
- Pounds formula

One of the most important steps in the water treatment process is the removal of suspended solids. The two-part process in water treatment involves chemical deactivation and physical removal of pathogenic organisms. The physical removal of pathogens is accomplished in several steps. The first two steps include the processes of **coagulation** and **flocculation**. In this process, colloidal particles are destabilized to gather all the suspended material together. They can also be referred to as **nonstable solids**. This process increases particle sizes which assists in removal during the filter process. The larger floc particles will be removed during the sedimentation and filtration process. Coagulation occurs very quickly in the rapid mix or flash mix process. The flash mix process only lasts several seconds as the coagulant rapidly mixes and reacts with the raw untreated water. The floc will gain in size during the second step of flocculation. Filtration cannot occur without proper coagulation and flocculation.

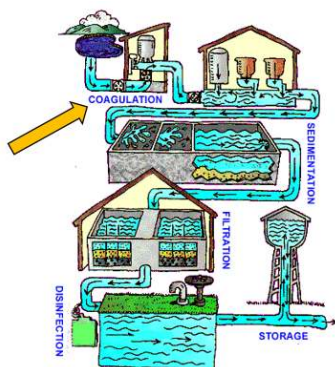


Figure 5.1: Image of water treatment by the EPA is in the public domain

Coagulant Types

In general, two types of coagulants are used during coagulation. A **primary coagulant** and a **coagulant aid** will be used during the rapid mix process. The colloidal surfaces are negative thus positively charged metal salts are used as primary coagulants. The coagulant dissolves in water and ionizes. To ionize is when a molecule loses or gains an electron to form an ion. The three most common coagulants used in water treatment are Aluminum Sulfate (Alum), Ferric Sulfate, and Ferric Chloride. The most commonly used primary coagulant in water treatment is Alum because of its wide availability and affordability.

Synthetic polymers are often used as a coagulant and filter aid but have also been used as a primary coagulant. Operators use different charged groups of polymers known as cationic (positive), anionic (negative), and nonionic (without ionizable groups). Coagulation aids assist in the building of settable floc.

Important things to consider when choosing a polymer:

- Overdosing a polymer can decrease the efficiency of the coagulation process and cause filter binding and increased headloss in filters
- Not all supply water is created equal. Every single water source has a different chemistry and jar tests must be performed to see what polymer works best with the specific source water.
- There is no widespread standard for choosing a polymer. Different states have chemical approval standards that must be met.
- The addition of chlorine can effect polymer effectiveness
- As with any chemical, there is a dosage limit

Chemistry

Like many processes in water treatment, the theory of coagulation is very complex. As an operator, you should have a basic understanding of the chemistry involved in each process. The coagulant added to the water will react with the alkalinity in the water to form insoluble floc. Insoluble is something that will not dissolve. If the floc is not formed properly, then operators cannot effectively remove pathogens from the treated water. Alkalinity is not the same as pH. This gets very confusing as “alkaline water” has become a more widely popularized marketing term. Alkalinity is the water’s ability to neutralize an acid based on its makeup of carbonate, bicarbonate, or hydroxide. The measure of alkalinity is the amount of acid that would have to be added to water to lower the pH to 4.5 and there is where the confusion arises.

Coagulation is most effective in the pH range of 5-7 because of the water’s ability to react with alkalinity. In this range, the water tends to buffer or stay in the same pH range and will allow the complete mixing of coagulant chemicals. If the raw water has a low pH, agents such as soda ash can be added to increase the pH.

Proper water quality tests must be performed by operators to promote the proper addition of coagulant chemicals. Underdosing coagulants will cause problems with floc formation while overdosing coagulants can cause clogging during the filter process. Clogged filters lead to head loss problems in the filters and increased filter washing.

Mixing

As noted earlier the coagulation process is completed within a matter of seconds. Mixing can be achieved by utilizing hydraulic mixers, mechanical mixing, diffusers, or pumped blenders. Hydraulic mixers use flow to achieve mixing. This kind of mixing requires enough flow to create a disturbance in the water to achieve proper mixing. Mechanical mixers require the greatest amount of energy because they require an electrical source to achieve mixing. Diffusers apply uniform flow during the coagulation process but may require many adjustments after flow changes. Finally, pumps can be used to push coagulant into water flow. Mixing can occur in a basin, channel, or pipeline.

Flocculation

Flocculation is the slow mixing process that causes smaller particles to merge into larger particles that settle more easily. The particles are then more easily removed in the sedimentation and filtration process. The process of flocculation is achieved by controlling the rate of impacts between particles as they gain size. Floc size can range between 0.1 mm-3 mm. The size of the floc produced depends on which type of treatment process is utilized at a specific plant. It is important that floc has good size but also density so the floc will not shear during the sedimentation and filtration process. This process is much longer than coagulation lasting roughly 15-45 minutes.

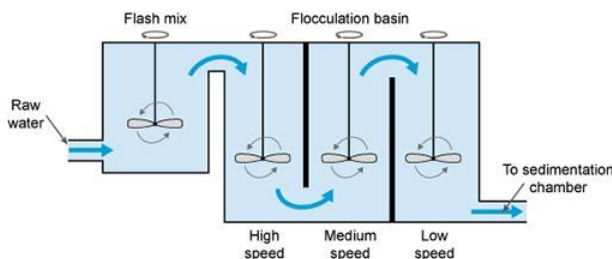


Figure 5.2: The coagulation–flocculation process by The Open University and partners is licensed under CC BY-NC-SA 4.0

Shown Above: The flash mix portion is also known as coagulation. The chemicals are added together and the process occurs within seconds. After flash mix, the water heads to the flocculation basins to allow floc particles to gather in size.

Mechanical flocculators can be installed both horizontally and vertically. The horizontal type utilizes paddle-style mixers while the vertical style mixers can include paddle, turbine, and propeller style mixers. The shape and size of a flocculation basin is determined by the type of mixing used and the adjacent structures such as the sedimentation basins. Flocculation basins are usually split into 3 compartments. The speed of the mixing is decreased in each compartment to prevent the particles from breaking apart as they become larger. If the particles break apart during flocculation, the particles will place a heavier burden on the filters during the filtration process causing lower filter run times. This phenomenon will be discussed in further detail in Chapter 6.

Monitoring and Process Control

The coagulation and flocculation process requires a great amount of attention to detail along the way. An operator cannot just set a dose and “hope” everything works out. Water quality can change frequently and operators must ensure they are on top of changing conditions. One way an operator can achieve this is through jar testing. This is a laboratory procedure that finds the best coagulant dose based on water quality conditions.



Figure 5.3: Image of a jar test by Jigchen L. Norbu is licensed under [CC BY-SA 4.0](#)

The efficiency of water treatment plants is determined by the combined effluent turbidity reading. Individual filter efficiency is also closely monitored. This will be discussed at length in the filtration chapter of the text, but it’s important to have a basic grasp of that concept at this point in the treatment process. Water treatment is a lengthy process. What’s occurring right now can have grave impacts down the line in the treatment process. Because of this consideration, jar testing and plant monitoring is all the more critical. Jar testing and laboratory grab sampling ensure the water which is theoretically being treated now, will be safe when entering the distribution system hours from now.

During abnormal conditions, it is important for operators to take notes and inform senior operators and/or a supervisor. Record keeping is important because operators can go back to notes used from previous experiences. For example, a large rain event has changed the influent turbidity entering the treatment plant. A similar event happened three years prior and the influent turbidity is very similar to the ones operators are seeing currently. Wouldn’t it be nice to have a record of what the operators did three years ago? Hopefully, the operations staff from three years ago noted changes in coagulant dose, mixing speeds, chlorine demand, and other significant plant changes.

Enhanced Coagulation

The **enhanced coagulation** process is used to remove natural organic matter by adjusting the pH and coagulant dose to remove the greatest amount of suspended matter during the treatment process. The addition of acid is used to achieve the proper pH unlike sweep treatment where the operator overdoses the coagulant to achieve the correct pH range. Enhanced coagulation occurs at a lower pH, and accordingly will see improvements in treatment such as:

- Humic and fulvic molecules separate better with lower pH. Humic and fulvic acids are organic acids commonly found in raw water sources
- Less coagulant is required for treatment
- Flocculation improves at a lower pH
- Sulfuric Acid addition before coagulant is added preconditions organic matter

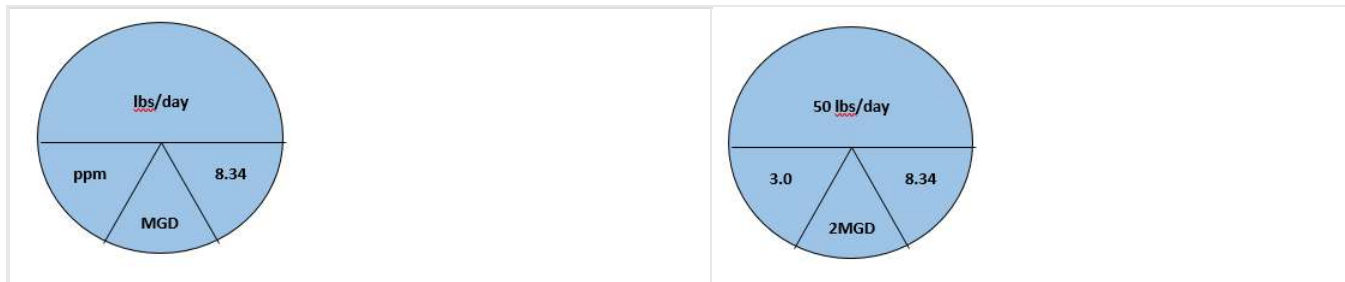
The Pounds Formula (Chemical Dosage Problems)

One of the most important calculations an operator will use is the “Pounds Formula.” The pounds formula can be used to solve water math problems including milligrams per liter to pounds per day, feed rate, chlorine dosage, percent strength, and dilution calculations. The formula for the pounds calculations is:

$$\text{Feed Rate (lbs/day)} = \text{dosage (parts per million)} \times \text{Flow Rate (million gallons per day)} \times 8.34 \text{ lbs/gal}$$

Below is the pounds formula expressed in a mechanical wheel. The mechanical wheel can be used to solve a chemical dosage problem by plugging in the given information and then multiplying or dividing as indicated to determine the solution. For example,

a treatment plant will produce 2 MG a day. The chlorine dose is 3.0 mg/L. How many pounds of chlorine will the operator use?



$$\text{Feed Rate} = 3.0 \times 2 \times 8.34$$

$$\text{Feed Rate} = 50 \text{ lbs/day}$$

It is important to note that when using the pounds formula, the water production is always expressed in MGD. For example, a problem may ask what the chlorine production is for a water treatment plant that produces 1,000,000 gallons a day. This is the easiest expression because $1,000,000 \div 1,000,000 = 1$.

EX: A water treatment operator must super chlorinate a 650,000-gallon tank at 50 ppm. How much chlorine must the operator add?

$$50 \times .65 \times 8.34 = 271 \text{ pounds}$$

* You must divide 650,000 by one million to get .65 to solve the equation. Round to the nearest tenth or hundredth place to get the most accurate answer. Remember you must show all work during exams to get credit but the format is still multiple choice so be careful when rounding.

Practice Pounds Formula Problems

Example 5.1

The dry alum dosage rate is 15 mg/L. The daily flow rate of a treatment plant is 5 MG. How many pounds of dry alum per day is required?

Solution

$$\text{Feed Rate} = 15 \text{ mg/L} \times 5 \text{ MGD} \times 8.34 \text{ lbs/gal}$$

$$\text{Feed Rate} = 625.5 \text{ lbs/Day}$$

* This sample problem is pretty simple. Plug the numbers into the equation and multiply to get your answer.

Example 5.2

A treatment plant uses 300 lbs of alum a day. The plant output is 2,500,000 gallons a day. What is the dose?

Solution

$$\text{Step 1: } 2,500,000 \div 1,000,000 = 2.5 \text{ MGD}$$

$$\text{Dose} = 300 \text{ lbs/day} / (2.5 \times 8.34)$$

$$\text{Dose} = 250/20.85$$

$$\text{Dose} = 12 \text{ mg/L}$$

Example 5.3

How many pounds of 65% available chlorine must an operator add to a treatment plant with a dose of 3.0 mg/L and a plant output of 5 MGD?

Solution

* Start the equation as you normally would. Since you are adding a solution of chlorine that is not 100 percent an extra step is needed to solve the problem. Remember, if the chlorine solution is not 100 percent available chlorine you are going to need more to dose properly!

$$\text{Feed Rate} = 3.0 \times 5 \times 8.34 = 125 \text{ lbs/day}$$

Next, you will need to make an adjustment based on chlorine strength.

$$(.65)(x \text{ lbs/day}) = 125 \text{ lbs/day}$$

$$x = 125.65$$

$$x = 192 \text{ lbs/Day}$$

Note: If the number is smaller than the original number, you multiplied instead of dividing. You will always need more chemical if it's not 100% strength.

Example 5.4

A water tank that is 30 ft high and 100 f in diameter must be dosed at 50 ppm for disinfection. How many pounds of 65% calcium hypochlorite must be added to dose the tank?

Solution

$$0.785 \times 100\text{ft} \times 100\text{ft} \times 30\text{ft} \times 7.48 \text{ Gal/ft}^3 = 1,761,540 \text{ gallons}$$

$$1,761,540 \text{ gallons}/1,000,000 = 1.76\text{MGD}$$

$$\text{Feed Rate} = 50 \times 1.76 \times 8.34 = 733.92 \text{ pounds/day}$$

$$(.65)(x) = 734$$

$$x = 734/.65$$

$$x = 1129 \text{ lbs}$$

Chapter Review

1. The optimal coagulant dose is determined by a _____.
 1. Chlorine test
 2. Flocculation test
 3. Jar test
 4. Coagulation test
2. The most common primary coagulant is _____.
 1. Alum
 2. Cationic polymer
 3. Fluoride
 4. Anionic polymer
3. Bacteria and viruses belong to a particle size known as _____.
 1. Suspended
 2. Dissolved
 3. Strained
 4. Colloidal
4. The purpose of coagulation is to _____.
 1. Increase filter run times
 2. Increase sludge
 3. Increase particle size
 4. Destabilize colloidal particles

5. The purpose of flocculation is to _____.
 1. Destabilize colloidal particles
 2. Increase particle size
 3. Decrease sludge
 4. Decrease filter run times
6. Primary coagulant aids used in the treatment process are _____.
 1. Poly-aluminum chloride
 2. Aluminum sulfate
 3. Ferric chloride
 4. All of the above
7. Flocculation is used to enhance _____.
 1. Number of particle collisions to increase floc
 2. Charge neutralization
 3. Dispersion of chemicals in water
 4. Settling speed of floc
8. If there is a problem with floc formation, what would you consider changing?
 1. Adjust coagulant dose
 2. Stay the course
 3. Adjust mixing intensity
 4. Both 1 and 3
9. Which step in the treatment process is the shortest?
 1. Filtration
 2. Sedimentation
 3. Flocculation
 4. Coagulation
10. To lower the pH for enhanced coagulation the operator will add _____.
 1. Chlorine
 2. Sulfuric acid
 3. Lime
 4. Caustic soda
11. The flocculation process lasts how long?
 1. Seconds
 2. 5-10 minutes
 3. 15-45 minutes
 4. Over an hour
12. The function of a flocculation basin is to _____.
 1. Settle colloidal particles
 2. Destabilize colloidal particles
 3. Mix chemicals
 4. Allow suspended particles to grow
13. A treatment plant has a maximum output of 30 MGD and doses ferric chloride at 75 mg/L. How many pounds of Ferric Chloride does the plant use in a day?
 1. 18,765
 2. 17,765
 3. 19,765
 4. 16,765

14. A treatment plant uses 750 pounds of alum a day as it treats 15 MGD. What was the dose rate?
1. 4 mg/L
 2. 5 mg/L
 3. 6 mg/L
 4. 7 mg/L
15. A treatment plant operates at 1,500 gallons a minute and uses 500 pounds of alum a day. What is the alum dose?
1. 18 mg/L
 2. 28 mg/L
 3. 8 mg/L
 4. 38 mg/L

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6: Sedimentation

Learning Objectives

After this chapter you should be able to identify and explain:

- Sedimentation process
- Zones of sedimentation basin
- Types of sedimentation basins
- Velocity math problems
- Detention time

Sedimentation is the 3rd step in a conventional treatment process. It occurs after coagulation and flocculation and before filtration. Sedimentation removes suspended solids with the use of gravity by slowing the flow of water down to allow material to settle. The settleable solids fall to the bottom of the sedimentation basin reducing the load on the filtration process. A sedimentation basin acts like a lake in the sense that it allows particles to settle naturally. Deeper lakes have much higher quality water entering the treatment plant because the water is able to “settle” for a longer period of time. Treatment plants that use imported water from higher turbidity water sources may be required to use conventional treatment with sedimentation for efficient treatment.

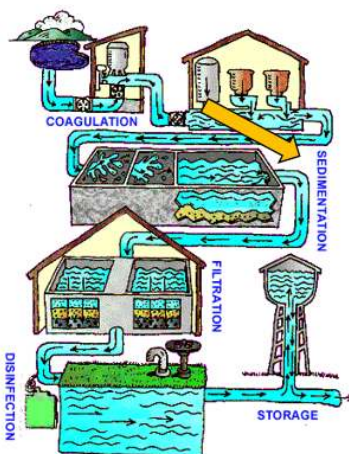


Figure 6.1: Image of water treatment by the EPA is in the public domain

Factors affecting Sedimentation

Several factors can affect the sedimentation process including physical and environmental conditions. Increased pretreatment may be necessary when adverse conditions are present. Factors that affect the sedimentation process include the shape and size of particles, the density of particles, water temperature, particle charge, dissolved substances in the water, environmental effects, and characteristics of the basin.

As discussed in the previous chapter, smaller particles do not settle out easily and their size must be increased with coagulation and flocculation. The larger, denser particles created are called floc. Particles greater than .01 millimeters will settle in the sedimentation process. The shape of particles is also a consideration. Smoother particles with less jagged edges settle out quicker and easier.

Temperature decreases will cause the settling rate to decrease. The settling rate or velocity decreases when the water temperature is colder. Chemical dosage rates need to be adjusted during colder periods of the year or lower flows are necessary in the flocculation basins.

There are three types of currents, surface, density, and eddy, found in a sedimentation basin. Surface currents are caused by wind while density currents are caused by temperature differences and the concentration of solids. Eddy currents are caused by the influent and effluent flow of water in the sedimentation basin. Currents can be beneficial as they can help to promote the building of floc, but they can also cause uneven disbursements of solids throughout a sedimentation basin reducing the efficiency.

Sedimentation Zones

The four zones of a Sedimentation basin include:

- **Inlet Zone:** Where the water enters seamlessly from the flocculation basin. Water is distributed evenly throughout the sedimentation basin to prevent short-circuiting. Short-circuiting occurs when water entering a treatment process tank or basin quickly moves from influent to effluent reducing the water's detention time in a given process.
- **Settling Zone:** Is the largest part of a sedimentation basin. The water will stay here undisturbed for three or more hours while particles settle to the bottom.
- **Sludge Zone:** Located at the bottom of the settling zone. It is where settled particles collect in the form of sludge. Velocities at the bottom of the sludge zone should be minimized to prevent solids from re-suspending.
- **Outlet Zone:** The location where water seamlessly enters a channel or conduit. Launderers also known as effluent troughs are used to collect the clarified or settled water.

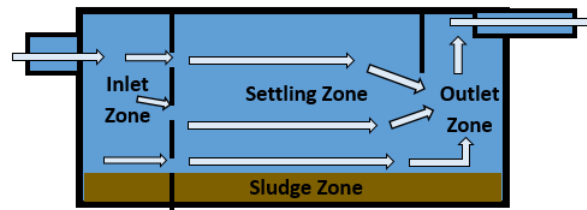


Figure 6.2: Image by COC OER is licensed under [CC BY](#)

Types of Sedimentation Basins

There are several different types of basin designs available for engineers and planners when building a water treatment facility. Rectangular basins are typically found at large scale water facilities because of their predictable treatment and high tolerance to turbidity, color, and algae. They are cost-efficient, require lower maintenance, and have less short-circuiting issues. The figure above representing the different zones of a sedimentation basin is an example of a rectangular sedimentation basin.

Circular and square basins are used in areas where space is limited. They are sometimes called clarifiers and are subject to short-circuiting in the corners. The circular variety can also include up-flow clarifiers or solids contact clarifiers. In these types of clarifiers, the coagulation, flocculation, and sedimentation process all occur in the same basin or clarifier.

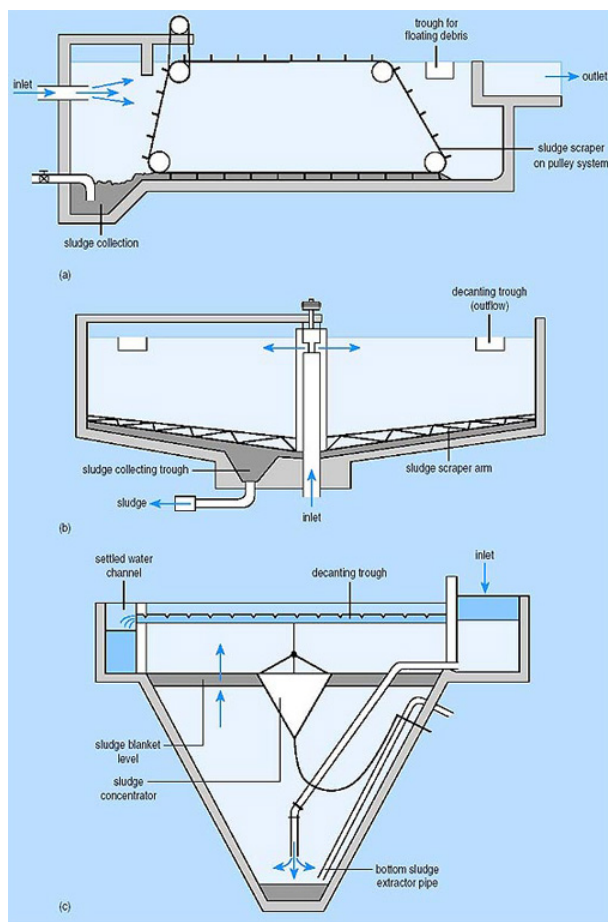


Figure 6.3: Typical sedimentation tanks: (a) rectangular horizontal flow tank; (b) circular, radial-flow tank; (c) hopper-bottomed, upward flow tank – [Sedimentation Basins Image](#) by [The Open University](#) is licensed under [CC BY-NC-SA](#)

Detention Time

Detention time is the amount of time it takes water to travel through a tank or sedimentation basin. It is also referred to as retention time. You will hear some operators use the terms CT and contact time interchangeably but it is incorrect. The term CT will be discussed in greater detail in Chapter 7. Detention time can be used to solve equations for time and flow. Formulas for both types of equations are listed below:

Detention Time (Hours) = $\text{Volume gal} / ((24 \text{ Hours/day}) / \text{Flow gal/day})$

Flow, gal/day = $\text{Volume} / ((24 \text{ Hours/day}) / \text{Detention Time, Hours})$

Reminder: Circular volume calculation $(.785)(D \text{ ft})^2(\text{Depth ft})$

Rectangular Volume calculation: $(L \text{ ft})(W \text{ ft})(D \text{ ft})$

Answers will be in cubic feet so it will be multiplied by 7.48 gal/ft³ to convert to gallons.

Example 6.1

A rectangular sedimentation basin is 80 ft long by 25 ft wide and is 25 ft deep. The flow per day is 2.0 MGD. What is the detention time in hours?

Solution

Step One: calculate the volume of the basin $(80 \text{ ft})(25 \text{ ft})(25 \text{ ft})(7.48 \text{ gal/cubic ft.}) = 374,000 \text{ gallons}$

Step two: $DT = (374,000 \text{ gal} \times 24 \text{ Hours}) / 2,000,000 \text{ gal day}$

Step Three: Detention Time = 4.5 hours

*Note: more complicated problems will have you solve for hours and minutes. Keep this in mind for future math equations that ask for more information.

.5 hours/1 x 60 min/1 Hour = 30 min

Example 6.2

A circular basin is 100 ft in diameter. The basin is 25 ft deep and has a detention time of 4 hours. What is the flow per day?

Solution

Step one: Calculate volume $.785(100 \text{ ft})^2(25 \text{ ft})(7.48 \text{ gal/cubic ft.}) = 1,467,950 \text{ gal}$

Step Two: Flow = $(1,467,950 \text{ gal} \times 24 \text{ Hours})/4 \text{ hours}$

Step three: Answer is 8,807,700 divide by 1,000,000 to get 8.807 or 8.8 MGD

Sludge Handling and Removal

Sludge that collects at the bottom of a sedimentation basin must be removed from time to time for several reasons. As discussed earlier in the chapter, sludge can become re-suspended after settling creating greater load on the downstream filter. Next, sludge buildup can cause the water source to become septic. In this scenario, microbiological growth occurs when oxygen supplies are depleted. Septic conditions can cause taste and odor problems in treated water and also require more chlorine during the disinfection process. Finally, the more sludge that builds up leads to decreased detention time because there is less area for the water to travel and for solids to settle out.

Larger plants will have to remove sludge at a greater rate and with the assistance of sludge removal equipment. Smaller plants may be able to remove sludge manually with portable sanitary pumps and squeegees and hoses to complete cleanup. The amount of time between cleanups can vary with the quality of the water source and the amount of water being treated. Consequently, shutdowns for sludge clean-up will vary dramatically from treatment plant to treatment plant.

Sludge removal equipment includes mechanical rakes, drag-chain and flights, and traveling bridges. The chain and flight system is used in rectangular basins. A chain with scrapers attached moves across the bottom of the basin collecting sludge and moving it to a sump. This system works well but has several moving parts. Additionally, the basin must be dewatered to perform maintenance. The traveling bridge system is also used in a rectangular basin. It travels the entire length of the basin. A pump is attached to the bottom of the system and sludge is pumped to a trough just below the top of the sedimentation basin. The bridge system is easier to perform maintenance on because the parts can be removed from the basin; therefore, dewatering the entire basin isn't required. Finally, mechanical rakes are used in rectangular or circular basins. A rake spans the entire length of the basin while spinning around the basin. Sludge is moved into a trough that can be pumped out or moved by gravity to a sludge collection tank.

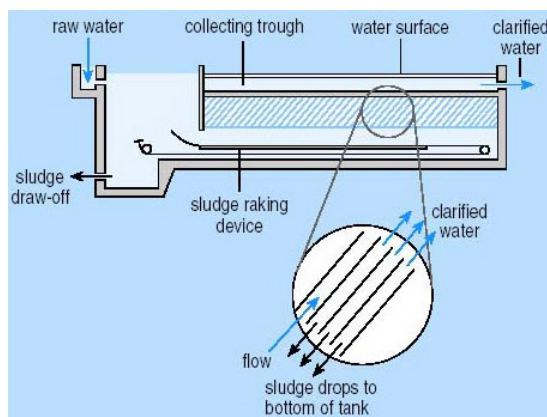


Figure 6.4: Sludge Removal Equipment Image by The Open University is licensed under CC BY-NC-SA

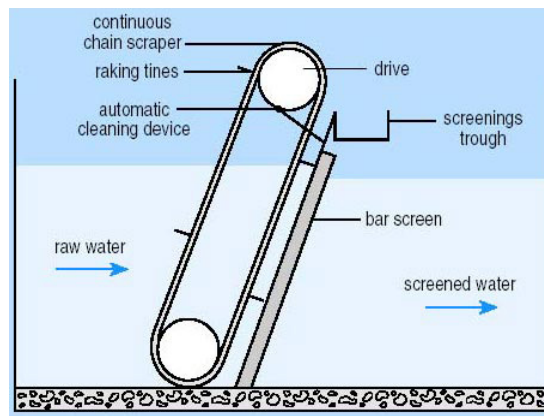


Figure 6.5: Sludge Cleaning Image by The Open University is licensed under CC BY-NC-SA

Review and Daily Operations of Sedimentation Basins

In review, the purpose of sedimentation is to allow suspended solids to settle through the physical process of gravity. The flow of water is slowed down to allow settling to occur. During the process, sedimentation can remove 95% or more of the total solid material operators remove during the entire water treatment process.

Water treatment plants with low turbidity or fewer than 10 turbidity units may find direct filtration or clarification to be a more cost-effective process. A reoccurring theme that should be noted throughout the text is the fact that all treatment facilities are different and the factor that dictates plant processes including facility building and day to day operation is the source water.

The settling characteristics of suspended material dictates how well sedimentation performs. Flow rate through the sedimentation basin drives performance as well as the control of sludge. High flow rates cause solids to carry over, and could also impact the sludge at the bottom of the sedimentation basin. Operators must perform regular jar testing and laboratory testing as well as operate sedimentation basins based on the designed capacity of the basin to ensure optimal performance. Improper operation of sedimentation basins will lead to increased load on downstream filters resulting in early filter washes, increased disinfectant use, and possibly tier violations.

Velocity Math

The movement of water is obviously an important thing to know if you are a water treatment operator. You will need Algebra to solve the next set of water math problems which will have two known values and one unknown value. Flow rates can be used to determine dosage rates, to identify daily averages, to dewater pipelines, to fill pipelines, and for future planning of distribution and treatment equipment.

To solve a flow problem, you need the diameter of the pipe and the velocity of the water (or liquid) in the pipe. Flow rate is expressed as Volume over Time. The equation will be:

$$Q(\text{Flow}) = A(\text{area}) \times V(\text{velocity}). \quad (6.1)$$

A velocity problem occurs when the known value is the pipe diameter and flow rate of pipe.

$$V(\text{Velocity}) = Q(\text{Flow}) \div A(\text{Area}). \quad (6.2)$$

Finally, pipe size is used when known value is pipe flow rate and velocity.

$$A(\text{Area}) = Q(\text{Flow}) \div V(\text{Velocity}). \quad (6.3)$$

Proper use of dimensional analysis is critical when attempting to solve velocity equations. These types of problems often take multiple steps because answers need to be converted to the appropriate units based on the situation.

Example 6.3

A pipeline is 18" in diameter and flowing at a velocity of 125 ft per minute. What is the flow in gallons per minute?

Solution

Step one: Convert pipeline diameter to feet: 18 inches/(12 inches/foot) = 1.5 ft

Step Two: $Q(\text{Flow}) = (.785)(1.5\text{ft.})^2(125 \text{ ft./min})$

Step Three: $221 \text{ ft}^3 \text{ per minute} \times 7.48 \text{ gal/ft}^3 = 1651 \text{ gpm}$

Example 6.4

The flow of a pipe is 2,000 gallons per minute. The diameter of pipe is 24". What is the velocity of the pipe in ft. per minute?

Solution

Step one: Convert units $2,000 \text{ gpm} \div 7.48 \text{ gal./ft}^3 = 267 \text{ ft}^3/\text{min}$. $24 \text{ inches} \div 12 \text{ inches/ft} = 2 \text{ feet}$

Step two: $V(\text{velocity}) = 267 \text{ ft}^3/\text{min} \div .785(2 \text{ ft.})^2$

Step three: $V = 85 \text{ feet per minute}$

Example 6.5

The velocity in a pipeline is 2 ft./sec. and the flow is 3,000 gpm. What is the diameter of the pipe in inches?

Solution

Step One: Convert flow to match units of measurement of velocity: $3,000 \text{ gal/min} \times 1 \text{ min}/60 \text{ sec} \times 1 \text{ ft}^3/7.48 \text{ gal} = 6.68 \text{ ft}^3/\text{sec}$

Step two: $\text{Area} = 6.68 \text{ ft}^3/\text{sec} \div 2 \text{ ft/sec} = 3.3 \text{ feet}$

Step three: Convert feet to inches: $3.3 \text{ ft}/1 \times 12 \text{ inches}/1 \text{ ft} = 40 \text{ inches}$

Chapter Review

1. The treatment process that involves coagulation, flocculation, sedimentation, and filtration is known as _____.
 1. Direct filtration
 2. Slow sand filtration
 3. Conventional treatment
 4. Pressure filtration
2. Sedimentation produces waste known as _____.
 1. Backwash water
 2. Sludge
 3. Wastewater
 4. Mud
3. What kind of process is the sedimentation step?
 1. Physical
 2. Chemical
 3. Biological
 4. Direct
4. The weirs at the effluent of a sedimentation basin are also called _____.
 1. Effluent weirs
 2. Baffling
 3. Launders
 4. Spokes
5. Sedimentation is used in water treatment plants to _____.
 1. Settle pathogenic material
 2. Destabilize particles
 3. Disinfect water

4. Reduce loading on filters
6. Scouring is a term that describes conditions in a sedimentation tank which _____.
 1. Could impact the rest of the treatment process
 2. Higher flow rates in the sludge zone
 3. Re-suspends settle sludge
 4. All of the above
7. The four zones in a Sedimentation basin include _____.
 1. Inlet, sedimentation, sludge, outlet
 2. Inlet, filter, waste, outlet
 3. Inlet, top, bottom, outlet
 4. Surface, sedimentation, sludge, outlet
8. Short-circuiting in a sedimentation basin could be caused by _____.
 1. Surface wind
 2. Ineffective weir placement, or weirs covered in algae
 3. Poor baffling in sedimentation inlet zone
 4. All of the above
9. How much solids should be removed during sedimentation?
 1. 95% or more
 2. 80-95%
 3. 70-80%
 4. 60-70%
10. The type of basin that includes coagulation and flocculation is _____.
 1. Rectangular
 2. Triangular
 3. Up-flow
 4. None of the above

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7: Filtration

Learning Objectives

After reading this chapter you should be able to identify and explain the following:

- Treatment technologies
- Filter media configurations and types
- Filter operation and backwashing
- Filtration math

Filtration is the final and most important removal requirement required by the Surface Water Treatment Rule (SWTR). Water passes through material such as sand, gravel, and anthracite coal to remove floc and disease-causing microorganisms from the finished water. Physical removal of colloids is also achieved during sedimentation but this filtration is the final step. This is the process where suspended colloidal particles are removed from the water. Along with removing possible pathogenic material in the water, removal of turbidity is also achieved which could hide pathogenic organisms and add color to the finished water.

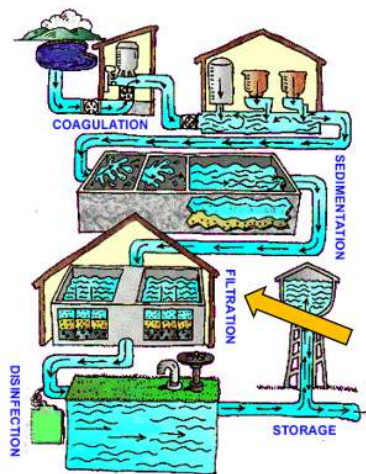


Figure 7.1: Image of water treatment by the EPA is in the public domain

The SWTR sets forth guidelines for all public water agencies that use surface water as a source. Surface water was covered in chapter #1 and is any water open to atmosphere that is susceptible to runoff. The minimum requirements for treatment is disinfection, but most water sources do not meet these very stringent guidelines.

The effectiveness of filtration is based on several important factors. Proper filtration occurs based on incoming source water quality. For example a storm event near the source water could cause higher incoming turbidity than the treatment plant is used to handling. Operational changes such as washing filters may be required. The physical and chemical characteristics of suspended material also come into play during treatment. Too much or too little chemical can lead to ineffective filtration. To follow the storm event example an operator may need to make changes to coagulant and polymer doses to account for increased turbidity and particulate entering the treatment plant. Finally the type of filtration used by a treatment facility is also very important. This decision is mostly out of operators hands as engineers and water quality experts will decide what the most effective treatment process is for the source water before building a treatment facility.

Pictured below is a filter in normal operation.



Figure 7.2

Treatment Technologies

There are four approved treatment technologies in the United States. The most widely used treatment technologies include conventional treatment, direct filtration, diatomaceous earth treatment, and slow sand filtration. Below are the descriptions:

Slow sand filtration facilities are becoming less common because of the large amount of time it takes to treat water and the large amount of space the facilities require. The filtration rates for slow sand filtration are .05 to .10 GPM/sq ft. Particles are adsorbed in a chemical layer known as a *schmutzdecke*. After an amount of time the biological layer must be manually removed by an operator or maintenance staff. The slow time and intensive labor make this treatment method the least ideal especially in areas with larger populations.

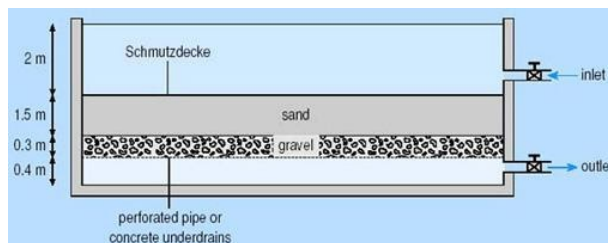


Figure 7.3: Image by The Open University is licensed under CC BY-NC-SA

Diatomaceous Earth filtration is accomplished through pressure filtration. It can also be referred to as precoat filtration. The filter media in this case is added as slurry to the treatment vessel. Within the vessel lays a pipe known as a septum. The slurry attaches itself to the septum and water is run through the vessel where pathogenic and suspended material is captured and strained out of the finished water. This kind of treatment process is very common for swimming pool treatment and beverage companies. This type of treatment method is generally not used by larger municipalities because of the large amount of disposal of sludge and the continuous purchasing of filter media.

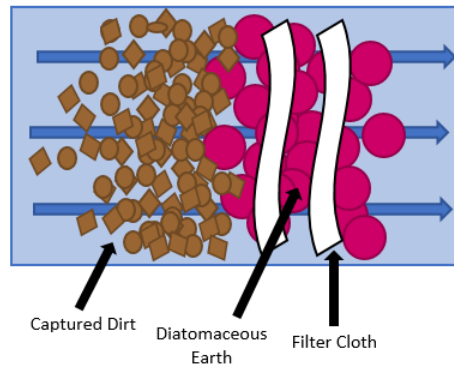


Figure 7.4: Image by COC OER is licensed under [CC BY](#)

Gravity filtration is comprised of the final two approved Water Treatment technologies. Direct Filtration and conventional treatment are the most widely used treatment technologies in the United States. It is described as gravity filtration because the head pressure of the water forces the water to travel through filtered media in order to remove impurities from the drinking water. Direct Filtration differs from conventional treatment because the sedimentation process is skipped. Areas with source water higher in quality may opt for direct filtration to reduce costs and the amount of land space for sedimentation basins can be substantial. The average filtration rate for gravity filtration beds is 3.0 GPM/Sq. ft.- 6.0 GPM/ sq. ft.

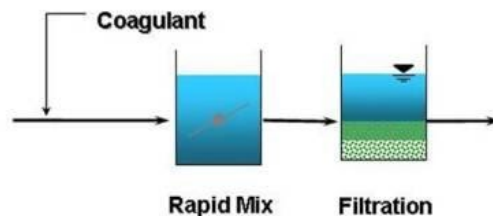


Figure 7.5: Direct Filtration - Image by the [EPA](#) is in the public domain

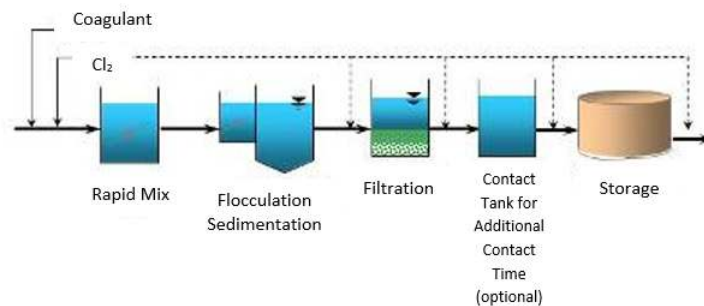


Figure 7.6: Conventional Treatment - Image by the [EPA](#) is in the public domain

Alternative treatment plant methods can be approved on a case by case basis. Newer technologies such as membrane filtration and reverse osmosis have been utilized as the technology has improved and the costs associated with running these particular operations have decreased. Santa Clarita Valley (SCV) Water utilizes an alternative technology known as Upflow Clarification. SCV Water is able to use this technology because of the very low turbidity levels in the water provided by Castaic Lake. It is a more condensed version of conventional treatment and requires much less space due to the lack of sedimentation basins.

Filter Media

The type of media used in gravity filters is sand, anthracite coal and garnet. Garnet is a reddish colored mineral sand comprised of silicates (calcium, iron, manganese, and magnesium) and its density is greater than sand. Gravel is also used as a filter under layer below the filter media being used. It has to be heavier than the filter media so it is able to settle back under media after a completed filter wash cycle.

When choosing filter media it is important to select media that has good hydraulic characteristics, is durable, has no impurities, is insoluble in water, will not dissolve, and does not react with constituents in the water supply. Media is classified by four parameters including its effective size, uniform efficiency, specific gravity, and the hardness of the media. The effective size is the sieve opening in the media that allows water to pass through while collecting the impurities in water. 90 percent of the particles must be bigger than the opening to filter out particles.

When deciding what kind of media to choose for the filter it is important to consider the amount of time it takes for filter turbidity to break through. For example, you operate a treatment plant that requires individual filters say below 0.3 NTU (or Nephelometric Turbidity Units). Once an individual filter goes above this limit it must be washed. Secondly, head loss must be a consideration. Head loss occurs after material builds up over time during the filtration process. The head loss will cause longer filtration times and cause the filter level to rise. Once a filter reaches terminal head loss it must be back washed. The media is not always the cause of head loss and turbidity breakthrough. As discussed earlier, operators must ensure they are dosing chemicals properly as improper dosage can cause the aforementioned head loss and breakthrough.

The uniform coefficient is the ratio between the different sizes of media comprised in the filter bed. The lower the uniform coefficient means the media is closer to the same size than if it were higher. The lower the efficiency number adds to the cost of the media.

Filter production is the amount of water that a given filter can produce in a day. This flow is usually accounted for in Million Gallons per Day (MGD). Individual filtration rates are calculated by dividing the flow rate of the filter in gallons per minute by the surface area of a filter. The filtration rate for gravity filters can be between 2gpm-10gpm/ square foot. This topic will be discussed in more detail during the math portion of this chapter.

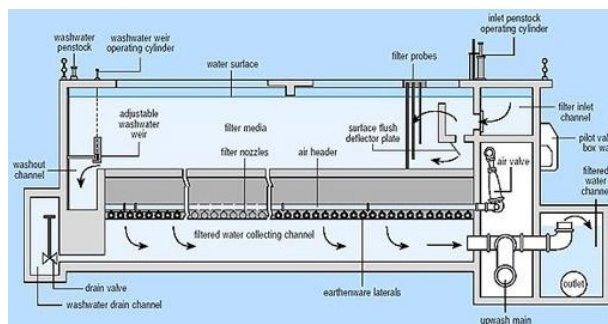
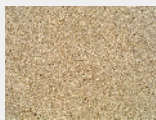


Figure 7.7: Rapid gravity sand filter - Image by The Open University is licensed under CC BY-NC-SA

Filter Media Types			
			
Gravel Image by Martin Olsson is licensed under CC BY-SA 3.0	Sand Image by Yug is in the public domain	Garnet Image by Siim is licensed under CC BY-SA 4.0	Anthracite Image is in the public domain

There are several different types of media configurations. Filtration plants can utilize monomedia, duel media, and multimedia configurations. A monomedia plant has only one type of media which could be coarse sand or anthracite coal. Single media filters may have to be washed more frequently as they tend to be smaller and have more frequent head loss issues. Dual media filters consist of a lower sand level and an upper anthracite layer with larger diameter pores that allow deeper solids penetration. Finally, multimedia filters are used in pressure vessel treatment applications. Multimedia filters have sand, anthracite, and an upper garnet layer. The drawback to pressure vessels is the inability to view filter media within the pressure vessel.

Filter Operation and Backwashing Filters

Throughout the chapter, the subject of “washing” has come up. Filter run times are dependent on three factors but first it is important to know how filter efficiency is measured. The efficiency of a filter is measured by the filter effluent turbidity. And the overall plant efficiency measures the combined effluent turbidity of all the filters. Improper coagulation and filtration could lead to turbidity spikes and in the worst-case scenario could lead to a public health crisis.

Pictured below is a photo of a typical filter deck at a water treatment plant.



Figure 7.8

The first factor that an operator uses to determine whether a filter wash is necessary is the individual filter turbidity. Each treatment plant will have its own operating conditions and permits to follow. If an individual filter fails to meet the turbidity goals or limits, the filter must be put in a backwash cycle. Filters that continue to have decreased run times may need a filter profile ran to figure out why the filters are not meeting standards.

High head loss is the second factor that will lead to a filter wash. After the filter is used for a certain period of time it becomes clogged with the solids the filter is removing. This condition will also lead to increased turbidity as the solids that should be getting captured “breakthrough” the effluent into the treated water supply. Finally, a filter wash will be performed after a certain period of time no matter what the operating conditions are. This is the ideal scenario for washing a filter.

The backwash procedure is the reverse flow of water through the filter. This process removes solids from the filter after breakthrough or the filter run time is hit. Operators must operate the backwash rates at an optimal range because inadequate rates will not properly remove the solids from the filter and excessive rates can cause mud balls and mounds to form within the filter.

Washed water is able to be recycled by sending the waste stream to collection basins. The water can then be returned to the head works of the plant and be mixed with raw water to be treated again. The filter backwash rule limits the amount of water that can be returned into the head works of the plant at a given time. Returning too much recycled water could increase the chances of allowing microorganisms such as *Cryptosporidium* into the treatment plant.



Figure 7.9

The Filter to the left is in the process of a filter back wash. The water is moving in the opposite direction and overflowing over the weir and heads to the waste basin where it is collected and eventually returned into the head works of the plant to be treated again.

Operator Actions

As a Water Treatment operator, you will be expected to have knowledge of how to properly use equipment related to filtration. While running a treatment plant you will routinely:

- Monitor filter performance
- Check turbidity levels with online analyzers and grab samples
- Adjust chemical flow rates
- Backwash filters
- Visually inspect filters

Filters run differently under changing water conditions. The plant will not always run the same as temperature differences and storm events will make operators examine important operational considerations from time to time. It is important to look at the weather and understand how it might affect the treatment plant. A severe rainstorm near your source water could increase turbidity levels coming into the plant. Under these conditions, operators may have to wash filters more frequently and make adjustments to chemical doses.

There are other abnormal conditions to consider when monitoring filter performance. Monitor filter washes to look for mud balls, excessive boiling in certain spots, and media being displaced and sent to waste basins. These conditions would indicate that the backwash flow is too high. Also identify shorter filter run times, filters that may not be coming clean, and algae growth. These factors may be due to improper chemical dosing and a backwash flow that is too low.

Filtration Math

Filtration Rate

Calculating the filtration rate of the filters in your plant is an important function. Operating plans and permits limit the amount of water a filter can produce so it is important to have an understanding of filter rates also known as loading rates. Filtration rates will also give an operator a basic understanding of the treatment plants daily average production.

The filtration rate formula is a velocity equation. These formulas are easily confused with flow problems so make sure you pay attention to the units you add into the formula and pay attention to what the problem is asking. The formula for filtration rate is:

$$\text{Filtration Rate} = \text{Flow Rate} \div \text{Area}$$

Filtration rate equations will use GPM and the area of a given filter. The area of the filter is length x width. A problem may give you the depth of a filter to confuse you. Pay very close attention to the wording in the problem.

Example 1

What is the filtration rate of a treatment plant that has 3 filters that are 20 ft. wide and 20 ft. in length in a plant that produces 1 MGD?

$$\text{Filtration Rate} = (20 \text{ ft} \times 20 \text{ ft} \times 3) \div 1 \text{ MGD}$$

$$1,000,000 \text{ gal/1 Day} \times 1 \text{ Day/24 hours} \times 24 \text{ hours/60 min} = 694 \text{ GPM}$$

**Note: Quick shortcut for future equations, there are 1,440 minutes in a day.*

$$\text{Filtration Rate} = 1,200 \text{ ft}^2 \div 694 \text{ GPM}$$

$$\text{Filtration Rate} = 1.73 \text{ gpm/ft}^2$$

Backwash Rate

As discussed in the chapter, after a filter reaches its capacity due to head loss, turbidity break through, or maximum amount of hours run, the filter must be washed. The filter wash cycle water velocity will be much greater than the amount of water that flows through the filter during normal operation. Many math equations will have the operator solve for rate of rise which is expressed as in/min. The backwash is the flow of water in the opposite direction where water is moving up instead of down.

Example 2

The maximum backwash rate for a filter is 5,000 GPM. The filter is 20 ft. wide and 20 ft in length. What is the rate of rise in the filter?

$$\text{Rise Rate} = \text{Flow Rate} \div \text{Area}$$

$$\text{Rise Rate} = 5,000 \text{ GPM} \div 400 \text{ ft}^2$$

$$\text{Rise Rate} = 12.5 \text{ GPM/sq.ft}$$

$$12.5 \text{ gal/min(ft)} / 1 \text{ ft} \times 12 \text{ inches} / 1 \text{ ft} \times 1 \text{ ft}^3 / 7.48 \text{ gal} = 20 \text{ inches rise/min}$$

Percent Back Wash

Water treatment plants are very efficient at recycling waste stream water. The water sent to waste basins and lagoons is able to be recycled but only a certain amount at a time. The percent backwash math problems compare the total plant production with the amount of finished water used to backwash a filter.

Example 3

A treatment plant treats 2 MGD. It has 2 filters that are washed each day and each uses 10,000 gallons during the wash. What is the percent backwash water?

$$2 \text{ filters} \times 10,000 \text{ gallons} = 20,000 \text{ gallons}$$

$$20,000 \text{ gal} / 2,000,000 \text{ gal} = 0.01$$

$$100 \times 0.01 = 1\%$$

1% of the water the plant uses is for backwash water.

Chapter Review

1. Solids removed from a filter are most commonly removed by what method?
 1. Adsorption
 2. Straining
 3. Deactivation
 4. Flocculation
2. What is a typical filtration rate for slow sand filters?
 1. 2.0-6.0 GPM/sq. ft
 2. 6.0-10.0 GPM/sq. ft

3. 1.0-2.0 GPM/sq. ft
4. 0.5-0.10 GPM/sq. ft
3. In a typical conventional treatment plant, the finished water turbidity for an individual filter should be less than _____.
 1. 1.0 NTUs
 2. 0.3 NTUs
 3. 5.0 NTUs
 4. 3.0 NTUs
4. A filter running under normal conditions will see head loss in a filter _____.
 1. Remain constant
 2. Increase slowly
 3. Rapidly increase
 4. Decrease slowly
5. A filter must be washed if this condition is met:
 1. Head loss
 2. Turbidity breakthrough
 3. Maximum filter run time
 4. All of the above
6. Filter performance is measured by the removal of _____.
 1. Oxygen
 2. Head loss
 3. Turbidity
 4. Chlorine
7. What is the biologically active layer of a slow sand filter called?
 1. Mixed media
 2. Dual media
 3. Sludge layer
 4. Schmutzdecke
8. The pressure drop in a filter is called _____.
 1. Turbidity breakthrough
 2. Head Loss
 3. Filtration
 4. Backwash
9. What is the most common reason for putting a filter into the wash cycle?
 1. Head loss
 2. Filter run time
 3. Turbidity breakthrough
 4. Water level decrease
10. Formation of mud balls and excessive boiling during a wash is an indicator of _____.
 1. Proper backwash rate
 2. Too low backwash rate
 3. Excessive backwash rate
 4. Improper chemical dose
11. Important processes which occur during filtration are _____.
 1. Sedimentation
 2. Adsorption
 3. Straining
 4. All of the above

12. Typical filtration rates for a conventional treatment plant are _____.
1. 0.2-0.6 GPM/sq.ft
 2. 2.0-10.0 GPM/sq.ft
 3. 10.0-20.0 GPM/sq.ft
 4. 200-400 GPM/sq.ft
13. There are four filters at a water treatment plant. The filters measure 20 feet wide by 30 feet in length. What is the filtration rate if the plant processes 8.0 MGD?
1. 1.51 GPM/sq.ft
 2. 2.31 GPM/sq.ft
 3. 2.61 GPM/sq.ft
 4. 2.91 GPM/sq.ft
14. A water treatment plant treats 6.0 MGD with four filters. The filters use 60,000 gallons per wash. What is the percent backwash at the plant?
1. 10%
 2. 8%
 3. 6%
 4. 4%
15. A treatment plant filter washes at a rate of 10,000 GPM. The filter measures 18ft. wide by 24ft. long. What is the rate of rise expressed in inches per minute?
1. 17 inch/min
 2. 27 inch/min
 3. 37 inch/min
 4. 47 inch/min

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8: Disinfection

Learning Objectives

After reading this chapter you should be able to identify and explain the following:

- Disinfection terminology
- Regulations
- Types of disinfectants used during water treatment

The final step in the water treatment process before finished or treated water enters a clearwell for storage is the disinfection process. Disinfection is the process where chemical agents are added to a water source to kill or inactivate pathogenic microorganisms. Pathogenic microorganisms are disease-causing and must be eliminated from treated water. As population sizes increase and freshwater sources become scarcer, the ability to remove and deactivate microorganisms becomes increasingly important. Another factor to consider, especially in California, is stricter regulations due to advancements in technology and water quality testing.

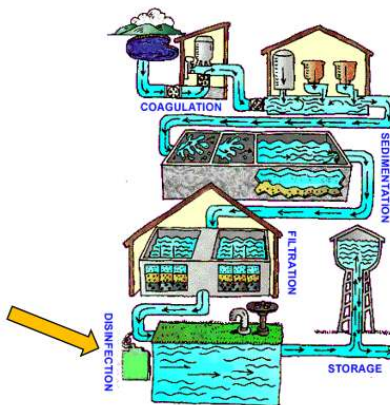


Figure 8.1: Image of water treatment by the EPA is in the public domain

Disinfection Basics

Treatment plant operators use the two-part process of removal and deactivation of microbiological constituents in water. Most of the pathogens water treatment professionals remove and deactivate from drinking water have adapted to living in the bodies of warm-blooded animals. Pathogens thrive and survive in those environments. Outside those environments, these pathogens can stay dormant until they are consumed. Even more frightening, some of the illnesses can cause death. In the United States, our water is generally safe to drink and we often take for granted that turning on a tap will produce a flow of potable water.

Because of limitations in testing, it is difficult to indicate the presence of specific waterborne illnesses caused by virus, bacteria, and *Giardia*. Water professionals use tests such as the total coliform test to look for the likely presence of waterborne disease. The Surface Water Treatment Rule sets specific guidelines for removal and treatment to ensure the removal and inactivation of pathogenic organisms.

Strict regulations set forth by the Safe Drinking Water Act were created to ensure the public's drinking water was safe to consume. To ensure drinking water is safe for human consumption, 3 log removal and deactivation or 99.9% of *Giardia lamblia* is required. For viruses, 4 log or 99.99% removal and deactivation is required. Bacteria fall in the middle of viruses and *Giardia* so the government determined it was not necessary to have regulations specifically regulating their inactivation and removal.

Below is a list of waterborne diseases and illnesses.

Table 7.1: Waterborne Diseases and Illnesses

Bacteria	Internal Parasite from Protozoa	Virus caused
• Anthrax • Dysentery • Cholera • Gastroenteritis (Stomach Flu) • Leptospirosis • Paratyphoid • Salmonella • Shigellosis (Shigella) • Typhoid fever	• Dysentery • Ascariasis (round worm) • Cryptosporidiosis from <i>Cryptosporidium</i> • Giardiasis from <i>Giardia</i>	• Gastroenteritis • Heart anomalies • Hepatitis A • Meningitis • Poliomyelitis

Purpose of Disinfection

Operators disinfect water to destroy the harmful organisms listed in the above chart. Filtration is used to remove the organisms while disinfection kills them or deactivates them. Operators do not sterilize water because sterilization would kill everything in the water. The process of disinfection relies heavily on everything that occurs downstream in the treatment process. As water enters the treatment plant in the form of raw water, the chemistry of that water affects how well the specific disinfectant will work at each stage of the treatment process.

What Affects Disinfection?

There are several characteristics of water that can affect treatment. Water is more easily disinfected with higher temperatures. In lower temperatures, longer contact times may be required and larger amounts of chemical must be used. Higher turbidity rates will decrease disinfection as well. The chapters before have covered how critical it is to remove suspended material early and efficiently. Excess turbidity will require greater amounts of chemical to properly disinfect the water supply. Chemicals such as chlorine can interact with organic and inorganic matter. Chlorine's ability to interact with these constituents may reduce or eliminate the effectiveness of the disinfectants.

Types of Disinfectants Physical

Physical disinfection is not widely used to treat potable water at this time. Ultraviolet rays are starting to be used more consistently but chemical means of disinfection must still be used as ultraviolet disinfection does not carry a disinfectant residual. The process is very expensive and thus is not used by large scale treatment operations in the United States. Other means of physical disinfection include boiling and ultrasonic wave production. Agencies will call for boil water notifications during emergencies and when there is a waterborne illness outbreak but it is not used as a primary means to disinfect drinking water.

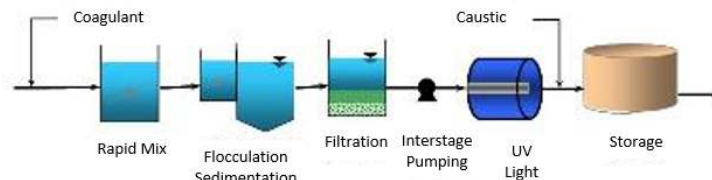


Figure 8.2: Image by the EPA is in the public domain

Types of Disinfectants Chemical

There are several chemical disinfectants available in drinking water applications. The most commonly used in the United States is Chlorine. The topic of Chlorine will be discussed in greater detail in the following chapter. The basics will be covered below as well as a background on the other chemicals available. The chemicals will be broken down into subcategories based on their practical usage in the United States.

Rare

- Iodine—It is commonly used for emergency treatment in the form of droplets or tablets. It is not used by the water treatment industry because of its cost and the potential health hazards to pregnant women and possible thyroid issues, which can develop

with frequent use.

- Bromine—It is not used by water treatment facilities as it is very corrosive and can cause severe skin burns. It is used more commonly as a disinfectant in swimming pools. When it reacts with choline (a common nutrient from plants and animals) in water, it can create disinfectant byproducts. It was used by the United States Navy for a time, but most systems have been removed because of bromine's corrosiveness.
- Sodium Hydroxide and Lime—More frequently used to sterilize pipes. They are not used as an everyday disinfectant because of the bitter taste that is left behind after application. Sodium hydroxide and lime are more often used to increase the pH of the water in the distribution system after treatment with gas chlorine.

More Common

- Chlorine Dioxide—Used as a water treatment disinfectant and oxidizer. It does not react with ammonia which is an issue with chlorine. Chlorine dioxide is used as a disinfectant but is also very effective at removing iron, manganese, taste, odor, and color from treated water. *Cryptosporidium* is resistant to chlorine but is not resistant to chlorine dioxide. Up to 70 percent of chlorine dioxide is converted to chlorite, which is a regulated disinfectant by-product so the dosage rates when using it as a disinfectant must be lower than 1.4 mg/L. Chlorine dioxide must also be made on site which necessitates higher operational and maintenance costs.
- Ozone—Ozone was first used in Europe in the early 1900's. It is a strong disinfectant that also reduces taste and odor issues. The drawback of ozone is that it is very expensive to produce, has high electrical costs, has limited solubility, and does not leave a residual in the treated water because it is so reactive. If bromide is present in the water, ozone can react with it to form bromate, an undesirable DBP (this is an issue for SCV Water). Ozone is very efficient at disinfecting *Cryptosporidium* so it is generally used as a secondary disinfectant along with chlorine or chloramines.

Most Common

- Chlorine—The most widely used disinfectant in the United States is free chlorine. Chlorine can be added as a gas in the form of chlorine gas, as a solid in the form of calcium hypochlorite, or as a liquid in the form of sodium hypochlorite. Most likely, you have a bottle of sodium hypochlorite in your house. We call it bleach. The use of free available chlorine has declined over the years because of the discovery of disinfectant by-products (DBPs). This topic will be discussed in further detail in the next chapter.
- Chloramines—The use of chloramines has become more common in recent years in order to reduce DBPs mainly trihalomethane (THM). Chloramines are also referred to as combined chlorine as it is the combination of chlorine and ammonia. Chloramines are also effective at eliminating taste and odor problems and the residual lasts longer in the distribution system. However, chloramine disinfection is not as strong as chlorine and the improper addition of ammonia can lead to excessive amounts of ammonia in the treated water which results in nitrification.



Figure 8.3: Image by olegschedrin0 is licensed under CC0

Regulations

The Safe Drinking Water Act (SDWA) is the basis of all drinking water regulations in the United States. It is the umbrella in which all new regulations and rules have subsequently been created and enacted. The SDWA regulates drinking water standards in the United States along with its territories. We take for granted all of the research and technology we have available that allows us to

never really be concerned about the quality of our drinking water. The SDWA was passed in 1974 and set forth standards to regulate public water sources. It was amended in 1986 to include some basic principle definitions:

- Defined regulated contaminants and approved treatment techniques
- Defined criteria for filtration of drinking water
- Defined criteria for disinfecting surface and groundwater
- Outlawed the use of lead material in drinking water facilities

After a large public health crisis in Milwaukee, Wisconsin, provisions were made to support drinking water programs through operator training and certification programs. All entities serving water to the public were required to meet program standards with regards to training and certification. In 1999, the government allowed states to hold primacy over drinking water certification programs as long as federal minimums were met. States such as California often have stricter standards than the federal standards.

Surface Water Treatment Rule

The Surface Water Treatment Rule (SWTR) was enacted in 1990 and sought to prevent waterborne illness from surface water sources. Water systems with supplies from surface sources which are susceptible to carrying viruses, legionella, and *Giardia lamblia*, have to follow new requirements with regards to filtration and disinfection known as the multiple barrier approach. The rule also required that systems that used groundwater as a source for drinking water had to adhere to SWTR standards if their water source could come into contact with surface water sources.

Water treatment plants would have to achieve removal and deactivation requirements through the combined efforts of filtration and disinfection. The removal and deactivation of 99.9% of *Giardia* or 3 Logs and the removal and deactivation of 99.99% of viruses or 4 Logs was the new standard set forth. This requirement is measured by monitoring combined effluent turbidities in the combined filters and meeting disinfection requirements through the CT calculation. The CT calculation will be covered in greater detail in the following chapter.

Groundwater Rule

The Groundwater Rule (GWR) was established in 2009 in response to the frequency of groundwater contamination from surface water runoff sources. The rule requires monitoring for systems that do not disinfect to make sure microbiological contamination is not occurring. If a groundwater supplier did use disinfection, they are to meet 99.99% virus inactivation much like groundwater sources.

Total Coliform Rule

The final rule which indirectly relates to water disinfection is the **Total Coliform Rule** (TCR) which was established in 1990. As stated several times within the text, it is nearly impossible and certainly too costly to test for every type of microbiological contaminant that could lead to a public health risk. Instead, the TCR uses a risk based process which tests for the “worst case scenario”. Coliforms grow in warm blooded animals just like viruses, bacteria, and *Cryptosporidium*. They pose no health risk to humans and they grow more abundantly than forms of microbiological agents that will do us harm. If coliforms are present in the water supply, there is a chance for a public health concern.

In the event of a positive coliform test, the downstream and upstream sampling sites as well as the site where the positive sampling occurred will be retested. System maps and sampling plans are a requirement of the TCR. The amount of samples the water supplier takes is based on the population served. Systems which collect less than 40 samples a month can only have one positive sample before notifying the public of a Maximum Contaminant Level (MCL) violation. Systems that collect more than 40 samples a month must not have positives in more than 5% of their coliform samples. If you work at a water supplier that takes 50 samples a month and had 3 positive samples are you in compliance with the TCR?

$$\begin{aligned} 47 \div 50 &= 0.94 \times 100 \\ &= 94\% \end{aligned}$$

So 6% of the samples were positive so you would not be in compliance) or...

$$\begin{aligned} \frac{3}{50} &= 0.06 \times 100 \\ &= 6\% \end{aligned}$$

Therefore, 3 positives are 6% of the total samples.

Disinfectant By-Product Rule

The disinfectant by-product rule (DBPR) was put in place to protect the public from cancer causing risks associated with disinfectants reacting with organic and inorganic matter in treated drinking water. Disinfectant by-products such as trihalomethanes are classified as volatile organic compounds. The Stage 1 DBPR established maximum contaminant levels for several DBPs including:

- Trihalomethane (TTHMs) - 80 ug/L or 80 ppb
- Haloacetic Acid (HAA5) - 60 ug/L or 60 ppb
- Bromate - 10 ug/L or 10 ppb
- Chlorite - 1.0 ppm

While it is all together possible to remove DBPs from treated water with activated carbon, it is a very expensive treatment process and not cost effective for large treatment operations. Other forms of disinfection are possible but will also cause DBP formation. Chlorite formation is associated with chlorine dioxide treatment, while bromate is associated with ozone treatment. The issue of DBP formation becomes even more problematic as chlorite and bromate are often found naturally in source water.

The stage II DBP Rule enhanced regulations on DBP formation by targeting water sources that are more vulnerable to DBP formation and the rule also requires monitoring for HAA5s and THMs. The number of samples taken and the number of sampling sites is based on the size of the population served by the water agency. The use of chloraminated water is being used more commonly to combat DBP formation but using chloramines instead of other disinfectants has other risks associated with its use which will be covered in greater detail in the next chapter.

This is the link to the [quick reference guide from the EPA](#) with information on the Stage I and Stage II DBP rule.

Chapter Review

1. What is residual chlorine?
 1. Chlorine used to disinfect
 2. The amount of chlorine after the demand has been satisfied
 3. The amount of chlorine added before disinfection
 4. Film left on DPD kit to measure residual
2. When Chlorine reacts with natural organic matter in water it can create _____.
 1. Disinfectant by-products
 2. Coliform bacteria
 3. Chloroform
 4. Calcium
3. What are trihalomethanes classified as?
 1. Salts
 2. Inorganic compounds
 3. Volatile organic compounds
 4. Radio
4. What disinfectant is used for emergency purposes and not utilized in the water treatment industry?
 1. Chlorine
 2. Iodine
 3. Ozone
 4. Chlorine dioxide
5. What is the disinfectant with the least killing power but that has the longest lasting residual?
 1. Chlorine
 2. Ozone
 3. Chlorine dioxide

4. Chloramines
6. The active ingredient in household bleach is _____.
 1. Calcium hypochlorite
 2. Calcium hydroxide
 3. Sodium hypochlorite
 4. Sodium hydroxide
7. *Cryptosporidium* is not resistant to this chemical:
 1. Ozone
 2. Chlorine dioxide
 3. Chlorine
 4. Both 1 and 2
8. The Removal and inactivation requirement for *Giardia* is?
 1. 99.9%
 2. 99.99%
 3. 99.00%
 4. 90%
9. If a coliform test is positive, how many repeat samples are required at a minimum?
 1. None
 2. 1
 3. 3
 4. Depends on the severity of the positive sample
10. Your water system takes 75 coliform tests per month. This month there were 6 positive samples. What is the percentage of samples which tested positive? Did your system violate regulations?
 1. 3% Yes
 2. 5 % No
 3. 8 % Yes
 4. 10 % No

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9: Chlorine

Learning Objectives

After reading this chapter you should be able to identify and explain the following concepts related to chlorine:

- Chlorine terminology
- Chlorine chemistry
- Chlorine safety
- Advanced chlorine calculations

As discussed in previous chapters, chlorine is the chemical most frequently used in the water treatment industry for disinfection purposes to meet the standards of the surface water treatment rule. Chlorine is used in several different forms and can be fed into the system in a variety of different methods. It is a very dangerous chemical, so proper safety and handling procedures must always be followed. In addition to being used as a disinfectant, Chlorine may also be used as a controlling agent for the removal of algae, for taste, and for odor. Other beneficial applications of Chlorine include disinfecting new water facility infrastructure such as pipes and tanks and the oxidation of iron, manganese, and hydrogen sulfide.

Chlorine Terminology

Chlorine is available in three forms: gaseous, solid, and liquid.

- Gaseous(chlorine)— Cl_2
- Solid (Calcium hypochlorite)— $\text{Ca}(\text{OCl})_2$
- Liquid (Sodium Hypochlorite)— NaOCl

It is not explicitly known how chlorine disinfection works. One explanation is that chlorine attacks a bacterial cell and destroys it. The other theory suggests that chlorine deactivates enzymes within the cell enabling the microorganisms to use their food supply. Adding chlorine to a water supply causes chemical reactions to take place between the water and the organic and inorganic molecules within the water. After chlorine is done combining with organic and inorganic material in the water, the demand has been satisfied.

The dose of chlorine minus the demand is the residual. The reason chlorine is used in the United States over other disinfecting chemicals is Chlorine's ability to leave a lasting residual within the framework of the water distribution system. This residual continues to fight potential disease-causing microorganisms after treatment has concluded.

Assume that you are working as an operator at a water treatment plant. Your chief operator would like to maintain a residual of 2.0 mg/L of chlorine residual in the distribution system. The demand is 1.5 mg/L. What is the dose you must add to achieve a residual of 2.0 mg/L?

$$\begin{aligned}\text{Dose} &= \text{Demand} + \text{Residual} \\ &= 2.0 \text{ mg/L} + 1.5 \text{ mg/L} \\ &= 3.5 \text{ mg/L}\end{aligned}$$

Therefore, you would need to maintain an average of a 3.5 mg/L dose of Chlorine to achieve the residual requested by your chief operator.

Other Chlorine terminology includes Free Chlorine, Combined Chlorine, and Total Chlorine. It is important to understand these terms before taking a deeper dive into the chemistry of chloramination. The term Free or Available Chlorine refers to the amount of Chlorine that is "free" or "available" in the system to kill or deactivate pathogenic organisms. Combined Chlorine is chlorine that has combined with other molecules and Total Chlorine is the combination of Free or Available Chlorine and Combined Chlorine. Although Combined Chlorine lasts longer in the distribution system, it is a far less effective disinfectant. It is also important to note, not all Chlorine has the same strength. This will be covered more thoroughly later in the chemistry and math portion of the chapter.

Chlorine Content

As mentioned earlier in the chapter, Chlorine is available in different states of matter. The amount of chlorine used to dose water in a treatment plant is determined by the compound used. Below is a chart illustrating the different compounds of chlorine. The percent column indicates the percentage of Chlorine in the compound. For example, Chlorine Gas is pure Chlorine and yields the highest available percentage at 100%.

Table 8.1: Chlorine Content in Different Chlorine Compounds

Chlorine Compound	Percent	Amount needed to attain 1lb
Chlorine Gas	100	1 lb
Calcium Hypochlorite	65	1.54 lbs
Sodium Hypochlorite	15	.8 gallons
Sodium Hypochlorite	12.5 (most common)	1.0 gallons
Sodium Hypochlorite	5 (household bleach)	2.4 gallons

In the math section of this chapter, the impact of the Chlorine percentage will become evident. It is important to read the question and understand what concentration of chlorine is being added to the treatment plant. The third column in the table provides the lb or gallon needed of the specific compound to provide 1 lb of chlorine. For example, if your plant requires a weight or quantity of 100 lbs of chlorine and you use gas chlorine, then you will be adding 100 lbs. of Chlorine Gas. If you are using calcium hypochlorite, you will need to calculate the number of pounds required. From the table, Calcium Hypochlorite is 65% Chlorine. To determine the total number of pounds of Calcium Hypochlorite needed to provide 1 lb of Chlorine, divide the dose required by the percent chlorine.

$$100 \div 0.65 = 153.8 \text{ lbs} \quad (9.1)$$

Therefore, you will need to use 153.8 lbs of Calcium Hypochlorite to obtain a 100 lbs dose of chlorine.

Factors of Chlorine Success

Several factors during the water treatment process will impact the effectiveness of the Chlorine. The five factors that affect chlorine treatment are:

- The concentration of chlorine, more specifically the dose
- The amount of time chlorine is in contact with the water
- The temperature of the source water
- The pH of the source water
- The constituents in the source water

The amount of time the chlorine is in contact with the water determines the effectiveness of the chlorine disinfection. The CT (Concentration multiplied by Contact Time) formula is used to calculate the time chlorine is in contact with the water. C is the concentration of chlorine residual, therefore CT is expressed as (mg/L-min). If water is leaving a drinking water storage tank, also referred to as a clearwell, at a rapid rate, then the chlorine concentration will have to be increased. If the concentration of chlorine is decreased, then the water will have to stay in contact with the disinfectant for a longer period of time. Combined Chlorine treatment associated with monochloramine disinfection will require longer holding periods due to its decreased effectiveness.

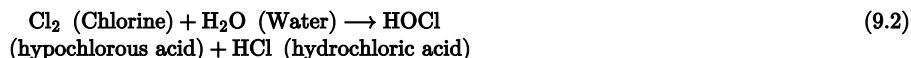
Temperature affects chlorine treatment in a variety of ways. Chlorine is more effective at killing pathogens at higher temperatures but at lower temperatures, the chlorine residual will last longer. Practically speaking, chlorine disinfection works better in warmer temperatures as more credit is given with the CT calculation. The pH level of the water is also a significant factor when treating with chlorine. The ratio of HOCl to OCl⁻ is affected based on the pH. HOCl will remain the dominant disinfectant in water with a lower pH while OCl⁻ will remain in higher quantities in water with a higher pH.

In the disinfection process, chlorine not only reacts with organisms that are to be killed, but it also reacts with the turbidity in the water and other substances such as ammonia. Reducing turbidity in treated water through coagulation, sedimentation, and filtration

ensures disinfection will be more effective. The maximum and minimum chlorine residual in the distribution system is 4.0 mg/L and 0.2 mg/L respectively. However, corrective measures must be taken when you see the residual in the distribution system dramatically decreasing. You would never want to see the minimum chlorine residual of 0.2 in the distribution system.

Chlorine Chemistry

Below is the reaction that occurs between water and free available chlorine:



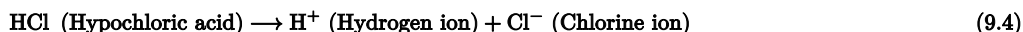
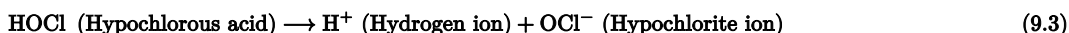
Hypochlorous acid is more effective of the two forms of available chlorine. First, it is important to understand how chlorine demand works. We can do this by examining the effects of adding free available chlorine to distilled water. If we were to dose distilled water with 1.0 mg/L of free available chlorine, the residual would be 1.0 mg/L chlorine because there is nothing in distilled water that will react with the chlorine other than the water itself. Distilled water lacks impurities.

It is clear that the raw source water is filled with impurities. Chlorine will have many constituents to react with during the treatment process. When the Chlorine reacts with the water and the impurities, five different types of chlorine residuals result. The following chart illustrates the effectiveness of each type of residual. The most effective is hypochlorous acid (HOCl) so the effectiveness of the remaining four types of residuals are in comparison to HOCl.

Table 8.2: Effectiveness of Different Residuals

Residual	Abbreviation	Effectiveness
Hypochlorous Acid	HOCl	1
Hypochlorite Ion	OCl ⁻	1%
Trichloramine	NCl ₃	More info later in the chapter
Dichloramine	NHCl ₂	1.25 %
Monochloramine	NH ₂ Cl	.667%

As the chart shows, dichloramines are a “more effective” disinfectant than monochloramines but their use may cause taste and odor problems. The effectiveness of trichloramines has not been extensively researched and as with dichloramines, a pungent taste and odor problem occurs with its use. Thus the water industry only uses monochloramines as a disinfectant. Below are the chemical formulas which illustrate when hypochlorous acid disassociates and becomes a weaker disinfectant.



Chlorine Handling and Safety

As a Treatment operator, you will come in contact with many dangerous chemicals used to treat water. One chemical that is widely used in the water treatment industry is chlorine. As discussed earlier in the chapter chlorine comes in three different states: gas, powder, and liquid. All three types of chlorine have risks associated with the handling and storage of the chemical. It is important that proper safety procedures are followed at all times.

Chlorine Gas

Chlorine gas is 2.5 times heavier than air. The odor is pungent and the color is greenish-yellow. Gas chlorine is only visible in very high concentrations and you don’t ever want to see it. Chlorine gas is irritating to eyes, nasal passages, and the respiratory system. It is a very dangerous substance and concentrations as low as 100 parts per million can kill a person.

Chlorine gas is available in three different types of containers: 150-pound containers, 1-ton containers, and, for very large plant operations, in railroad containers. A majority of treatment operations will use chlorine tanks. The amount of chlorine used in a given day will determine which type of chlorine container your plant will use.

When delivered to a treatment plant, the “150-pound” chlorine gas cylinder container weighs roughly 250-280 pounds. A chlorine cylinder consists of the cylinder body, neck ring, valve, and protective hood.

Cylinders are transported around the facility with the use of a dolly or hand truck. The use of a safety chain or strap is mandatory at all times. Cylinders should never be rolled because it could lead to employee injury or the shearing off of the valve. The maximum daily withdrawal rate for a chlorine cylinder is 40 pounds.

Chlorine “ton” containers hold 2000 pounds of chlorine and usually weigh around 3700 pounds when filled with chlorine. The Containers are shipped and stored horizontally. The edge of the chlorine container has a ring, which enables cranes and hoists to move them from the truck to the storage or withdrawal area. Containers are stored on trunnions that allow operators to rotate containers with the use of a special tool. Each container has two valves, one at the top and one at the bottom. The top valve allows the chlorine to be withdrawn as a gas while the bottom valve allows chlorine to be withdrawn as a liquid. When putting containers in storage, they should be placed such that the liquid chlorine is allowed to settle in the bottom of the container. The maximum daily withdrawal rate for a 1-ton tank is 400 pounds.

Gas chlorine cylinders and 1-ton containers use similar methods to feed chlorine into the water. Feeding systems include a scale, valves and piping, a chlorinator, and an injector or diffuser. The weighing scales are used to keep track of how much chlorine has been used or is left in the cylinder or tank. Record keeping is critical for all chemical use. Monitoring and reviewing the recorded data can help identify problems with the chlorine system and manage costs by reducing chemical waste.



Figure 9.1

Pictured above is a standard chlorine scale and trunnion system. The trunnion acts as a form of storage while also keeping an accurate measure of the amount of chemical left in the tanks. In California, the tanks must also be secured with straps due to earthquakes.

Valves and piping are just as important in the chlorine system as they are in the water system. Each chlorine tank or cylinder is equipped with valves which allow or prohibit the flow of the chlorine during the transfer and storage of the product. In an emergency, the valves can also be used to quickly shutoff the flow of chlorine. Large scale systems will include piping manifolds that allow for the transfer of chlorine from multiple tanks. The associated feed system will include valves and piping to provide chlorine to different feed points in the system.

The chlorinator is the piece of equipment that feeds the chlorine directly into the system. Chlorine is dispersed evenly into the system based on a dosage set point using vacuum and pressure regulators. For safety, a vacuum must be maintained in the line. The vacuum ensures that the chlorine is always being pulled into the chlorine feed system. Most modern systems have emergency shutoff valves which prevent the flow of chlorine if a leak is detected. Chlorine systems will also include alarms, leak detectors, and repair kits.

Other safety measures are also required when working around chlorine gas. For example, when switching tanks, the operator should be wearing a self-contained breathing apparatus (SCBA). As noted earlier, a 0.1 part per million dose of chlorine gas can be immediately fatal.

Hypochlorination

Gaseous chlorine is the strongest and least expensive form of chlorine even with the required use of bases such as caustic soda to increase the pH of the finished water. However, due to safety concerns associated with the use and transportation of gaseous chlorine, Hypochlorination is becoming more common in water treatment. When building or redesigning a treatment plant, an analysis of the costs and benefits of each type of chlorine will determine which type the facility will use.

The two types of hypochlorite used are calcium and sodium. Calcium hypochlorite is a whiteish-yellow dry chemical. It contains 65% available chlorine. Since it is highly reactive with organic compounds, special requirements must be used when storing it. Additionally, it is flammable if enough heat and oxygen are added. Calcium hypochlorite should always be added to water and not the opposite. Generally, Calcium hypochlorite is primarily used for disinfecting new and repaired water mains and storage tanks. It is not used for the day-to-day treatment of finished water.

Sodium hypochlorite is a clear to yellowish liquid available in a variety of different concentrations. Household sodium hypochlorite, known as bleach, is available in a 5% solution. Treatment plant operations will use an industrial-strength concentration of 12.5% available chlorine. Sodium hypochlorite does not have special storage requirements, but it is a strong base at 9-11 on the pH scale so it is highly corrosive. Additionally, the chemical loses its effectiveness over time in storage. Stored for a month, the chemical can lose 2% to 4% of its chlorine content. Direct exposure to sun and/or heat will further the loss of available chlorine. Therefore, it is recommended that sodium hypochlorite be stored no more than two weeks (in the event the treatment plant is offline) in a temperature-controlled room to prevent excessive strength loss.

Cylinder Tank Safety and Connecting

When attaching a chlorine regulator to a tank to go into service, safety precautions must be used. Any time you are dealing with gas chlorine the operator must wear a respirator, but a SCBA is recommended. The SCBA has a 30-60 minute supply of oxygen. SCBA's operate under positive pressure, so in the event of a leak, no chlorine gas will be able to enter your mask, assuming a proper seal of the mask has been established. Along with a respirator or SCBA the operator should wear gloves and a long sleeve shirt when changing cylinders or tanks.

A new lead washer should be used every time a new tank is put into service. Inspect the lead washer for any deformities, cracks, or bends. Washers must be thrown out after one use. When the chlorine tank valve is first opened, rapidly open and close the valve. Use an ammonia solution near the valve and tubing to check for leaks. The ammonia solution used is commercial Be ammonia. If chlorine is present, you will see a white cloud. In the event of a white cloud, a leak is present. The operator will have to remove the regulator from the valve and use a new lead washer.

Lastly, be vigilant. Changing chlorine cylinders will become routine as it is something you will have to do often if your facility uses chlorine gas. Chlorine gas is a highly dangerous chemical that must be respected. Forgetting how dangerous it is, could lead to serious harm or death for you or your co-workers.

Chapter Review

1. The form of Chlorine which is 100% available chlorine is?
 1. Sodium hypochlorite
 2. Calcium hypochlorite
 3. Calcium hydroxide
 4. Gaseous chlorine
2. What is the minimum amount of chlorine residual required in the distribution system?
 1. There is no minimum
 2. mg/L
 3. 0.2 mg/L
 4. mg/L
3. What is the approximate pH range of sodium hypochlorite?
 1. 4-5
 2. 6-7
 3. 9-11
 4. 12-14
4. What is the typical concentration of sodium hypochlorite utilized by water treatment professionals?
 1. 5%
 2. 65%

3. 100%
4. 12.5%
5. Chlorine demand refers to _____.
 1. Chlorine in the system for a given time
 2. The difference between chlorine applied and chlorine residual—usually caused by inorganics, organics, bacteria, algae, ammonia, etc.
 3. Chlorine needed to produce a higher pH
 4. None of the above
6. What is the most effective chlorine disinfectant?
 1. Dichloramine
 2. Trichloramine
 3. Hypochlorite ion
 4. Hypochlorous acid
7. What can form when chlorine reacts with natural organic matter in source water?
 1. Disinfectant by-products
 2. Sulfur
 3. Algae
 4. Coliform bacteria
8. What is the maximum withdrawal rate per day for a 150-pound chlorine cylinder?
 1. There is no maximum
 2. 20 pounds
 3. 40 pounds
 4. 50 pounds
9. What kind of solution is used to check for a gas chlorine leak?
 1. Sodium hydroxide
 2. Ozone
 3. Ammonia
 4. Calcium hypochlorite
10. Chlorine is _____.
 1. Heavier than air
 2. Lighter than air
 3. Brown in color
 4. Not harmful to your health
11. Chlorine demand may vary due to _____.
 1. Chlorine demand always stays the same
 2. Temperature
 3. pH
 4. Both 2 and 3
12. What effect does high turbidity have on disinfection?
 1. It can increase chlorine demand
 2. It has no effect
 3. It gives the water a milky appearance that will clear out after some time
 4. You must increase the temperature of the water

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Index

C

chlorine

[9: Chlorine](#)

coagulation

[5: Coagulation and Flocculation](#)

confined aquifers

[1.2: Source Water Quality](#)

E

enhanced coagulation

[5: Coagulation and Flocculation](#)

F

floc

[5: Coagulation and Flocculation](#)

flocculation

[5: Coagulation and Flocculation](#)

H

hydrologic cycle

[1.2: Source Water Quality](#)

N

nonstable solids

[5: Coagulation and Flocculation](#)

S

sedimentation

[6: Sedimentation](#)

sedimentation basins

[6: Sedimentation](#)

sedimentation zones

[6: Sedimentation](#)

specific capacity

[1.2: Source Water Quality](#)

surface water treatment rule

[2.3: Microorganisms](#)

T

total coliform rule

[2.3: Microorganisms](#)

[8: Disinfection](#)

U

unconfined aquifers

[1.2: Source Water Quality](#)

W

water quality monitoring

[2.2: Water Chemistry and Standards](#)

wells

[1.2: Source Water Quality](#)

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